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1916.

COMMONWEALTH OF AUSTRALIA



Clerk of the Senate.

27-9-16.

PARLIAMENTARY STANDING COMMITTEE ON PUBLIC WORKS.

REPORT

TOGETHER WITH

MINUTES OF EVIDENCE

AND AN

APPENDIX

IN REGARD TO THE

PROPOSED EXTENSION OF THE PINE CREEK-KATHERINE RIVER RAILWAY SOUTHWARDS SO FAR AS BITTER (MATARANKA) SPRINGS.

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MEMBERS OF THE PARLIAMENTARY STANDING COMMITTEE ON PUBLIC WORKS.

(First Committee.)

EDWARD RILEY, Esquire, M.P., Chairman.

Senate.

Senator the Honorable John Henry Keating,
Senator Patrick Joseph Lynch (Vice-
Chairman),
Senator William Harrison Story.

House of Representatives.

James Edward Fenton, Esquire, M.P.,
William Fyfe Finlayson, Esquire, M.P.,
The Honorable Henry Gregory, M.P.,
Sydney Simpson, Esquire, M.P.,
William Henry Laird Smith, Esquire, M.P.

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PARLIAMENTARY STANDING COMMITTEE ON PUBLIC
WORKS.

PROPOSED EXTENSION OF THE PINE CREEK-KATHERINE RIVER
RAILWAY SOUTHWARDS SO FAR AS BITTER SPRINGS.

REPORT.

EXTRACT FROM THE VOTES AND PROCEEDINGS OF THE HOUSE OF REPRESENTATIVES.

No. 96 of 5TH NOVEMBER, 1915.

5. PUBLIC WORKS COMMITTEE—REFERENCE OF WORK.—KATHERINE RIVER TO BITTER SPRINGS RAILWAY.—
The Order of the Day having been read for the resumption of the debate on the following motion of
Mr. O'Malley, That, in accordance with the provisions of the *Commonwealth Public Works Committee
Act 1913-1914*, the following work be referred to the Parliamentary Standing Committee on Public
Works for their Report, viz.:—

Extension of the Pine Creek-Katherine River Railway southwards so far as Bitter Springs.

Debate resumed.

Mr. Poynton moved an amendment, That the words "and from Oodnadatta to Alice Springs in a
northerly direction" be added to the motion.

Debate continued.

Amendment ruled out of order.

Debate on original question continued.

Original question—put and passed.

THE PARLIAMENTARY STANDING COMMITTEE ON PUBLIC WORKS, to which
the House of Representatives referred for consideration and report
the question of the proposed extension of the Pine Creek-Katherine
River Railway southwards so far as Bitter Springs, has the honour
to Report as follows:—

INTRODUCTORY.

1. On the 1st October, 1889, the State of South Australia, as custodian of the
Northern Territory, opened for traffic about 145½ miles of railway from Darwin to Pine
Creek as the first section of the line which it was intended should link up the Northern
Territory with South Australia.

2. When the Commonwealth took over the Northern Territory on the 1st
January, 1911, one of the conditions of the transfer was that the Commonwealth
should—

"Construct or cause to be constructed a railway line from Port Darwin
southwards to a point on the northern boundary of South Australia
proper"

and a section of 54½ miles to carry the line southwards from Pine Creek to Katherine
River is being proceeded with. In November, 1913, the field survey work was completed
and the line is now under construction.

PRESENT PROPOSAL.

3. The present proposal is that the line now under construction between Pine
Creek and Katherine River ~~should be~~ continued from the left or southerly bank of the ^{shally}
Katherine River for a further distance of 63 miles 65 chains in a south-easterly direction
which will carry it to Bitter Springs.

4. It is the intention to use 60-lb. rails and fastenings, and to construct the line
so that the earthworks, bridges, water-ways, and sleepers will be suitable for the
4 ft. 8½ in. gauge, and will permit of the rails, though laid temporarily to a 3 ft. 6 in.
gauge, being widened at any time to the 4 ft. 8½ in. gauge.

5. The proposed line is to be a single track with all stations on the rail level;
the ruling grade is to be 1 in 100, and the sharpest curve will have a radius of 40 chains.

ESTIMATED COST.

6. The Departmental estimate of the cost of the work is set down at approximately £320,000, including provision for station accommodation, telegraph line, and water supply; and it is stated that the work can be completed in about 18 months from the date of commencement.

ESTIMATED REVENUE.

7. The revenue which the Department estimate will be derived from the construction of the proposed railway amounts to approximately £12,000 per annum; the estimated cost of traffic, locomotive, working, and general supervision would be approximately £6,500 per annum; while the estimated cost of maintenance is set down at approximately £5,300 per annum.

8. The total revenue expenditure including maintenance would thus amount to about £11,300 per annum, so that not taking into consideration interest on capital cost the revenue would approximately balance the expenditure.

ALTERATION OF NAME—"BITTER SPRINGS."

9. On the 12th January, 1916, a notification appeared in the Press to the effect that, on the recommendation of the Administrator of the Northern Territory, the Minister for External Affairs had approved of the alteration of the name "Bitter Springs" to "Matranka Springs." The reason for the change was stated to be that the Springs were not bitter but fresh, and the name therefore was misleading.

10. As the locality was called "Bitter Springs" in the reference of the proposed work to the Committee, and referred to as such by witnesses throughout the inquiry, the old name has been retained in this Report to avoid confusion.

COMMITTEE'S INVESTIGATIONS.

11. In the course of its inquiries the Committee ascertained that the Royal Commission on Northern Territory Railways and Ports was favorable to the railway taking a westerly swing from the Katherine to the neighbourhood of Willeroo, subsequently returning to the overland telegraph route in the vicinity of Renner Springs.

A further suggestion was made that the railway should follow a "central route" about mid-way between the westerly swing and the so-called easterly swing *via* Bitter Springs.

12. The Committee studied the various reports dealing with the Northern Territory which have been issued from time to time, and took evidence in Melbourne and Adelaide from persons acquainted with the Northern Territory generally, and with that particular section of the country proposed to be traversed by the railway under consideration.

13. As there were still several important matters upon which the Committee had no satisfactory information, a Sectional Committee, comprising three members of the Committee, was constituted for the purpose of visiting the Northern Territory and reached Darwin on 12th July, 1916.

14. Advertisements were inserted in the Darwin Press intimating that the Sectional Committee was desirous of obtaining information in regard to the proposed railway and the country to be traversed, evidence was taken wherever practicable, and in addition the several members of the Sectional Committee by informal inquiry and conversation at all times sought to acquaint themselves with the condition of affairs in the Northern Territory and the nature of the country in the various parts.

15. A journey was undertaken over the whole of the railway open to traffic and continued to the head of the line under construction, and an inspection was subsequently made of the country through which would pass each of the suggested routes for the railway from the Katherine River southwards.

16. The Sectional Committee returned to Sydney on the 20th August, 1916, and on the 21st idem presented its report to the General Committee.

SLEEPERS.

17. In the course of its inquiry the Committee ascertained that when the first section of the line from Darwin to Pine Creek was constructed by the South Australian Government steel sleepers were used, and, with the exception of some within a short distance of the coast, are still in good condition. It was the intention of the Commonwealth to continue the use of steel sleepers, and portion of the section from Pine Creek to Katherine River was so laid. Owing, however, to the rapid increase in the price of steel sleepers from 10s. to 19s. each, it was decided by the Department to use wooden sleepers for the remainder of the section, and a similar type of sleeper is proposed for the section now under consideration.

18. These sleepers will be 8 ft. 6 in. long x 9 in. x 5 in.—a size sufficient to take a 4 ft. 8½ in. track—and it is intended that they shall be laid in such a manner that the gauge may in future be altered from 3 ft. 6 in. to 4 ft. 8½ in. by moving one line of rails only.

RAILS.

19. From Darwin to Pine Creek the track is laid with 41-lb. steel rails, but it is intended to use 60-lb. rails on the Katherine River-Bitter Springs section.

20. Eventually, when the gauge is altered to 4 ft. 8½ in., it is proposed to have the whole line laid with 60-lb. rails and to utilize the lighter rails for any branch lines which may be built.

BRIDGES.

21. The estimate set down for the construction of bridges and culverts is £13,362. It is stated, however, that there are but few engineering difficulties on the proposed route and that there will be only one bridge of any consequence, namely, that across the King River. This bridge will consist of three 40-ft. spans and be constructed on concrete piers with steel girders at an estimated cost of £1,920.

22. The balance of the amount estimated is to cover provision for the remaining smaller bridges and concrete box drains.

EMPLOYEES.

23. It was stated in evidence that there are 302 men working on the line to the Katherine River who will be available for work on the extension to Bitter Springs if the construction of that extension be not delayed too long.

24. Of this number the majority are foreigners, consisting principally of Greeks, Italians, Maltese, Patagonians, Russians, and Spaniards.

25. Inquiries were made as to the possibility of these men remaining permanently in the Northern Territory, and the Committee was informed that there is little likelihood of any number of the men employed on the railway becoming permanent settlers.

ROUTES SUGGESTED.

26. Possibly few people realize that this immense Territory, comprising an area of 523,620 square miles, is more than four times the size of the United Kingdom, and greater than the combined areas of the United Kingdom, Belgium, France, Italy, Roumania, and Switzerland. Further, that the total population of this area, exclusive of aborigines, is less than one person per 100 square miles.

27. It should, therefore, be understood in dealing with this proposal that an extension of 60 or 70 miles of railway from the Katherine southwards, whether on the central route or the easterly or westerly swing, will, of itself, contribute little towards the early development of the country, and in addition it is very improbable that such a line will become a payable proposition for many years.

28. The Committee, therefore, considered the matter chiefly from the point of view of the main North-South line, and endeavoured to satisfy itself as to the best route which such a line should follow for the first 60 or 70 miles after leaving the Katherine.

29. Briefly, the class of country along the several routes as observed by the Sectional Committee may be outlined as follows:—

- (a) *Westerly Swing*.—Medium to good pastoral country, soils of excellent quality excepting on the quartzite ridges, some stretches of sand and sandy loam merging into basalt country in the vicinity of Willeroo. This district contains a good deal of broken country, and a few miles east of Willeroo can be seen a line of cliffs, often sheer, which stretch in a continuous line along a general north and south direction. The country is dotted with table-topped and conical-shaped hills, probably the outliers of a former extensive tableland.

The westerly route shown on the plan issued by the Royal Commission on Northern Territory Railways and Ports is impracticable, excepting at very heavy cost. By making certain deviations, however, it would be possible to construct a westerly swing for the main line, but this would increase the length of the line owing to the broken nature of the country traversed in certain parts within the first 100 miles. Additional bridging would also be required, so that the construction of this swing would not only be longer but much more costly than on the route of the easterly or overland telegraph line to Mataranka. No permanent surface waters of any extent were found between the Katherine River and Willeroo.

- (b) *The Central Route*.—Would traverse country generally limestone and quartzite, medium pastoral land, with patches of black soil, hard and uneven during the dry season, but which would seem to be subject to periodical inundations and be soft and boggy during the wet season. Several deep dry creeks have to be crossed and no large supply of permanent surface water exists.

- (c) *The Easterly Swing*.—Would traverse country generally limestone and quartzite, with stretches of open forest and park-like lands, and good pastoral country near Mataranka. The route is fairly well watered; only one bridge of any magnitude would be required, and the country generally appears to be free from engineering difficulties.

PROSPECTIVE REVENUE.

30. The Committee made inquiries as to the class of industry likely to be most successful in developing the Territory, and discussed the bearing such industry might have on the prospective receipts from the railway, and is inclined to the opinion that for many years to come the railway must depend on pastoral holdings and mining for its revenue.

31. It was stated in evidence that under existing conditions a man would require 300 or 400 square miles of the best pastoral land in the Territory to be assured of a reasonable return for his labours, although naturally—with the erection of fences, the granting of railway facilities, and the provision of water supplies—the minimum size of a payable pastoral holding could be substantially reduced. Agriculture will probably develop later as the country becomes more settled and a market is available for the products, as the Committee had ample proofs that with a good supply of water the soil in parts of the country visited is good enough to grow almost anything.

BORES.

32. With the extension of a definite plan of the policy now adopted to some extent of sinking bores much might be done with country which is at the present time precluded from serving any useful purpose by reason of its lack of water during the dry season. South of Mataranka three bores already completed have, at depths of 108, 175, and 190 feet respectively, struck ample supplies of water of excellent quality—discoveries which must materially aid the development of that portion of the country.

MINING.

33. Apart from several mines on which more or less developmental work is being carried out, there is every indication of the existence of good tin-bearing stone over an area of perhaps 20 square miles at Marranboy. Although some of the mines are even now being worked profitably, the cost of transport under present conditions makes it impossible that the field should be developed as it should be.

34. It was stated in evidence that the price of carting to the railway at Pine Creek is £12 10s. per ton, while it costs approximately £20 per ton to place material on board at Darwin. It is possible that the rate to the railway may fall to £5 or £6 per ton upon completion of the railway to Katherine, while if the section to Bitter Springs be constructed and come within 12 miles of Marranboy the rate to the railway should be reduced to £2 or £3 per ton.

TIMBER.

35. Inquiries made convinced the Committee that no suitable local timber in any quantity can be economically obtained for sleepers. Plenty of good sand and gravel, however, are available for concrete work.

COMMITTEE'S RECOMMENDATIONS.

SLEEPERS.

36. The Committee is of opinion that no good purpose would be served by laying wooden sleepers 8 ft. 6 in. x 9 in. x 5 in. as now proposed with a view to their being utilized when the gauge is widened to 4 ft. 8½ in.

37. As a number of years will probably elapse before the gauge of the railway is altered from 3 ft. 6 in. to 4 ft. 8½ in., and assuming that the life of a wooden sleeper in this district is about 10 years, it is unlikely that any wooden sleepers now laid would be in a suitable condition to be utilized for the broader gauge.

38. It is therefore recommended that if wooden sleepers be used the size 7 ft. x 9 in. x 4½ in., which is suitable for the 3 ft. 6 in. gauge, be adopted at a saving of approximately £180 per mile.

39. It was stated in evidence that some 50,000 8 ft. 6 in. x 9 in. x 5 in. powdered karri sleepers have been supplied for the construction of the Pine Creek to Katherine River line. As it was ascertained that white ants abound along the route under construction, and it is known that untreated karri is speedily affected by dry rot when used as sleepers, and as the evidence submitted showed that no effective time test had yet been made as to the life of wooden sleepers in the Territory, the Committee considers that, if timber sleepers are used, only the best and most durable hardwood timbers should be utilized.

40. The Committee was, however, greatly impressed with the samples of concrete sleepers which it inspected in the tramway track in Adelaide, the railway track in the vicinity of North Adelaide, and in the railway track in the vicinity of Leichhardt, New South Wales, where they are undergoing tests under working conditions.

41. As it was stated in evidence by the Engineer-in-Chief, Commonwealth Railways, that about six months would elapse from the time this section is authorized until the time when any sleepers are laid, it is recommended that a supply of concrete sleepers be made at once, using sand and gravel obtainable locally, and that experiments be made with these sleepers on the section now under construction, particularly on steep banks and sharp curves, with a view to the adoption of this class of sleeper if suitable on the Katherine River-Bitter Springs section. In conducting these experiments it is suggested that the manufacture be carried out on such a scale as will not only give a good idea of the utility of the sleepers but of the cost of such sleepers with a view to their use on subsequent sections of the main line.

DEFENCE ASPECT.

42. In the event of Australia ever being called upon to resist an enemy attack on her northern coast, it is generally admitted that a railway connecting Darwin with the south, even if it be only a single narrow-gauge track, would be of considerable

advantage from a defence point of view. The greatest safeguard against aggression in that quarter, however, would be the existence of a considerable permanent white population, and for that reason the Committee is of opinion that it is of vital importance that settlement of the country should be, as far as practicable, contemporaneous with the construction of the railway.

EMPLOYEES.

43. The Committee is aware that a number of the foreigners employed on the line at present under construction, by working under what is known as the butty gang system, are carrying out the work more satisfactorily and earning more money than those employed on day work, and, as the result means better and quicker work and a better class of employee, suggests that it would be an advantage to the Commonwealth and to the men themselves if this system were extended wherever practicable, and the Australian and British workmen encouraged to adopt it.

Further, in order that the construction of the railway might be made to proceed hand in hand with the settlement of the country, it is suggested that an endeavour should be made to arrange with the majority of the men brought to the Northern Territory for work on the railway to bring their wives and families with them, and that they should be encouraged to take up holdings so that their families could remain on such holdings while they are working on the railway—thus forming the beginning of a permanent settlement.

At the present time the vast majority of the men employed are single, and even if they are saving money it is obvious that the money saved is of no use to the Territory. Some men work for six or eight months or a year and then go away and take their savings with them, while it is well known that others are sending large sums of money to foreign countries every month.

As every Australian State is encouraging immigration, it is improbable that the Territory can be settled by an overflow from the south, and in any case it is doubtful policy to encourage the transfer of people from one part of an empty country to another part of an empty country. It is therefore suggested that people from European countries be encouraged to come to the Territory by guaranteeing them work if suitable, by paying their passages in part and making it a debt against them until they are settled in the country, but at the same time holding out hopes that if they settle permanently the debt may be foregone. In the best interests of the Territory, however, and of the men themselves, it is advisable that all assisted immigrants be married and accompanied by their wives and families, or be closely related to those already in the Territory.

CONCLUSION.

44. A study of the evidence given, supplemented by the Sectional Committee's investigations and inspections, led to the conclusion that the central route offered no advantage over either an easterly or westerly swing, and it was consequently abandoned.

45. The line on the westerly swing offers some immediate temporary advantage by reason of the fact that it taps portion of the Victoria River land, which is excellent pastoral country, and at the present time carries the bulk of the stock of the Territory. There are no serious difficulties, however, in driving cattle from the Victoria River country to the railway, and these difficulties will be lessened by the action of the Government in providing a new stock route with sub-artesian and surface-water supplies at regular intervals.

46. On the other hand, the easterly swing commended itself to the Committee for the reasons that it would traverse comparatively well-watered country with fewer engineering difficulties than either of the other suggested routes; it would approach the mining field at Marranboy, and some good pastoral and agricultural land in the vicinity of Matamanka; and would open up possibilities of developing the country to the south, which would not be practicable in the absence of a railway.

47. The Committee is therefore of opinion that, as a section of the main line, any extension of the Pine Creek-Katherine River railway southwards should proceed via Bitter Springs.

RESOLUTION PASSED.

48. The decision arrived at by the Committee is shown in the following extract from its Minutes of Proceedings:—

Mr. Finlayson moved—That the proposed extension be approved.
Seconded by Senator Story.

Mr. Sampson moved as an amendment—That all the words after "that" be struck out with a view to the insertion of the words—"the Committee approves of the construction of the line to Bitter Springs as a part of the main line, but considers that as this section does not promise to give any largely increased traffic in the near future or to serve as a feeder to the Darwin Katherine line, its construction on local grounds is not urgent."

Seconded by Mr. Gregory.

The Committee divided on the amendment.

Ayes (3).
Mr. Fenton,
Mr. Gregory,
Mr. Sampson.

Noes (5).
Senator Story,
Senator Lynch,
Mr. Finlayson,
Mr. Riley,
Mr. Laird Smith.

And so it passed in the negative.

The Committee then divided on the motion.

Ayes (5).
Senator Story,
Senator Lynch,
Mr. Finlayson,
Mr. Riley,
Mr. Laird Smith.

Noes (3).
Mr. Fenton,
Mr. Gregory,
Mr. Sampson.

And so it was resolved in the affirmative.

Edward Riley

EDWARD RILEY,
Chairman.

Office of the Parliamentary Standing Committee on Public Works,
120 King-street, Melbourne, 30th August, 1916.

MINUTES OF EVIDENCE

(Taken at Melbourne.)

WEDNESDAY, 10TH NOVEMBER, 1915.

Present:

Mr. RILEY, Chairman;
Senator Keating, Mr. Finlayson,
Senator Story, Mr. Laird Smith.
Mr. Foulton,

Norris Garrett Bell, Engineer-in-Chief and Acting
Commissioner Commonwealth Railways, sworn
and examined.

1. To the Chairman.—I produce a tracing showing the general direction of the proposed extension of the railway from Katherine River to Bitter Springs. I have never been in the Northern Territory, and the information I have is obtained from the reports of my officers. These reports show that the proposed extension will be 63 miles 65 chains in length, and the estimated cost £320,000, or at the rate of about £5,000 per mile, of which nearly half will be for rails and sleepers. It will be possible to build the line in, say, eighteen months. There are no engineering difficulties, and there will be only one bridge of any consequence over the King River. This bridge, to consist of three 40-foot spans, will be constructed on concrete piers with steel girders. It will be a permanent structure, and is estimated to cost about £1,920. The extension from Pine Creek to Katherine River is a single line with 60-lb. rails. It is being built in accordance with the provisions of the Act, so that it may readily be converted from the 3-ft. 6-in. gauge to a 4-ft. 8½-in. gauge. All earthworks and waterways have been taken out in sufficient width, and the sleepers are sufficiently long to enable the line to be widened to the Commonwealth gauge. The traffic on the existing railway from Darwin to Pine Creek has been unduly inflated of late owing to the carriage of material required for the section now under construction. I estimate the probable earnings of the new section by taking the earnings per mile of the existing line, and making a slight allowance for cattle traffic, which we expect when the most works at Darwin are finished. I went into the matter, and I anticipate that the expenditure will be £11,800 per annum (traffic, locomotive, and maintenance charges), and that the earnings will be about £12,000 per annum, so that revenue and expenditure will just about balance. This does not provide for interest on capital. After the next section is completed, I expect the earning capacity of the railway will improve so that the extension will probably increase the traffic very considerably, as it will open up cattle and mining country. I cannot speak about the mining prospects, because I have no knowledge on that subject; but I know mines do exist, and I anticipate that they will be developed as a result of this railway construction. My reports do not show that there is any special scarcity of water, as the country to be traversed is comparatively well watered, compared with the east-west railway, at all events. The material for the railway, rails and sleepers, are procurable in Australia now, rails being manufactured either at

Newcastle or Lathgow. We call public tenders for sleepers, and get them from the different States. All this material is sent to Darwin by steamer. The first section of the railway from Darwin to Pine Creek is laid with steel sleepers, and also portion of the second section, which we are now building, but after war broke out we were prevented from getting any more steel sleepers, because the price has nearly doubled. It is probable, therefore, that we shall be compelled to use wooden sleepers unless we can get the steel sleepers made for us at Newcastle. I do not think the difficulty with regard to white ants will be so serious on this section as on the first, because the rainfall is less. I know that several railway lines have been built in Queensland in country where the white ants are bad, and I understand that, although timber sleepers are used, they have not been destroyed to any extent by the white ants. As I have already said, I have no personal knowledge of the country, and I have to depend on reports from the surveyors, and Mr. Hobler, who went over the ground. We have about 500 men working on the section between Katherine River and Pine Creek, and we desire to keep those men employed, if possible. We do not wish to see any break in the work between the two sections. Except in the case of certain Welshmen and Spaniards, the Department did not pay the fares of the men who are working there, nor was a guarantee given as to the length of time they would be working. I expect that we will finish the earthworks of the section now in hand by the end of the year, and if no further work is authorized perhaps 150 men will be displaced, as we could not put them on work elsewhere, but the balance of the men will be engaged in plate-laying and finishing up the line generally.

2. To Mr. Finlayson.—I anticipate that the section of the railway now under construction will be finished by about June, and if we were able to go on with the earthworks for the new section the plate-laying gang would be able to go right ahead. The terminus of the present section is across the Katherine River, because it would not be wise to stop the line on the northern bank of the river, as the people to the south would not have access to the line. This line is on the 3-ft. 6-in. gauge, and it forms a section of the north-south railway being projected to connect South Australia with the Northern Territory. In this proposed section, there is no deviation of the north-south line. The most suitable point for a connexion with the Queensland system will probably be at Daly Waters, 367 miles from Darwin, and 100 miles south of Bitter Springs. Steel sleepers, of course, remove all difficulty with white ants, but they are fairly costly, and since the present section was started they have increased in price from 10s. to 19s. each. We are now laying 50,000 powdered karri sleepers in the road, and in a little while I shall be able to state definitely how far powdered karri sleepers as a remedy for the white ant difficulty. I know of no more expeditious way of transporting the materials for this railway than by steamer to Darwin. We have sufficient rolling stock to handle the existing traffic, but as we come further south with the railway we will have to increase the stock.

As a matter of fact, a number of cattle trucks and a half trucks are being built in Queensland for this railway, and I think there will then be sufficient to handle all the increased traffic on the extra 60 miles of line. These are the only additions to rolling stock which at present have been arranged for. I am informed that the railway will pass through pastoral and mining country chiefly, and that there is little prospect of agricultural development. It appears, therefore, that we will have to depend on the cattle traffic, and the development that may take place in the mining industry. At present, the cattle traffic over the existing line is very small, chiefly for the butchers at Darwin, and Messrs Vestey Brothers are building meat works at Darwin, and they expect to be killing in March. While they are carrying out freezing operations they expect to truck about 320 head per day, and we are making preparations, as far as possible, to handle that traffic. When I said it would take eighteen months to build the proposed section of the railway, I meant that it would take eighteen months from the time we start work. If I have authority to start this section in the new year it will take eighteen months from that date to finish it. I am relying upon Mr. Hobbler's reports in regard to the final route of the line. We have a railway surveyor there, a fully qualified man, and Mr. Hobbler, who has been over the survey, is quite satisfied with the route. Mr. Hobbler will arrive in Melbourne to-day so that, if the Committee desires, he will be able to give the latest information as to the route and the nature of the country.

3 To Mr. Lord Smith.—At the Katherine River, the telegraph line will be about 2½ miles away from the proposed extension. I do not know whether King River is a permanent stream or not, but I imagine it is not. The line from Darwin to Pine Creek was laid down with steel sleepers 30 odd years ago, but the price recently rose too high to enable us to finish the second portion of the Pine Creek to Katherine extension with steel sleepers also. There are several hundreds of miles of steel sleepers in South Australia, and I think about 100 miles also in Queensland. Where steel sleepers are used the line does not "creep" very much. The proposed section from Katherine River to Bitter Springs will be a comparatively straight line, and what curves there are will be easy curves. The anticipated traffic will not warrant the use of heavier rails, as we are satisfied that the 60-lb rails will carry the traffic for very many years. I might mention that the line to Bourke, in New South Wales, is laid with 60 lb rails, and that a similar weight of rail is used for a considerable distance on the line between Adelaide and Melbourne.

4 To Senator Story.—We have had some difficulty in getting supplies of rails and sleepers landed at Darwin. Freight rates are very high, and shipping is scarce. I anticipate this difficulty will continue while the war lasts. A steamer runs from Fremantle to Darwin, and we can get a certain quantity of supplies up that way. Then there are the Japanese boats trading to Darwin, but it is somewhat difficult nowadays to get a ship with a full cargo. I dare say it would be worth while considering whether we could charter a steamer for this purpose. I am sorry I cannot give the Committee accurate details as to the ruling grade, but Mr. Hobbler will be able to supply this information. I know the grade is an easy one. I made inquiries as to whether steel sleepers could be obtained from the Broken Hill Proprietary Works at

Newcastle, and learned that they have more orders than they can cope with for steel rails. I understand the paper bark tree—known as the giant linc—growing in portions of the Territory is not ant-proof, and therefore it would not be suitable for sleepers. The ironwood tree gives fine timber, only it is very scarce. I have been making inquiries as to whether sleepers could be got from this tree, and the engineer in charge of the line has endeavoured to get people to quote for a supply of these sleepers, but, so far, he has not been successful. I know that the white ants got into the paper bark tree quickly, and that it also becomes affected with dry rot. I think 60 lb rails are heavy enough to enable express trains to run over the line, but this will depend entirely on the weight of engine used. Provided that the locomotive is not too heavy, there is nothing to prevent express trains running over a line laid with 60 lb rails. For several years I have been looking into the question of using reinforced concrete for sleepers, and nearly every engineering periodical published makes some reference to this class of sleeper, but, so far, a successful design has not been perfected. It has been found that the vibration shatters the sleepers. Italy went in largely for this class of sleeper several years ago, and the other day I read in an engineering periodical that they had been taken out because they had proved unsatisfactory. I understand, however, they are used with success on street tramways, but it seems they will not stand heavy and fast railway traffic. They are also very heavy. An ordinary maintenance gang of three or four men would not be able to handle them, so a special gang would have to be employed to lift the line and pack it. In the estimate of cost of the section under construction, as well as the proposed section, there is included a charge of 1d per ton per mile for the carriage of the material, so that the further we get away from Darwin the more expensive will the cost of construction become. On the east-west railway we are charging only 3d per ton per mile, and that just about pays the cost of working. I cannot tell the Committee what is the freight by steamer from Newcastle to Darwin, but I will ascertain and let members know what we paid for the last coal we obtained from that port. I anticipate that the freight on rails from Newcastle to the head of the road at Pine Creek would be rather heavier than on coal. If you ask me for an opinion as to which would be the cheapest and best way to construct a line, whether from south to north, or from north to south, I would say that it could be constructed more cheaply from the southern end, but I have not gone into the matter. The rate of wages is very high in the north, and, of course, if the line were started from the southern end, wages would increase the farther north we proceeded. At present, we are paying 14s a day in the Northern Territory. We have not had very much sickness up to the present among the men. A few cases of malaria have been reported, but they have not been serious. If we started the line at the southern end we could land material more cheaply at Port Augusta than at Darwin, because vessels call at Iron Knob for ore for the Newcastle Steel Works, and the material would, in a sense, be back-loading for them. I think, however, that the line constructed from the northern end will earn revenue more quickly than would the line from the southern end, because the meat works are established at Darwin, and as a result, there should be a big cattle traffic. I have no personal knowledge of the traffic in the vicinity of Macdonnell Ranges. I have heard that it is a survey

from Oodnadatta northwards to Pine Creek, and I am going to have an inspection of the country made. I hope to be able to go myself.

5 To Mr. Fenton.—The rolling stock employed on the existing railway and that which we are having constructed in Queensland to cope with the increased traffic could not be converted to the 4 ft 8½ in gauge, but probably it could be sold to the Queensland Government again, as there is always a demand for that class of rolling stock. The section under construction is being carried out by day labour, which is proving satisfactory. We have had very little trouble with the men. Recently their wages were increased very considerably. It is proposed to construct the section from Katherine River to Bitter Springs also under the day labour system. The section under construction includes the cost of a bridge over the Katherine River, so it would not be correct to say that the total of £320,000 which the proposed section is estimated to cost will include nearly £10,000 for a bridge over the Katherine. There is a deviation of the line there, and it is proposed to put in a low-level bridge until the line is converted to the 4 ft 8½ in gauge. This low level bridge will carry the traffic in the meantime, and it will save about £40,000. If the high level bridge were constructed, it would be on concrete pillars with steel girders, because the banks at that point are very steep. By a deviation, however, we can cross the river where the banks are lower, although the bridge will then be subject to occasional floods. It will be a wooden structure. There is provision only for single line, and though the bridge could readily be duplicated, this will not be required for many years. At the point where the high-level bridge will cross the river the distance is only about 12 chains, and the deviation will increase the distance by about 2 miles. I have not heard that the Katherine River has double banks, and the plans show one big channel with a high level bridge across. The present section of the railway takes a south easterly direction, as the country directly south is rather poor, the good country lying more in a south easterly direction.

6 To Senator Keating.—The present terminus of the line is at Pine Creek, 146 miles from Darwin. We are building 57 miles extension at present, and the proposed further extension from Katherine River to Bitter Springs is about 64 miles in length. No provision is being made for future duplications. All platforms are on the rail level. There are several sidings and loops for passing trains, and, as I have said, there is provision for widening the gauge to 4 ft 8½ in. All the necessary loops and sidings will be provided for in the proposed extension. The bridge over the King River will be constructed of concrete pillars with steel girders. This could be duplicated, if necessary. Where they have been tried, reinforced concrete sleepers have not been successful. I am told that a number have been tried in the Adelaide street tramway system. They have been tried also in America and other countries, including Italy, but have not been a success. I read recently that they had been taken out of the railways in Italy. I do not know of any special conditions in Italy, as compared with Australia, that would make them unsuitable for railway traffic. Steel sleepers have been largely used in South Australia, and, to some extent, in Queensland. The steel sleepers used on the line from Darwin to Pine Creek are trapezoidal, and those laid from Pine Creek to Katherine River are H-shaped, almost like a rolled steel joist. In India, cast iron sleepers, known as pot sleepers, are used. There are no grades of any consequence on the proposed

extension, nothing steeper than 1 in 100. It is all even country, and therefore the curves are also very flat. From the information I have received I gather that the country from the present terminus to the end of the proposed extension is lightly timbered. The estimated cost is based on the actual cost of the section which we are now constructing, so the figures are quite reliable. I get a monthly statement showing the cost of that work. From Darwin to Pine Creek the line is laid with 41 lb steel rails. Eventually, when the gauge is altered, it is proposed to have the whole line laid with 60 lb rails. We are using 60 lb rails on the section from Pine Creek to Katherine River. When the line from Darwin to Pine Creek is relaid with 60 lb rails the rails taken out might be used on branch lines, and for the loops and sidings, but the steel sleepers could not be so used, as they were made for the narrow gauge line.

7 To Senator Story.—The estimated cost includes ballasting with gravel. We have allowed for 1,450 yards per mile.

8 To Mr. Tuley.—The tracing I have exhibited indicates a branch line to the Marrabyn tin fields, but that is not included in the estimate. That branch line has nothing whatever to do with the proposal before the Committee, and it is shown on the tracing only as a possible route for a line to the Marrabyn fields, a suggestion being made that there should be a 2 feet line to assist in the development of the fields. There is no proposition before us at present for the construction of that railway.

Alfred Arthur Hunt, C.M.G., Secretary Department of External Affairs, sworn and examined.

9 To the Chairman.—I travelled through the Northern Territory recently, but I did not go over the route of the proposed line. I left the railway line at Katherine River, and followed the track to the Marrabyn tin fields, and thence I travelled to Bitter Springs, so I was not far distant from the route of the proposed extension. The impression I formed of the country was that it was good for steel raising, but, so far as I could judge, not suitable for agriculture under present conditions in the Territory. The country improves very much as one gets down towards Bitter Springs, in the neighbourhood of which there is the Government sheep farm at Manrakina. The Government desired to ascertain whether sheep would do well in that part of the Territory, because no sheep had been raised nearer than Avon Downs, which is away down on the Queensland border, at about latitude 21 south. They purchased 2,000 sheep from Avon Downs and these were taken in light stages overland to Bitter Springs, where land was resumed for the purpose of an experimental sheep station. The sheep arrived there in March last year, so they have been eighteen months on the property. So far as I could judge, they are likely to do very well, as they were in excellent condition, and the manager reported that, as far as quality of the wool is concerned, the experiment so far has been entirely satisfactory. When I was there they were lambing, and he expected a very fair percentage. It is hardly fair, however, to compare the conditions of that sheep station with the conditions of a properly organized station in more settled country, because there are no fences, and the sheep have to be shepherded. They are therefore moved about a good deal, and are not left to themselves as they would be on a properly equipped station,

but when I was there they seemed to be doing very well. In the neighbourhood of the Roper and Waterhouse Rivers I should say that there are admirable opportunities for small stock settlements, but I do not know that the country is so much better than agricultural land in other parts of Australia that the settlers could afford to pay the difference in the rates of wages and cost of transport. The country generally from the Katherine River to Bitter Springs is open, and presents no engineering difficulties. Water can be obtained in most places by sinking. All the streams run only in wet weather, but in most of them there are holes which provide permanent water. I think there is permanent water in such holes in Maudie's Creek.

10. *To Mr. Laird Smith.*—The King River is not a permanent stream, but like the others it contains waterholes where permanent water may be obtained. We camped for fifteen or eighteen hours at the King River alongside a good big waterhole, probably 50 yards long by 15 or 20 yards wide, and about 4 feet deep.

11. *To Mr. Fenton.*—We were at the King River about five months after last rains had fallen.

12. *To the Chairman.*—The Katherine River was a very tiny stream indeed when I was there, and it could be crossed on half-a-dozen stepping stones at the point where the road crosses it, the water-course then being only 12 to 15 feet wide and a few inches deep.

13. *To Senator Keating.*—The rainy season extends from about the middle of November to the end of March or middle of April.

14. *To Mr. Fenton.*—Judging by its banks, the Katherine River must be an enormous stream in flood-time and at the point where the tracks cross I should say it would be about 90 feet deep, and at least a quarter of a mile wide.

15. *To the Chairman.*—In my opinion the extension of the line from the Katherine River to Bitter Springs would only be worth considering as part of some bigger scheme to connect the Northern Territory with South Australia or with Queensland. The line from Pine Creek to the Katherine River was essential to get stock from all the country south of the river into Darwin, because the country from the Katherine to Pine Creek is difficult for drovers, there being practically no food. In normal seasons, however, there is good feed for drovers south of the Katherine, so that the bulk of the stock at present pastured in the country to the south of the latitude of the Katherine could be driven to the terminus of the section under construction. During my travels through the Northern Territory I went from Darwin to Herbert's station, on the Adelaide River, from Darwin to Laurie's and Sayle's stations, also on the Adelaide River; from Batchelor to Stapleton, and through the country in that locality; thence I went to the Daly River, and back to Brock's Creek, and on to Mount Bonny. Afterwards I went by sea to Bathurst and Melville Islands, thence to the East Alligator River to Cahill's station, and then across to Burundi; and later from Pine Creek to the Roper mission station. The Matarranka country struck me as being the best, the timber there is of a finer type than elsewhere. I inspected several mining fields, particularly that of Marranboy. 60 miles south-east of the Katherine River. This is quite an extensive tin-field, so far as I could judge, and the local mining people seemed very well satisfied with its prospects of

permanence. However, it is difficult country, and a good deal of capital will be required to develop it. The Government have erected a battery there for concentration purposes, and are prepared to deal with any quantities of ore that may be brought in.

16. *To Mr. Fenton.*—The miners on the Marranboy field are chiefly Australians, but, generally speaking, they are not a good class of miners, being mostly of the old prospector type—men who discover fields and then start somewhere else.

17. *To Mr. Finlayson.*—There is practically no settlement in the country I traversed except at Marranboy, where there are a good number of miners, and the men connected with the Government battery.

18. *To the Chairman.*—Many of the miners who have holdings at Marranboy have secured exemptions until the battery is in full working order, and in the meantime they are working in Darwin or on railway construction. Probably there are about 60 holdings, representing from 120 to 160 men. The proposed extension of the railway will assist the mining industry, because it will then be possible to get supplies up there more cheaply, especially if a tramway is constructed from the line to the field.

19. *To Mr. Fenton.*—The distance from the proposed extension of the railway to the mining field is about 12 miles. The cost of transport to Marranboy is one of the difficulties that prevent its development. There is an extensive stretch of sand to cover, and teamsters have the utmost difficulty in getting through it.

20. *To the Chairman.*—I gather from our experts that there is excellent ore at Marranboy, and that it probably goes down to a considerable depth, so I should say that the field will increase in importance in the future. It is not fair to compare the section at present under construction with that proposed to be constructed, because in the latter case the country is very much better in every way. The country improves immediately you get over the Katherine River.

21. *To Mr. Laird Smith.*—I am hoping, of course, that settlement will follow the line of the railway, but that has not been the experience in the Territory, and right from Pine Creek to Darwin—a distance of 160 miles—you will hardly see a dozen houses along the route of the railway.

22. *To the Chairman.*—I think it will be the policy of the Department to extend the sheep farming experiment, though no decision on the point has been given yet. We are going slowly. I think it is desirable that we should have another couple of years' experience to ascertain if there is anything in the soil or the grasses that will interfere with the development of the industry. The loss of the ironwood tree contains a rather bad poison, and sheep have to be kept off it. There may be other plants of a similar character, and of which we have no knowledge at present.

23. *To Mr. Laird Smith.*—The health of the men employed on the railway will probably bear very favorable comparison with the health of men employed on railway construction anywhere else in Australia. The bulk of the men are Australians, but they are a mixed community along the line. There are about sixty Maltese, and probably eighty to 100 Russians and Greeks. The majority of them, I should think, are not likely to become permanent settlers. The Australians are merely railway men who follow railway jobs from one place to another. They are mostly

single men who have their family connexions in the south. They work in that hot, uncomfortable country, and they get back south as soon as they have got together a decent choique. The Greeks and the Russians will probably leave as soon as they have made enough money also, but the Maltese may settle if some arrangement could be made to bring out their wives. Then we have some settlers from South America, including Spaniards and Italians. These people seem to offer the best prospect of becoming permanent settlers.

They seem to be keen about getting on the land, and they are working on the railway with the idea of accumulating a little money to enable them to do so. Most of them had very limited capital when they came, and the Department is encouraging them to work in this way so as to get money together, because from experience we know that men who go on the land without capital usually become Government pensioners. Some of these people from South America have brought their wives and families with them, and at present there are about fourteen married women and sixty children among them. I assume that when the proposed extension is put in hand these people will be transferred from the present work to the new section, and after they have earned sufficient money, probably they will settle on the land. They receive the same rate of wages as other men. They are not members of the union before they leave Darwin. We are not making any provision along the railway line for villages, because the country at present being traversed is too poor. The land improves from the Katherine River to Bitter Springs, but the men who are likely to be proposed later on to arrange for a small delegation from these people to visit different places, and there is a possibility of a settlement being established over on the Alligator River, because there the transport will be by sea, and, consequently, cheaper than by rail. I might mention that I am preparing a memorandum for the Minister expressing some general views about the prospects of settlement in the Territory. I do not see much hope for agricultural settlement. The existing line touches only one of the Government experimental farms. I refer to the Batchelor farm, the other farm is on the Daly River. The Batchelor farm is giving more encouraging results now than previously, but we have had no sort of difficulties with the men. We have a staff of only eight or ten men, but in the three years that have elapsed since the farm was established over 200 men have passed through it. Men come and take up work for a little while, then they go away again, or they get drunk and have to be discharged. We would gladly encourage married men to go there, but they do not seem to be any inclination on their part to do so. The country is healthy enough, but it is certainly uncomfortably hot. The products of the farm are chiefly hay and fodder, and we are allowed to sell to anybody. Vestor Brothers are taking a fair quantity.

24. *To Mr. Finlayson.*—I was in the Territory for about two months, from about the middle of August till the middle of October. During that time the weather conditions generally were suitable for travelling. We were blocked by rain once only for a few hours, but the weather was always very hot. From May or June to September the conditions are reasonably good for travelling, I think, but it depends where you want to go to. From September to December the water begins to get very scarce, but from about the latter month, there is abundance; and besides

there is a lot of black soil country over which it is almost impossible to travel. We used almost every known means of locomotion on my trip—train, motor-car, launch, horseback, and on foot. Unless the Territory is different from every other country in the world, I should say that the building of the railway will lead to development. Regarded as a main trunk line bisecting the Territory, or as a main trunk line connecting with some place in Queensland—Commonwealth has been suggested—I should say that the route of the proposed extension is the best; but unless regarded as part of the bigger scheme referred to I would say that the extension to Bitter Springs would not be warranted. As a link in the chain the proposed extension is in its proper place. I do not think that the proposed extension could with advantage be diverted in any other direction. In view of the fact that the country south of the Katherine River is better for droving than the country between Pine Creek and Katherine River, and that stock could travel the distance to Katherine River, the extension of the line to Bitter Springs is unnecessary for the development of the cattle industry to the west. There is land along the route of the projected railway suitable for settlement, but I do not know if it is available. I know, however, it is occupied on conditions that will allow of its acquisition. The whole of the property at Matarranka, the area of which is about 760 square miles, could be made available. I would not like to say, without looking at the detailed plan, exactly what land along the route of the railway would be suitable for small pastoral holdings. There does not appear to be much prospect of coal deposits being discovered in the locality. The Territory has been examined by geologists in many directions, but the only coal deposits I have ever heard of were over on the west near Anson Bay. None of the geologists' reports speak of the existence of coal along the railway line or near it, so all the coal required for the working of the line will have to be shipped to Darwin. A certain amount of work is being carried out with the object of discovering minerals in the Northern Territory, but our mining advisers recommend that we should work on the existing fields, and see if we cannot develop them to a lower depth rather than assist prospectors generally. In the neighbourhood of Pine Creek a lot of gold has been taken out, but the mines have mostly been abandoned at about 200 feet. The Government are now putting down bores, and are sinking shafts, for the purpose of testing whether the lodes continue or not. I am preparing a report for the Minister giving my impressions, and this will probably be available in two or three weeks.

25. *To Senator Storey.*—During my travels in the Northern Territory I saw very little timber that would be suitable for sleepers, as most of it is liable to be affected by white ants as soon as it is cut. Only two timbers—the cypress pine and the ironwood—are not affected by white ants, but they are not found in great quantities. The paper-bark tree, I am sure, is not white-ant resistant. It is a hard wood, and well suited for building, if it is kept off the ground to prevent the ants from getting in. I had no opportunity of seeing the country south-west of the Katherine River, where large numbers of cattle are raised; but from what I could gather from the settlers it is not suitable for closer settlement, owing to the absence of permanent water. However, I saw some ideal country for dairying far away to the east, on the Alligator River, but I did not

hear of any suitable country near Wave Hill or Victoria Downs. I met a number of people who came from there, and they told me they had difficulty about the water supply. Vestey Brothers, I know, are spending a lot of money in making wells. I do not know of any country on the railway line suitable for co-operative dairying, though it is possible this might be successful near Warburton. Attempts have been made to secure water by boring in the country south-west of the Katherine. It would be quite possible to have a branch line west from, say, Daly Waters or Newcastle Waters, to accommodate the Wave Hill or Victoria Downs country, and I know this line has been talked about.

26 To Senator Keating—In my travels I was close to the route of the proposed extension along its entire length, and having in view the ultimate linking up of the north-south railway, I have no doubt that the route selected is the best one. The cypress pine is not grown to any extent in the country over which I travelled. I saw only occasional patches of it. It is indigenous to the Territory. Up to the present nothing has been done in the way of afforestation in the Northern Territory, but that subject has been mentioned as one to be dealt with in the future. It has not been overlooked, but up to the present we have hardly thought the time had come to deal with it. Inquiries have been made in tropical countries concerning the cultivation of white ant-foresting trees, but I do not think our reports are in yet. The cypress pine is a very strong smelling wood, and it has a remarkable odour when burning, but it does not seem to me so highly impregnated with turpentine as does the Ironwood of Tasmania. The rainfall in the coastal areas is fairly heavy, averaging about 65 inches, but it gets lighter inland. At Bitter Springs the gauging shows 30 inches. Generally speaking, the rainfall is not nearly so heavy as in the greater part of Papua, which is in nearly the same latitude as the north of the Territory. The rainfall at Bitter Springs would be good if it were more evenly distributed. The question of water conservation has been discussed, but the evaporation in the Northern Territory is tremendous, and it would practically be necessary to cover all the reservoirs. The question of water conservation has been looked into by some of our surveys during the past two or three years, especially in the Adelaide River district, but we have never had water conservation experts there. The land in the Adelaide River district is very good, and no doubt there are possibilities of agricultural development, which would be worth considering at an early date. The country I travelled over was not heavily timbered except in a few patches, and there is very little jungle country similar to that met with in Ceylon or in Papua. Generally speaking, it is more like our ordinary open highly timbered country. Not 1 per cent of the distance I covered was over formed roads. I suppose that for fully three quarters of the distance I travelled by motor car, and there ought to be no considerable engineering difficulties in railway construction. The creeks along the proposed section are small, and the country is fairly level. You could go in some districts for probably 25 miles, without a rise or fall of 15 feet. On the existing railway there are two trains each way per week running regularly, and what are called special freight trains, perhaps on two or three days per week.

27 To Senator Story.—Mr. T. Day has recently completed a journey from Oodnadatta to Darwin. His particular mission was to examine the country north-east of Alice Springs, but he continued the journey right through. I do not think he could give you any more information than could a dozen other men regarding the country between Katherine River and Bitter Springs. I know, however, that the manager of the sheep farm, Mr. Watson, could give reliable information.

28 To Mr. Finlayson.—The water at Bitter Springs is highly mineralized, but is suitable for stock and domestic purposes.

(Taken at Melbourne)

MONDAY, 15th NOVEMBER, 1915

Present

Mr. RILEY, Chairman,	Mr. Gregory,
Senator Keating,	Mr. Sampson,
Senator Story,	Mr. Tanton,
Mr. Finlayson,	Mr. Laird Smith,

Rowland James Evans, Superintendent of Existing Lines, Northern Territory, sworn and examined.

29 To the Chairman—I have been in the Territory two and a half years. All I have to do with the construction branch is to forward supplies. We stack in the yard material arriving at Darwin, not having enough rolling stock to send it straight through. The material for the construction on the Katherine now going on is all in the yard or on the spot. There is none there yet for the proposed extension to Bitter Springs. I have been over the new section to the Katherine, and as far as Marrabon on the west, but not to Bitter Springs. I am told that the country is much the same as I saw as far as Marrabon, comparatively level, very small grades, and only three water courses of any size to cross—King River, Maudie's Creek, and another. No great engineering difficulties are presented. On the section to the Katherine there are plenty of water courses close together, so that the water trouble is not at all acute. That does not apply to the next section, as the streams are further apart. The most comfortable time for the Committee to visit the Territory will be about March, avoiding November, December, and January, which are the wettest months. May is the coolest month. The Katherine, which is the worst stream, can be and is crossed by boat even in the height of the floods. Near Maudie's Creek is a 6 mile stretch of black soil, impassable in rain, but soon drying. The heat in the wet months is very moist, but the clouds temper the sun's rays. The Burns, Philip Hue is the best to use from Brisbane to Darwin. Their boats leave Sydney on the first of each month. The Committee could hardly visit Bitter Springs and return in time to catch the summer break—that would be three weeks. It would take from Darwin to Pine Creek one day, to the Katherine two days, and to Bitter Springs (by buckboard), two days. The line is being built with steel sleepers on the straight runs, but on the curves we are putting in powdered karrri. The Traffic Superintendent at Darwin has put some of these on ant beds. The ants attacked the sleeper underneath at once, and for some time it looked as if they would go through it, but

they eventually abandoned it. All the effect was a very slight piercing where they tried to get through it. In my opinion the powdered karrri is impervious to white ants. Their attacks do not affect its stability in any way. It could not be said that it was even properly mixed. I am very much in favour of the steel sleeper, which is practically everlasting, especially the newest pattern, made on the girder principle, with a wide flange at the bottom and a narrower one at the top, enabling effective packing. The older sleeper was concave and thinner. Steel sleepers in the existing line, where well covered and away from the sea air, are as good now as on the day they were put in. Those exposed to the sea air near Darwin have had to be replaced. The steel sleepers come from America direct to Darwin, and the rails from England. The steel sleepers cost 11s 6d without freight, arriving December and February last, on contracts let prior to the war. They are priced for the 3 ft. 6 in. and 4 ft. 8 in. gauges, and as from $\frac{1}{2}$ in. to $\frac{3}{4}$ in. extra is wanted on the curves, wooden sleepers have to be used there. The landed cost of the powdered karrri was about 7s 6d. If the cost of the steel sleeper has gone up to 19s each, that would be prohibitive. They are not bulky, and do not contain very much steel. They could easily be made in the Commonwealth. The new girder type is the best I have ever seen. The Territory railways show a loss every year, thus—

Year	Revenue	Working Expenses
1910-11	31,523	16,320
1911-12	33,337	18,708
1912-13	31,793	17,000
1913-14	17,810	23,991
1914-15	22,140	26,992
1915-16 (estimate)	33,000	44,742

We have 14 in. lbs in use, and 65 to 60 under construction.

30 To Mr. Gregory.—The revenue includes freight for the new construction. In 1914-15 is included about six months' traffic in construction material, bringing in about £6,000. We charge a special rate of 1d per mile per ton for this.

31 To the Chairman.—But for replacing obsolescent engines, which might have been done years ago, we would have shown an excess of revenue over working expenses. The interest bill is £46,299. Under South Australian control the line never showed a profit. The working expenses per train mile had not been calculated for 1914-15 before I left the Territory. The previous year they were 10s 8d, and it is anticipated they will be reduced to 6s for the current year. Previously the line was starved, within the last year the rolling stock has been improved. We had Class W Beyer Peacock engines, a number of which were condemned by the boiler inspector, and replaced by Queensland type B 13. We now have three B 13 and four W locomotives. These are enough to do the present work, but not to handle the extra meat traffic for which thirty Queensland boggy cattle trucks are ordered. Twenty of these have been already delivered. When the meat works are erected I think the railways will show a profit over working expenses. The estimated revenue from the meat traffic in the first year is £12,000 for five or six months. Besides cattle traffic there is traffic for the mining fields. The line to the Katherine will bring in a little revenue, but we would not expect to have got beyond the Katherine outside of the Marrabon tin field. I do not think there will be any traffic coming in at all. The proposed line will help to develop the Marrabon field, of

which I have a high opinion. It is a poor man's field. The battery is about ready to start crushing, and a fair population, running to several hundreds, can be maintained. Many of the miners are now working on the railway line waiting for the battery to start. There are more than thirty shafts registered there, all working on lead ore. It is essentially a healthy district.

32 To Mr. Laird Smith.—The Darwin Pine Creek line was built thirty years ago, and some of the steel sleepers are as good as new. As a self-contained proposition it would be absurd to build the new sections, but they should be built as a part of the through line from north to south. No deviation towards Queensland is suggested or possible until after Bitter Springs. We are now running flying maintenance gangs of four men for 25 mile lengths. The work is light. The water is good, and does not affect the boilers. The engineer expects the Pine Creek Katherine River section to be open for traffic next June, but that does not include the temporary bridge over the Katherine. The station will be south of the Katherine. The bridge, which is a low level one, is part of the Pine Creek Katherine scheme. The meat works are at Darwin, to which the live stock will be railed, some from Mountain Head and some from the Katherine. The meat works proprietors say they will kill 250 head a day during the coming season, when the Katherine section is open. They will probably kill 500. We are getting now the full ton mileage out of the engines. There is no back loading, but the working expenses are bound to drop with the extra train mileage. The powdered sleepers are bound to give better results in ground which is not known to be infested with white ants than when put on an ant bed, but no one could prefer them to steel sleepers, which are everlasting, except that higher speeds can be got on wooden sleepers. In the Argentine, where cattle trains are run through at 60 miles an hour, the steel were replaced with wood. There are no sand hills or clay beds on the new route. We have had trouble with the Queensland B 13 engines on account of the wheel bases being bigger than on the W engines, because on the 10 chain curves the brasses grind on the curves. We cannot get over that trouble, which happens every trip. In every other essential the Queensland engines are good. The Beyer Peacock engines were built for the line.

33 To Senator Story.—The Marrabon field has been proved tin bearing, 6 miles by 3 with payable propositions all over it. Samples like this one now on view in the Department of External Affairs are common. About 16 per cent tin is wanted to make the field payable. Later, with the railway it is possible a lower percentage will pay. The field will probably support several hundred men. The cypress pine ironbark and ironwood all more or less ant-proof timbers grow in the Territory. The paper bark is good, but not absolutely ant proof, and the quantities available are not great. There are forests of cypress pine, some distance from the line. A saw mill there would pay only if a line was run out to the forests. Cypress pine is too short in the grain and splits when the dog spikes are put in. Ironwood makes an ideal sleeper, and we have used it as a crossing timber. The freight on the first lot of rails and sleepers from England was 19s per ton, that was a specially low freight. The powdered sleepers came direct from Bunbury round the north-west coast. The increase in the railway revenue is due to the operations of Vestey

Brothers, and the construction of the Pine Creek-Katherine line. The cost, in my opinion, would be approximately the same to construct the line from the south northwards as from the north southwards. When the line from Oodnadatta reached the Macdonnell Ranges it would be reproductive, but the country between is very poor. To start from the south would develop the Macdonnell Ranges country quicker. Freight for a long sea journey to Darwin or Adelaide would be about the same. If Broken Hill rails were used it would be cheaper to build from the south. I could not say whether the Oodnadatta climate is better to work in than the Katherine, but I know that the climate at Bitter Springs is fairly good, because as you go inland from Darwin you get altitude.

34. To Mr. Sampson.—The country is only grazing along the new line. Everything goes to prove that, except for the Marranboy field, there will be practically no new loading for the new line. The cattle will be driven in to the Katherine from the stations, the lowest of which is Wave Hill. I have heard discussion as to the advantages of a line to the cattle country around the Victoria River. It is good cattle country all through there, but it would mean a very wide deviation. The line to the Katherine has not given any impetus that I know of to cattle raising. As soon as it is completed the stations will send the stock to the Katherine, and they will be railled thence to Darwin. I can supply the Committee with figures showing the revenue from the different classes of goods carried over the existing lines for the last three years. We have had to take some concrete sleepers out of the road near the jetty already. They are too unresponsive. These were an imported make.

35. To Mr. Finlayson.—There is no urgency for the construction of the new section, except as a means to the completion of the through line, and incidentally to keep men employed. It is not justifiable from a traffic point of view at present, but it is imperative to build it in order to get the through line. Very few interests will be served by the Bitter Springs extension. I doubt if it will cause any increase in the cattle traffic. The miners at Marranboy are all looking forward to the building of the line, which will give some impetus to the mining industry. If steel sleepers could be manufactured in Australia at a satisfactory price they would meet our purpose.

36. To Mr. Gregory.—I cannot see where any new settlement is to come from to the south-east of the new line. I do not know that it will open up any further cattle country. That will all depend upon the extent to which the meat works are able to take the cattle. Later on Vestey Brothers may kill 700 head a day. They expect to be killing next March. It may not be generally known that they have bought up all the station properties round about, and if any one else went there to start cattle-growing they would probably buy them up also. Some of the shafts on Marranboy are down 40 feet or 50 feet. The 10-head battery is quite big enough to cope with any probable population that the field will carry. There is not a sufficient quantity of timber to use as fuel on the locomotives. There is not sufficient ironwood available for sleepers. I would not say that ironwood was impervious to white ants, but it is ant-resistant. For a recent contract we had to go 2 or 3 miles away from the railway line to get the quantity. Engineers say the steel sleeper is too rigid for high speeds, but

our speed in the Territory does not give us an opportunity of testing this point. I should say that steel and concrete sleepers were on a par as regards rigidity. Concrete sleepers could be made at certain centres along the track. Forty-eight thousand powdered karri sleepers have already been sent to the Territory. They are invoiced as karri, and the engineers have inspected them and declared them to be karri. I was surprised to hear that karri sleepers had been chosen, because I know that jarrah is a better wood to put in the ground. They arrived in February, and those I mentioned have been lying on the ant beds for about eight months, and have been inspected once a month since. We have not tried to treat sleepers locally to make them ant-resistant, but Mr. Gerald Hill is making experiments in the Department now in regard to various timbers.

37. To Mr. Fenton.—Practically none of the 57,000 cattle exported in 1913 was carried on the railways. They evidently went into Western Australia from the Territory, and were shipped at Derby or Wyndham. The proposed section will touch some of the buffalo country, but the buffalo are not there in great numbers. There are more of them in the country along the open line. Some parts of the buffalo are fairly good eating, and I believe the proprietors of the meat works contemplate putting some of them through for frozen meat, but they are mostly killed for their hides.

38. To Mr. Finlayson.—The grinding of the brasses on the B 13 engine is considerable where a grade and a curve occur together. Our grades and curves are not very severe, and it is only with that part of the engine that we have had trouble, but it was solely on account of the length of the wheel base.

39. To the Chairman.—The country round Mataranka, where the experimental sheep farm has been established, is very good sheep country, and will be tapped by the new line.

(Taken at Melbourne.)

MONDAY, 22ND NOVEMBER, 1916.

Present:

Mr. RILEY, Chairman;

Senator Keating;

Mr. Finlayson;

Senator Story;

Mr. Gregory;

Mr. Fenton;

Harry Vivian Francis, Assistant Engineer, Commonwealth Railways, sworn and examined.

40. To the Chairman.—I was in the Northern Territory as Superintendent and Resident Engineer of the Darwin to Pine Creek railway for three years. I have been back in Melbourne over twelve months. I travelled over the proposed new section from Katherine River to Bitter Springs; my furthest south is Milner's Lagoon. With reference to the "westerly swing" shown on the map, as recommended by Messrs. Clarke and Lindsay, two members of the Royal Commission for the Northern Territory railways and ports, starting from Katherine River, passing through the fringes of the Victoria River district, and rejoining the telegraph line at Renner Springs, I understood that the majority of the Commission recommended as direct a line as possible, which would branch off from Pine Creek or

Drock's Creek direct south, but the usual impression in the Territory is that the country immediately south of the Katherine telegraph station is of little value, as no one has attempted to take it up. I know of only two blocks taken up: one occupied by a man who calls himself Duggan, but whose real name is Goughan, and another taken up by a man named Kruger. In 1902, Allan Davidson worked his way out from Barrow's Creek, and found Tanami, and back to Kelly's Well, and he gives anything but a bright description of the country. The route of the telegraph line is the one most nearly adopted by everybody concerned as a railway route, giving some chance of settlement, and along which there would be water for at least the greater part of the way. From Newcastle Waters across to the Victoria River Downs station is a very dry track, with 70 and 80 mile stages without water in a dry year. Some part of the country is spinifex, which is supposed to be good for stock. The Hon. John Lewis has a large place at Newcastle Waters with good water supply, but this, and places like it, are all on the telegraph line, or east of the line. There is no one settled on the western side until you reach the Victoria Downs. The direct south route, therefore, has never appealed to persons in the Territory. The Commission, which I met out-back, prepared a map showing where they had been. According to this, they went to the east, and were never in the western part of the country at all. Neither Messrs. Clarke and Lindsay, who were the majority, nor Mr. Combes, who was the minority, saw that country at all. It is not understood why they recommended that deviation, unless for directness, and certainly the westerly swing, as shown on their map, is by no means direct. A scientific expedition consisting of Dr. Gilchrist, Dr. Breinl, Dr. Woolnough, Professor Spencer, and others, some years ago examined the Roper River country lying to the east of Bitter Springs, and made a report on it. Their guide, Mr. Alfred Giles, also wrote a series of articles, but the greater part of that country is still not taken up. It is permanently watered all the way along by the Roper River, and offers an opportunity for the laying of a large number of blocks, all well watered on at least one fringe for a start. It is supposed to be a good class of country, and all who have been over it speak well of it. Bitter Springs, being the source of the Roper, is the head of that country, so that the Katherine to Bitter Springs extension would undoubtedly tap it. Another point in favour of the Bitter Springs line is that it will serve the Marranboy tin-field, which, as soon as the battery is started, will probably be the biggest mining field in the Territory. When I was there the prospects were very promising, and a large number of men were working on the field. The Bitter Springs line goes within about 9 miles of Marranboy, but the swing to the west would put that place right out of court. The Bitter Springs line catches the Roper River country, Hodgson Downs, Nutwood Downs, and good country extending down to Anthony Lagoon. Most of the country south of Anthony Lagoon would send cattle through Camooweal to Charleville and other railway stations, Queensland. If water was available, cattle could be driven from Anthony Lagoon up to Bitter Springs. We ought to draw cattle from country for at least 300 miles south-east of Bitter Springs. I think that trade would make the Bitter Springs railway a better paying proposition than the Katherine River sec-

tion is likely to be. Bitter Springs is also closer to Wave Hill station and the Victoria River district than Katherine is, and Wave Hill is practically the home station of Vestey Brothers, who are putting up the meat freezing works in Darwin. From Bitter Springs to Wave Hill is about 220 miles. At present, cattle are driven from Wave Hill to Wyndham, Western Australia. There is a good deal of inducement for fresh settlement on the Darkly Tablelands, where a large amount of the land is held under permit, while, as I said, the Roper country is practically unsettled. These are points in favour of the more easterly route, whereas the best part of the Victoria River Downs country is but on forty-two-year leases, which have a long while to run, although I presume the Government would have powers of resumption there as elsewhere. The Bitter Springs extension is certainly more reasonable than that from Pine Creek to Katherine, which is a necessary length in the scheme for a north-south railway, but which passes through what is well known as poor country. The Bitter Springs extension takes the railway into country that offers better prospects. After the rains there is a fine growth of grass at Bitter Springs, and it is more like what appears to be good agricultural country further south. We had no wooden sleepers in the Territory when I was there. Steel sleepers were being sent there after I left. The trough-shaped steel sleepers have been in the old Darwin-Pine Creek line for thirty years, and have stood well, except near the coast, where some of them have corroded. It is hard to say whether the ants would destroy untreated wooden sleepers. There is always the chance that the vibration of the trains will keep them from going through the sleepers. In a line running from Cooktown, in Queensland, to the Roper, wooden sleepers have stood very well in similar country, yet I believe only about two or three trains a week run on the line. Ironwood is not absolutely immune to white ants. If there is a pipe in the centre, the ants will start on it there. We used to fill the centre with concrete. The ants do not care about ironwood, but will eat it. If steel sleepers can be obtained at a reasonable price, there is no question which we should use. The Katherine to Bitter Springs section is very easy from an engineering point of view, but just after leaving the Katherine there is a large belt of limestone country consisting of high limestone rocks with sharp pinnacles. This presents considerable difficulty. We had difficulty with this country in our survey as far as Bacon Swamp, after which we get into easy country. We got the survey through there before I left the Territory. That limestone would be very suitable for ballast, if blasted out, and will be probably used there.

41. To Senator Story.—I know Mr. Pearce, who suggested to the Commission the direct south line from Brock's Creek. He is the owner of Wilberoo Station, and the line goes straight through his property. He is an estimable and honest man, but that consideration would undoubtedly influence him, for a man would be foolish to fly in the face of a possible railway through his land. He has lived on Wilberoo for a good many years, and I dare say he would not mind getting out well. This may not be a charitable suggestion, but it counts. If the Commonwealth and the State of South Australia is binding, as we have always been given to understand, we must either follow the telegraph line, or endeavour to find a route which will secure for the Commonwealth one of the two best

patches in the Territory—the one on the east, consisting of the Barkly Tablelands, with Anthony Lagoon in the centre, or the one on the west consisting of the Victoria River district. Some of the Victoria River district trade is already secured by Wyndham, Western Australia, and practically all that district is already taken up. My opinion, therefore, is that the nearer we go to Anthony Lagoon the better chances we have to secure the best portion of the Territory. If we take our line close to the Victoria River district we shall come into competition with Wyndham, but at present no one else is after the Barkly Tablelands trade, and the Katherine to Bitter Springs section must be part of any line intended to tap that country. A line towards Anthony Lagoon would get the Alroy Downs country, and all the country lying south of that to Alice Springs. A flying survey was made by Lawrence and Chalmers, from Anthony Lagoon to Camooweal. An excellent alternative suggestion has been made for a line midway between the telegraph route and Anthony Lagoon, and running through the Frow River station. This would not divert too far from the direct route, and would still tap the Barkly Tablelands. If the only object was the connexion of Port Augusta and Darwin by as short a route as possible, I should say go as straight as you can, but we must try to pick up some trade on that big length of track, and we should not get the best revenue returns if we went through the unoccupied country to the west of the telegraph line. I would recommend the shortest possible route compatible with the idea of developing the country, if this is to be a trans-Australian line carrying express trains, but the more easterly route is still the better, especially in view of the fact that Port Augusta is more to the east, and to reach it it is not necessary to go straight through Alice Springs. The line from Port Augusta to Oodnadatta is nearly on the 138th parallel of longitude as far as Hergott Springs. A line from Port Augusta or Hergott Springs, coming up through the tableland, is still more nearly direct, leaving the western country further out than over I do not think there would be greater engineering difficulties in getting through the eastern side of the Macdonnell Ranges. These ranges tail out on the eastern end as shown by the run of the watercourses towards Lake Eyre, which is below sea level. The telegraph line goes through nearly the highest portion of the Macdonnell Ranges. A line from Kingoonya to Boorthanna is a fairly direct north-south line. That is one of which we have had a trial survey running from 210 miles from Port Augusta on the east-west line in an almost direct northerly direction. Boorthanna is 60 or 70 miles south of Oodnadatta. That line would offer the most direct route. If that was adopted the telegraph line would be the nearest to make a direct connexion. I believe the greater number of the advocates of a through line would be well pleased with the telegraph route rather than to see it moved further to the westward. There is a general belief in the Territory that the westerly land is not very good, as shown by the fact that it is almost as lately unoccupied. Dr. Chalmers went to the west of the line, and his account of it is no better than Davidson's. We are not likely to secure the trade of that part of the Victoria River district which adjoins the Western Australian boundary, for that cattle country is within fair walking distance (for cattle) of Wyndham. I have heard it said that

Vestey Bros. have bought Copley and Patterson's and Buchanan's station properties in the Victoria River District, but even though they own the most works at Darwin, I do not suppose they would run their stock from that country to Darwin if it paid them better to send it to Wyndham. If the track is good, and the grass good, owners will not entrain their stock until the last moment. If they want to get them to the freezing works, and the grass is not good, they will put them on the train at Bitter Springs rather than at the Katherine. For at least four or five months of the dry season, they would prefer to put the cattle on the train at Bitter Springs. The freezing will be done in Darwin during the dry season, and the cattle will be coming in as long as the works are running. During the heavy wet the cattle could not be got to the railway station. The cattle trade will mean considerable revenue to the Bitter Springs section. So far as revenue from the Marranboy field is concerned, I thought very highly of the field when I was there. It was an extremely promising proportion, but it had been hung up for a long time owing to the slow arrival and erection of the battery. The man had been started out, although the storekeepers had been backing them for a long time. One storekeeper at Darwin was £1,000 out of pocket when I left there. Now that the battery is nearly completed, Marranboy should go ahead. A number of the men on the field would get employment on the railway line if they could secure exemptions, which I understood that Dr. Jensen was willing to recommend. It is a big field, comprising 40 square miles of land, all prospecting, and with shows all over it. They talked of supporting from 500 to 1,000 men there. If it is likely that 150 men now employed on the Pine Creek-Katherine River section will be thrown out of employment in the event of the Bitter Springs section not being gone on with for the next six months, I do not think that they can be profitably absorbed by the Marranboy field, for I doubt very much if any of the miners can afford to employ labour at the present time. When I was there they were all working miners, and practically devoid of money. Those who have been making wages on the railway will go back to their claims as soon as possible, but the trouble is that the men who will be put off the railway works would probably not be the best class of workmen, and it would be difficult for other industries to absorb them. The battery should give employment to about a dozen men. There are other fields between Bitter Springs and Darwin worth prospecting, but, unfortunately, not during this period, as the "net" has started. If these 150 men were turned off in June or July, I should say yes, but not now. Between Pine Creek and the Katherine the grass is about 12 feet high about the time when these fellows would be let go. Not much prospecting can be done under those conditions. The Committee could travel to this part of the country in the middle of March, although the weather will still be hot and March fires and other pests troublesome. I put concrete sleepers in at Darwin, but they were not a success. The manufacture was not good, and they failed where we expected them to fail. They did not stand under the rail at the rail seat. I am aware that concrete sleepers have been tried so successfully in the Adelaide from time, but the axle load on a train line are much lighter than we try on our railways. Concrete sleepers have

been tried a great deal in America and in Italy. I know they have not yet been successful in America.

49. To Mr. Gregory.—It is hard to say whether the added traffic on the line to Bitter Springs will make the railway pay working expenses; but there will be undoubtedly an improvement at Marranboy, and more pastoral land will be leased. As soon as people know the line is going to Bitter Springs, I am certain there will be considerable endeavour to lease land along the Roper and in the vicinity of Bitter Springs. All this means pastoral, as well as mining, traffic, and back loading for the stations. I framed all the freight charges for the Territory. Our charges are 25 per cent. in excess of the southern States, for both fares and freights. When the railway is completed to Bitter Springs, I anticipate considerable cattle traffic. The Katherine is what we call bad holding country during the dry season. For several months in the year the mailman cannot feed his horses there. That is not the case at Bitter Springs, which is on the Elsey Station. This was at one time a fair-sized cattle station, and was held for years. Droving to Bitter Springs instead of Katherine will save at least ten days in the journey from Wave Hill, and that makes a great deal of difference in the dry season. The country between Katherine and Pine Creek is very poor. In the old days cattle were loaded at Mountain View. Nearly all the existing mining in the Territory is just on the eastern side of the Pine Creek to Katherine section. The question whether the line recommended by Mr. Combes, from Anthony Lagoon to Port McArthur and the Pellow Islands is more likely to develop the Territory than the north-south railway has never cropped up, because we have always been under the impression that we had to make the north-south connexion in accordance with the agreement with South Australia. To open the Barkly Tablelands by a line to Port McArthur and the Victoria River country by connexion with Blunder Bay would develop the country quicker, but the extension to Bitter Springs is still justified. In stopping at the Katherine we are not stopping at a good place, nor are we opening up any good country. When Mr. Thomas was Minister of External Affairs he asked me to map out a railway construction policy for the Territory—a very difficult thing to do, unless you can put the north-south railway aside as a matter for the next generation. If we must connect Port Augusta and Darwin by virtue of the agreement, we must find the best route to serve as much country as we know will help the line. If the Territory could be developed in an entirely different manner, and without considering the agreement, I would develop the Barkly Tablelands, and then develop the Victoria River country, if we could resume sufficient land there; but I would not go into the Victoria River country at the present time while it is held, as it is now, in enormous areas on long leases by people who are not doing a tap to develop it. There are certain development conditions in the new leases, but not in the old ones. The present holders of the Victoria River district have an extremely good proposition at a very cheap rate. I was very sanguine about the Marranboy field when I saw it, and I have had considerable experience as a mining engineer. I walked over the field 8 miles one way and 5 miles the other. I saw at least fifty claims opened up. These were our quartzite lodes, and some of

the better-class tin ran very high. On examination, it was estimated that a large part of the lodes should run up to 10 per cent. of tin, which, with tin at its present value, means a very payable proposition. The rock country is very good, and I doubt if much timbering will be wanted. If there is, there are good forests of the best cypress pine I have seen, near Marranboy. Sinking on the lodes means sinking in hard country. A 10 head mill will not keep many big shows going, but if a number of good claims were proved, it would not be long before another battery was provided. The lodes I saw were of considerable length. It is easy in places to see from 500 to 300 yards of tin lode in front of you with shafts sunk at intervals. Unfortunately, the experience of concrete sleepers everywhere has been bad. The Americans have not yet got the concrete sleepers out of their railway yards, and they have been trying to make a good concrete sleeper for a long time. The Bessemer and Lake Erie line had 1,600,000 steel sleepers in two years ago, of the Carnegie pattern, which we are using on the new line. The ironwood sleeper is very hard; but the trees do not grow to any dimensions, and are nearly all piped. I made two tests with pole-laid sleepers when I was at Darwin. In the first I laid several test specimens made up of a block of untreated and a block of treated timber on an ant's nest, with a space between them. The ants went through the pole-laid block, but did not eat it. They cleaned out the untreated block, leaving it a mere shell. The second test consisted in laying pole-laid jarrah crossing timbers in the road, partly in Darwin and partly at Boomleera, along the route. So far the crossing timbers have not been touched by the ants. We have tried a large number of species which will keep the ants away so long as the specific lasts on the outside of the timber. If you can get impregnation that will stand, the ants will leave the timber alone.

48. To Mr. Fenton.—I am quite prepared to believe that all the Victoria River country is much under stocked. It is not fenced, and has practically no improvements in the way of water. If they have a bad year, the stock must be all in on the water. Cattle can be driven from the lower portion of that district to Wyndham for 5s 6d per head, and put on the boat there for another 1s 6d. The Wyndham works are a State enterprise, but there is at present some difficulty over the contract. From Wyndham to the boundary is about 100 miles, and to Wave Hill about 200 miles. We might get the traffic from the Victoria River Downs Station, which belongs to the Boral Estates Limited, and per- which Vestey Brothers are understood to have purchased. It is reported that Vestey Brothers have spent about £2,000,000 in the Territory. If it paid them better to send their cattle to Wyndham, I do not suppose they would consider their own works in Darwin; but Darwin, as a port, is incomparably superior to Wyndham. Ships of 30 ft draught go into Wyndham, but they have to sit on the mud for twelve hours out of the twenty four. At Darwin we have a depth of water running from as high as 40 feet at the top of the spring tides, to as low as 21 feet at the bottom of the neap tides. The captain of the Empire reported to me on one occasion that he was on the bottom drawing 21 feet; but I think that was on sill. I have measured as low a depth

as 6 feet at the wharf at Wyndham. This is very much against the big boats, until the mariners learn to sit the boats on the mud, which does them no harm. A dredging proposition at Darwin would not be difficult. I do not know that the better shipping facilities at Darwin would induce stock-owners to drive their cattle further in order to reach that port; but the driving would not be so bad to Bitter Springs as it is to the Katherine or Fountain Head. Cattle can arrive at Bitter Springs in good condition. The Port McArthur railway proposition requires a careful survey of the bottom between the mainland and the Pellow Islands. Mr. Stretton, at one time Chief Inspector of Aborigines, Sub-Collector of Customs, and Harbourmaster, knows Port McArthur very well, and he would not say that it was the best port in the Territory. To take the main trunk line from Katherine River to Oodnadatta, via Renner Springs and Alice Springs, would be to follow as nearly direct a route as is consistent with engineering considerations, and would give a length of about 1,020 miles. I would be agreeable to taking the line that way if the country to the west of the telegraph line could be shown to be of any value. The fringe of the Victoria River district touched by Clarke and Lindsay's westerly swing is an expensive country for railway construction, as the heads of a number of rivers have to be crossed. The Pine Creek to Katherine River section will cost, in the long run, a little over £3,000 a mile to construct. It comprises some fairly heavy work and heavy bridging. That estimate will, I think, cover the low-level bridge across the Katherine. I think the line should be taken to the south side of the Katherine for that amount. I told the Commission that if we were building that railway from Katherine to Bitter Springs in the south, in the same class of country, we would build it for £3,000 a mile in the Southern States; but we have to carry all our materials, and at the same time pay 16s. a day in wages. Mr. Combes' estimate of £9,000 a mile was for a 4-ft. 3-in. gauge. The Katherine to Bitter Springs section will be constructed for much less than £3,000 a mile. There is only the King River to cross, requiring about 120 feet of bridging.

44. To Mr. Finlayson.—The lower portion of the Victoria River country belongs, geographically, to Wyndham. The Vestoyts will not let the whole of the Victoria River district trade go to Wyndham if it means impoverishing their freezing works enterprise at Darwin. When the line is built to Bitter Springs, mules will be laid there which cannot be held at the Katherine after June. Much of the trade of the Roper River and Barkly Tableland country has to go at present through Queensland. A line running a little further east than the telegraph line would catch that trade. The suggested swing through Anthony Lagoon would be a little too wide, and would leave a greater portion of the Territory out of the track; while the telegraph route will be too far away from the Barkly Tableland, much of which is good sheep country. The Government experimental sheep station, on the Waterhouse near Bitter Springs, is a little too high in latitude for sheep, and the country round there is not good sheep country. You do not get to good sheep country until you get below Anthony Lagoon. I believe that the leaseholders would surrender portion of their land for railway purposes, as one of them told

the Commission. A large number of men holding country there have had bad times for many years. Some time ago, in good years, stock was so low in price that it did not pay. I see no objection to the Bitter Springs extension. It is a necessary link in the southern connection, and the better part of the country that we can tap lies to the south-east. Half the Victoria River country is lost to us already, and the other half does not come back to us for many years. The immediate construction of the Bitter Springs extension is a financial question. If there are good financial reasons for not going on with it, that stops it; but if we stop railway construction at the Katherine, we stop the north-south railway, which we believe must come eventually, and which is quite as justified as an east-west railway. If it has any justification at all, the north-south railway should be built, even if it has to be done a little at a time; but if we stop it at the Katherine, we lose the lot. If it is extended to Bitter Springs it will be a more payable proposition.

45. To the Chairman.—If there is no prospect of the gauge being altered for the next fifteen or twenty years, I should advise the use of 3-ft. 6-in. sleepers, that is, if wood sleepers are used. The difference in the length between 4-ft. 3-in. and 3-ft. 6-in. would probably save £150 a mile. It would be wise to construct the earthworks, bridges, and culverts wide enough to take the wider gauge at the start. The saving in length would probably be from 1s. 6d. to 2s. per sleeper, and we use about 2,040 sleepers to the mile. The life of the Carnegie steel sleeper should be at least from 60 to 100 years inland, but I understand that the price has become almost prohibitive.

46. To Senator Story.—The Roper River is navigable for a short distance for boats of small draught. It could be made navigable at great cost. If this through line is completed the traffic from north of the Macdonnell Ranges will probably go to Darwin, and that from the south to Port Augusta, but a considerable area north of the ranges about Barrow Creek is fairly barren. It is hard to say whether the line should be constructed from north to south or from south to north. The development of the Macdonnell Range country from the south would give the Oodnadatta line some mining and pastoral traffic. We anticipate a little mining development in the Davenport country, about 150 miles north of the Macdonnell Ranges, but a large area further north would for some time be a drag until we got in touch with the downs to the south of Anthony Lagoon. Then we might expect a return going to Port Augusta. If we stopped the line at Katherine, we would be repeating the mistake that was made at the South Australian end by stopping at Oodnadatta, which is practically the Salafra of South Australia.

47. To Mr. Gregory.—If necessary, we could bring our rates for trucking cattle down to the Queensland rates. On our present rates, it would cost about 30s. a ton on Queensland rates about 18s. per head for 300 miles of trucking, speaking from memory. The Queensland rates might be a bit less with their big trucks. They carry cattle in that State between 400 and 600 miles. Two or three years ago a beast worth £3 on the station cost about 6s. to put on the boat at Wyndham. The same beast would be worth £3 at the freezing works at Wyndham or Darwin. Owners can therefore afford to pay freight for cattle to be run fast into freezing works.

TUESDAY, 23rd NOVEMBER, 1915

Present:

Mr. RILEY, Chairman;

Senator Keating,	Mr. Finlayson,
Senator Story,	Mr. Gregory,
Mr. Fenton,	Mr. Sampson.

George Alexander Mobler, Construction and Maintenance Engineer, Commonwealth Railways, sworn and examined.

48. By the Chairman.—Last Christmas, I was over the country covered by the section of this line. I have not gone into the question of revenue. The country throughout, from Katherine to Bitter Springs, is easy construction country, with no engineering difficulties of any consequence. The earthworks generally are light. The only bridge of any size is over the King River. That is a bridge of three 40-foot spans on concrete piers. There are a few minor bridges and a few concrete box drains. The country from Katherine to within a few miles of Bitter Springs is, I should say fairly good pastoral country, improving as it nears Bitter Springs, and around Bitter Springs itself. It also passes within 12 miles of Marranboy tin-field. I went through the Marranboy tin-field, and from what I could hear there appears to be a large extent of tin-bearing country with very good prospects. The Government have there erected a 10-horse stamp battery. From conversations I have had with several of the leaseholders of mines, I should be inclined to consider that the prospects of Marranboy are very good. The great complaint was the difficulty and the high price of cartage for supplies, and the consequent high price of cartage of tin from the mines to Katherine River. At that time the Katherine River railway was under construction, and it was known that the line would eventually come to Katherine; but the road from Marranboy to Katherine River is not too good for teams. In the wet weather, for several months in the year, it is practically impassable for teams. During my visit, there was very good feed for horses between the Katherine and Marranboy, but I was there at the best time of the year. I was told that for eight months in the year there is very little feed indeed on the road, and feed is scarce at the time of the year when the road is most fit for team traffic. In my opinion, this section of railway will have to be constructed no matter what deviations may take place afterwards. If the line is to connect with South Australia, the section between Katherine River and Bitter Springs is undoubtedly a part of that route. I have seen the map and report by Messrs. Clarke and Lindsay, but, personally, I have not seen the country on that route. I consider that it would be inadvisable to bend the line to the west. The Victoria River and Sturt's Creek district would be far better served with a separate railway system or branch line later on. It would be better to have that than to bend the main line, which would mean only a partial development of the country; it would not develop it so well as would a branch line. The question is still undecided whether Wyndham might not prove to be the natural port of the district. The Department does not propose to follow the alterations as shown on the map to which you have referred. It is proposed to follow the present overland telegraph line to Bitter Springs, and probably to Daly Waters. After Daly Waters, the question will arise whether the route should

bend towards the east from there, or whether it should still continue along the overland telegraph line to Newcastle Waters. At present, from information I have, and from what I have seen, I am inclined to think that, on investigation, it will be found that Daly Waters will probably be the point at which the line should bend to the east, with the idea of keeping closer to the proposed connection with the Queensland system at Camooswell and also to keep in better country on its way south. I consider it absolutely certain that we are on the right route of the main line so far as Bitter Springs. I have only made one visit to the Territory, and that was last year. I travelled from Pine Creek to Bitter Springs along the proposed railway route and back through the Marranboy district, and thence to Katherine River. I went about 10 miles south of Bitter Springs. As to which part of the Territory should be developed first, the local development of the Territory must necessarily be considered to a certain extent in connexion with the main route south. Still, if one were confined to the local development of the Territory itself by a railway system, it would entail a considerable amount of investigation which, at the present time, I have not personally gone into. I have not the personal knowledge to enable me to express an opinion. On the Katherine to Bitter Springs extension it is proposed to make the earthworks of sufficient width to take a 4-ft. 8½-in. road, and put sleepers in of sufficient length to widen out from the 3-ft. 6-in. gauge later on. So far, we have had no absolute tests of the life of wooden sleepers in the Territory. But if the best class of hardwood is used, and the attacks of white ants are not worse than they are in the white ant infested country in Queensland, I should consider that wooden sleepers would have an average life of between fifteen and twenty years. I should not like to say, or express an opinion, when the construction of a 4-ft. 8½-in. line would be completed. Under the present circumstances, in view of the war, such a railway might be demanded at a very early date, but, on the other hand, it might not be required for a considerable number of years. The difference in the cost of using the 3-ft. 6-in. sleeper as against the 4-ft. 8½-in. sleeper, would be about £180 per mile. This would make no difference in the rate of construction. The difference means about 1s. 9d. a sleeper. That would mean 7 feet x 9 inches x 4½ inches for the 3 ft. 6-in. gauge as against 8-ft. 6-in. x 9 inches x 5 inches for the 4 ft. 8½-in. gauge, allowing 2,080 sleepers to the mile. I prefer not to express an opinion whether the long one or the short should be used, as that is a matter of policy. As to a branch line to the tin field, the country from the junction with the main line to Marranboy is exceedingly easy. It has a good sandy formation, and there are practically very few waterways. There is only one bridge over the Roper Creek, and a line could be laid with light rails it could be easily and cheaply constructed. I think that such a line would be a great help to the development of the field. I have no estimate as to the number of people likely to be employed on the line, or to live about the neighbourhood. It would take about eighteen months to build a line from Katherine to Bitter Springs, and that would mean that the men now employed there could be kept steadily on. There is no work that I know of in which 150 men could be engaged between the end of the year, when the section under construction is expected to be

completed, and when the new section would be started. So far as I know, the men there at the present time employed on the railway would be entirely dependent on such work if they are to continue in the Territory. From a railway point of view it would be advisable to keep these men employed, but I am unable to say whether there is any work other than railway work for them in case of a temporary cessation of the latter. I could not say whether Messrs. Vealey Brothers could absorb some of the men. That depends entirely on whether Vealey Brothers have sufficient men at the present time, or whether they intend to add to the buildings they have in hand. This firm had only just started when I was there, and very little work had been done to the building at the time. Most of the timber used by Vealey Brothers for the buildings was being imported from the south, but I understand they were getting local timber for the fencing.

40 To Senator Story.—It is still a question whether powdered sleepers are effective against attacks by white ants. There has been much controversy on this point. We have not had an opportunity of thoroughly testing the powdered sleeper in the white-ant country. We have a good number of such sleepers laid on the east-west line, but they have been there such a short time that it is impossible to express an opinion as to their resistance to white ants or dry rot. In my opinion the Northern Territory is infinitely worse for white ants and dry rot than the country through which the east-west railway goes. The use of these powdered sleepers is really an experiment. We are using about 40,000 powdered karru sleepers on the section between Pine Creek and Katherine River. Steel sleepers are very hard to procure, and the price is about double that of the normal market, presumably on account of the war. The life of the steel sleeper is a great deal longer than that of a timber sleeper, even though it is powdered. For instance steel sleepers at present used on the Darwin-Pine Creek line have been on the road just about thirty years, and, with the exception of about 5 miles of the line from near Darwin and the sea atmosphere, the sleepers are still in an excellent state of preservation. No one could say whether it would be possible to obtain steel sleepers made in Australia within the next twelve months. These sleepers and powdered karru sleepers are being used on the present section from Pine Creek to Katherine River. The first portion of the road is laid with steel sleepers, and after that come the timber sleepers. We procured a certain number of steel sleepers, which were used first, and then the karru sleepers followed. It is my opinion that the section of the line now under investigation is necessary as a part of the transcontinental line, and it would be advisable to get a report from the Engineer in Chief as to the desirability of first constructing a line to the Marranboy tin fields, and then continuing the main line when steel sleepers are procurable. There are many circumstances to be considered and I think the answer to the question asked requires some preparation. I should be inclined to say myself that the line should not stop short of Bitter Springs. The line now proposed is 12 miles from the Marranboy tin fields and this of course requires 12 miles of cartage, unless a branch line is built. In any case, the necessity for a branch line into Marranboy should be considered in connection with the further development of the

mine after the main line to Bitter Springs is actually built to the point of junction. The building of the line from Katherine towards Bitter Springs will necessarily bring it within 12 miles of Marranboy, whereas with the line finished at Katherine it will be 45 miles from Marranboy. Necessarily the extension of the line from Katherine to Bitter Springs will bring the line so much nearer to Marranboy, and that in itself, without a branch line, will assist very considerably in the development of the tin-fields. The question then arises whether the expenditure on a cheap branch line to the tin field itself would be warranted. At Bitter Springs there are better facilities for entraining cattle, and there is a good water supply for a township. The surrounding country is better, and the line is brought nearer to the cattle stations south and south-east of Bitter Springs. Further, the line is in a better position to assist in the settlement of Roper River lands. Roper River starts at Port Roper, in the Gulf, runs due west right to Bitter Springs, where it turns round and runs north-west for some distance, following within a few miles of the route of the railway from Katherine to Bitter Springs. I have been on the Marranboy tin-fields, and I can say that they are fairly extensive. From what I could see there was a considerable area of tin-bearing country. I think that if a line were run into the tin-fields it would assist greatly in the development of the field. I am not, however, in a position to say that the extent and value of the tin-field at the present time would be sufficient to justify the construction of a branch railway, when a main line of railway can be constructed within 12 miles, allowing future developments to show whether the line can run on a branch line is warranted. I do not consider that any great advantage would be gained by the construction of a section to Marranboy as a terminus. In my opinion, there is land there suitable for agriculture later on. Until the railway is opened right through from either Queensland or South Australia there is no market. As to the class of agriculture I should think that maize, for one thing, might be grown, but I do not think it is a wheat country, seeing that the rain comes at the wrong time of the year. But tobacco, and other tropical and semi-tropical produce, might be grown. I know nothing personally of the land north of Oodnadatta.

50 To Mr Finlayson.—There is need for this railway extension, for one thing, as part of the north and south transcontinental line. There has also to be considered the development of the country to the south of Bitter Springs and along the Roper River. As an individual proposition, I think that the extension of the Katherine River to Bitter Springs railway is justifiable, apart from the transcontinental railway. At present the railway stops at an inconvenient place. In my opinion, the country is much better from Bitter Springs southward than it is at Katherine itself, and, quite independent of any junction with the north-south railway, this railway ought to be built. We have also to take into consideration the assistance it will prove to the development of the Marranboy tin field. From Katherine southwards I consider the country fairly good pastoral country, and there is an improvement when we get within a few miles of Bitter Springs. The country at Bitter Springs is better than it is along the route. It would cost considerably more to divert the line via

Marranboy to Bitter Springs than it would to build direct to the latter place, and subsequently build a branch line to the tin field. The distance would be increased very considerably, and the line constructed would have to be a standard one, thus increasing the cost, while later a cheap branch line could be made to the tin-field. Such a deviation would also bring the line into country requiring very heavy earthworks and expensive bridges. Altogether, the deviation would be exceedingly expensive as compared with the present route. I consider that the country to the south-west of Katherine River could be better served by a branch railway, because that branch could be extended in any direction and to any length that would be desired to serve the country. On the other hand, the bending of the main trunk route would only partially serve the country, and could never be carried any further except by starting, from the point reached, a branch railway. I cannot say whether the general character of the Victoria River country is easy or otherwise for construction purposes, because I have not been there. I should prefer steel sleepers to the 13 pattern, which we are using in the Northern Territory. I am impressed with that sleeper. I would not say that I prefer them to any class of wooden sleeper, but I am impressed with the fact that it is most suitable for a climate like that of the Northern Territory, where we have to contend with white ants. I would prefer steel sleepers at the present time, if they could be got at a reasonable price, until we got further south. I do not think that anything would be gained by putting down a cheaper line with wooden sleepers as a temporary measure until steel sleepers are available. I do not think that the half-round sleepers used in Queensland could be got, practically, any cheaper than those we have already taken into consideration. There is very little difference in the cost of laying such sleepers, but there would be an extra cost of adzing in the case of the half-round sleeper. In any case there would be very little difference in the matter of price by using the half-round sleeper. It is very hard to say what that "little difference" might be, but I should have to call tenders in order to find out. Personally, I consider that the use of half-round sleepers should not be allowed for any railway in Australia, because it has caused the destruction of millions of feet of our very best timber. It has caused the destruction of young trees that will have eventually produced immense quantities of timber. The use of half-round sleepers means the destruction of the best trees in the timber forests of the States. I estimate the saving on the square sleeper for 3 ft 6 in gages at about 1s 9d, but I doubt whether the half-round sleeper would prove to be any cheaper at all. I do not think that any of the States, except Queensland, have used the half-round sleeper. I consider the 11 steel sleeper an improvement on the trough sleeper. These sleepers are not available just now, except at a very high price, and, so far as I am aware, they could not be manufactured in Australia at the present time. I think it would be impossible to get such sleepers at the present time, especially in this country. A permanent survey has been made for the extension from Katherine to Bitter Springs. A trial survey has been made from Bitter Springs towards Daly Waters for a distance of about 25 miles, and the permanent survey for

about 12 miles. The permanent survey is proceeding on towards Daly Waters. We have approval to go only as far as Daly Waters. It will take some months from now to complete the permanent survey with the present strength of the party. I should say that the Pine Creek to Katherine River section will be open for traffic soon after the middle of next year, and in this I include the construction of the low level bridge now to be built over the Katherine River. The idea is that the staff should continue with the work on the extension to Bitter Springs, as the work finishes on the Pine Creek Katherine section.

51 To Mr Gregory.—In addition to considering the route of the railway, I also got a general idea of the possibilities of traffic and the opening up of the country—as to the possible places of production and the likelihood of making the railway a paying proposition. I have only gone into the question of other ports than Darwin as a general way, I have not had an opportunity to consider the matter to a sufficient extent to enable me to give the Committee any detailed information. So far as I know at the present time, only small craft of light draught go up the Roper with provisions for the settlers. The Victoria River, I should say, is navigable for boats of from 5 to 10 tons. I think that the building of this line, with a view to opening up new country, would be justifiable. It would depend considerably on the class of country how far we would be justified in carrying railways inland for the purpose of transporting stock to the seaboard. In some cases stock owners might prefer to travel stock in good seasons, and take them by train over poorer parts of the country. I think, however, that it would be quite justifiable to deal with cattle country situated 500 miles from Darwin. It would be opening up a very large question to talk of extending that to 1,000 miles. A great deal depends on local circumstances when you begin to negotiate such long distances. I should say that, in view of the country I have seen as far as Bitter Springs, it would be quite reasonable to truck cattle for 500 or 600 miles for the sake of bringing them to market quickly. Approximately, the distance from Barkly Tablelands to Darwin would be 500 or 600 miles, and I take it that stock could be trucked that far. If a high price is given for stock, that is an inducement to carry it a long distance, and I think you would be quite justified in anticipating a cattle traffic of the kind I have indicated. So far as I know there is no possibility of oil or coal for fuel being found locally in the Northern Territory, the only prospects I have heard of are for copper, tin, and gold. We could get a considerable quantity of firewood, but in ordinary railway working it is somewhat doubtful whether it would pay to use firewood as against coal. On the ballast train we are using principally coal and firewood mixed. At the present time I am not able to say whether the return is cheaper than coal, but we have the matter under consideration. If it is possible to use the rise and fall of the tides for the purpose of generating electricity, it ought to be applicable at Darwin, because there the rise and fall is considerable. That is a matter for electrical experts. For the purpose of cheap freights and making railways pay, it is essential in the case of an ordinary railway that the line should be built as cheaply as possible, but

that is in the case of a line which it is not anticipated will be a main trunk line. As trade warrants, such lines can be built up, and I would adopt that principle to a reasonable extent in a main trunk line, in the way, for instance, of using low-level bridges over big rivers. It may be found economical to use low-level bridges over the big rivers in constructing the main trunk line from north to south. It is a very doubtful point whether, given a life of fifteen to seventeen years for wooden sleepers, it would pay to lay 7-ft sleepers instead of 8-ft 6 in sleepers. That is really more a question of policy. The ballast does not come right out to the end of the sleeper, one end is covered, but the other is uncovered. The widening of the line will eventually be done with one rail only, we shall not shift both rails. The sleepers will not warp, because so much will be covered with ballast. This does not mean much heavier ballasting than for the ordinary 3 ft 6 in line, because the ballasting is kept the same width as for that line. 60 lb rails are being laid. With the present traffic, 45 lb rails would be ample, but we should have to use more sleepers. We should approximately have to use 3,640 sleepers with 41 lb rail—it is not a 16 lb rail—as against 2,080 sleepers. It would not be advisable to go in for the lighter rail. Using the heavier rails, we decrease the cost of maintenance considerably. I do not consider there will be any saving in using a 41 lb rail, with a 60 lb rail we use fewer sleepers, and the maintenance is cheaper. Further, when it becomes necessary to widen the gauge, you would have to lay a big proportion of the cost of the lighter rails. I think that the weight of the rails used on the Clarence railway, in Queensland, is 49 lbs. On a good many of the lines in Queensland they are using 60 lb rails, which were considered the best for the Queensland gauge. In fact, when I left Queensland it had been decided not to import any more 42 lb rails. They were then taking up the 42 lb rails on the more busy lines and putting down 60 lb rails, utilizing the former for very light agricultural branch lines where there was no anticipation of their becoming main lines. I have heard of the concrete sleepers used on the Adelaide tramways, but I have not seen them, and so far as I know, no investigation has yet been made by the Commonwealth Railway Department. I have investigated some concrete sleepers that were laid down on the Northern Territory line and they were an absolute failure. The trouble is that the concrete no matter what the reinforcement may be, appears to fracture under the running of the trains. I certainly think it would be worth while to investigate the effect of the concrete sleepers laid down by Mr. Fenton in South Australia some eighteen months ago, and we shall probably get reports regarding them. Sleepers on a railway have to stand much greater shock and strain than on a tramway. I very much doubt whether it will be possible to make concrete sleepers more cheaply than wooden sleepers. The reinforcement costs money, and, apart from that, the sleepers are very heavy to handle. Concrete sleepers have been very much discussed of late years in relation to all railway operations but up to the present there is nothing to show that they have been an absolute success anywhere. We have reduced the size of the sleepers from 9 ft x 10 in x 5 in to 8 ft 6 in x 9 in x 5 in. The

present price of the powdered sleeper and jarral sleeper runs from 7s 6d to 7s 8d, and up to 8s 4d; but I should like to make myself sure as to that, because the prices vary. As to my figures in regard to the two different gauges, I may say they are the result of actual tenders called. The prices altered very much, and of recent years they have gone up. The powdered sleeper generally runs to 1s or 1s 2d more than the unpowdered. We have some thousands of powdered sleepers up there, and I take it that Mr. Hill recommended that they should be sent. I remember hearing of a report about some untreated karri sleepers being shelled off the Great Southern line in Western Australia. There has not been sufficient time to show whether the powdering process will prevent dry rot. The sooner we have some decisive experiments the better, because in the meantime a great deal of controversy exists. Nothing will prove the value of the powdered sleeper except an actual test. There are already about 40,000 powdered sleepers up there. When the estimate was made out for this railway we proposed to put either steel sleepers or powdered karri sleepers in, but what will be recommended is a matter that lies with the Engineer-in-Chief.

62 To Mr Fenton.—The construction of the Pine Creek to Katherine River line, exclusive of rolling stock, will cost approximately £20,000 per mile. That is slightly less than the original estimate, because we have eliminated high-level bridges over the Cullen and the Edith Rivers, thus effecting a saving of about £20,000. The Katherine River to Bitter Springs extension will cost at about £4,938 per mile or £1,000 per mile less than the Pine Creek to Katherine River extension. I have £6,000 per mile which the Pine Creek to Katherine River extension is costing includes the cost of erecting a low-level bridge over the Katherine. The height of this low-level bridge has been fixed at just above the normal flood mark, and when floods exceed that normal height they will go right over the top of the bridge. Low-level propositions of this kind have been used very considerably in Queensland for a great many years, and experience has proved that they have withstood floods very well indeed. We anticipate that the low-level bridge over the Katherine will be equally successful. It is a somewhat more difficult proposition than is the usual low-level bridge, but I have no doubt that it will stand equally well and that it is well worth the risk. From what I have seen of Mr. Combes' report, I take it that he looked at this railway rather from the standpoint of the local development of the Northern Territory, and without considering it as a through connexion to South Australia. In paragraph 7 of his report he states—

The railways routes that could best develop the northern portion of the Territory, and also give most promise of (in time) turning out payable propositions, are as under—

1 New Islands to Anthony Tagoon and Chinmooval	400 miles
Colliery to Anthony Tagoon westward, passing south of Wave Hill to the Western Australian border	470 miles

In saying that, he was confining the railway propositions entirely to the northern part of the Northern Territory, and was regarding only local development, and evidently not looking at the

matter as a through line proposition. I know that he also says—

The proposal to extend the Darwin Pine Creek railway through the Katherine, Bitter Springs and Newcastle Waters to Anthony Tagoon (and Camooswell) will not stand investigation.

In doing so he appears to eliminate the idea of a line going to South Australia. He has not indicated a route to South Australia. If he had in mind a line from Darwin to Camooswell, with a branch from Anthony Tagoon to the mouth of the McArthur River, his proposition would then fit in fairly well. But if he wished to connect the Darwin line with a line from Anthony Tagoon to Camooswell, he would have to branch off somewhere—not at Newcastle Waters, probably at Daly Waters, 100 miles north. In my consideration of the Katherine River to Bitter Springs line, I always had in view the fact that a transcontinental railway has to be constructed from north to south, in accordance with the agreement entered into between the Commonwealth and South Australia. I do not say that that line must be from Pine Creek to Oodnadatta, but it must be from Pine Creek to South Australia. I thoroughly believe that the construction of a line from the Katherine River to Bitter Springs will not only be in conformity with that agreement, but will assist development of the Northern Territory. I consider that the Commonwealth will be also largely on the safe side, so far as the route is concerned, in constructing the line to Bitter Springs.

63 To Mr Sampson.—The country surrounding Bitter Springs, though not of the best, is fairly good pastoral country. Within 4 miles of Bitter Springs an experimental sheep station has been established, under the care of the Administrator, and when I was there had 1,000 sheep on it. About 10 miles south is the Lisey Creek, and towards the western border of Queensland the country has been more or less taken up, and is stocked. The old Lisey station is only a few hundred yards off the route of the proposed railway. Stock could be travelled from Bitter Springs to the Katherine in a good season, but in the dry portion of the year it would not be good business. The feed would be scarce. The best part of the year in which to drive stock covers two or three months immediately succeeding the wet season. As far as I could ascertain from inquiry, the grass lasts for two or three months after the wet season. It commences to grow vigorously with the first rains of the year, which generally take the form of thunderstorms, and set in about November. From that time onward it continues to make rapid progress during the wet season, which ends about the middle of March. For a couple of months subsequently there is good feed. From that time on wards, however, the grass begins to dry up. It can be burnt about May. It will be seen, therefore, that stock driving would be practicable for say, four to six months of the year. From what I have seen of the country near Murrumbidgee, there is some good agricultural soil, but it is patchy. Owing to its isolation, however, there has never been any agricultural operations in that locality which would form a guide as to its production. My own idea is that tropical and semi-tropical agriculture could be carried on there if markets were available. The markets which I have in my mind are those of Queensland and South Australia, but the best overseas markets for agricultural and pastoral products would probably be those of China, Japan, and India.

Leaving out of consideration the question of the construction of a main trunk line to South Australia, I consider that the extension of the line from the Katherine River to Bitter Springs is thoroughly justified on its merits. The building of that extension will bring the line closer to the Barkly Tablelands, which, I am told, is the best portion of the Northern Territory, and also to the settled land on the Roper River, and on the Lisey Creek. I cannot say off-hand what revenue the proposed extension is likely to earn, but certainly that information could be obtained by ascertaining the number of stock on the different stations in the vicinity and estimating the proportion of that stock which would be attracted to the railway. The line would undoubtedly attract new traffic, as well as assist in new settlement. If I were supplied with details as to the number of stock in that portion of the country, and the area of occupied lands, I could form an estimate of the probable traffic. This information could be obtained from various officials in the Territory, although its acquisition would take a little time. I have read the report of Messrs Clarke and Lindsay on the Northern Territory railways and ports, and I am aware that they recommended—

(1) That the main trunk line to continue from the Katherine River to Oodnadatta and thence to Alice Springs as nearly direct as is consistent with the best engineering and at revenue producing connections; and that the construction be commenced from both ends.

(2) That the Victoria River district should be served by railway connexion with Darwin as its nearest and most convenient port. Preferably the connexion should be made by a western swing, of the trunk line south westerly from the Katherine passing near Wilberforce, thence to the McArthur line at about Bitter Springs.

I have not seen the country to which they refer, but, judging from the map, and keeping in mind the country to be traversed on the eastern side of the overland telegraph line, I have no hesitation in saying that it would not be a good proposition to bend the line to the west in order to bring it nearer to the Victoria River country. I would prefer to bend the main trunk line to the east of the overland telegraph line, and to serve the country in the neighbourhood of the Victoria River and Sturt Creek by means of a separate or branch railway system. The bending of the line to the west would give only a partial connexion with that country, whereas a branch line could be extended to any distance that might be required to develop it. I consider that an extension of the line from Bitter Springs to pass somewhere in the locality of Anthony Tagoon should be thoroughly investigated, as I believe that the bending of it east to that point, and thence following as nearly as possible a due south line to Oodnadatta, or perhaps to Herget Springs, would take it through the greatest area of good land. That route should be thoroughly investigated before the Commonwealth is committed to following the route of the overland telegraph line.

64 To Mr Gregory.—There is a very extensive cypress pine forest in the vicinity of the Marram berrine, so that there would be a fair amount of timber traffic until the forest was cut out. The area embraced by this forest is several square miles.

65 To Mr Gregory.—The line could not be taken further east just there, because the route is on the edge of a very large area of flooded country.

66 *To Mr. Fitzgibbon*—The cypress pine to which I have referred is of first rate milling quality. The trees are of fine size, and the timber area could be advantageously tapped at the present time. Roughly speaking, it would support a saw mill for a few years. I have no great hopes of afforestation in the Northern Territory. The timber there does not grow very symmetrically solidly, or of great size. When one gets away from Darwin, it is of too crooked and stunted a growth. The ruling grade on the line is 1 in 100, and the sharpest curve has a radius of 40 chains.

67 *To Senator Story*—The earthworks on the Pine Creek to Katherine River line will be completed within the estimated cost. I recommend that the trunk line should be a reasonably straight one from the Katherine River to the South Australian border, and that spur lines should be constructed to develop the country on either side. I mentioned previously that the proposal that the main trunk line instead of being linked up with Oodnadatta should be linked up with Hergeot Springs, is one that should be investigated. We have had a survey made, with a view to linking up the north south line with the east west line at a suitable point, instead of taking it south from Oodnadatta to Port Augusta. Whether that is a more desirable proposition than a connection with Oodnadatta or Hergeot Springs, which would necessitate an alteration of the existing gauge on to Port Augusta, is a matter which is receiving consideration at the hands of the Railway Department at the present time. Under existing conditions it is impossible to say which is the better course to adopt. But that phase of the matter is being kept carefully in view in connexion with the determination of the route north.

68 *To Mr. Gregory*—In regard to the quantity of powdered timber that would be required to permit of a thorough and effective test being made of it, in proceeding with the line sleepers of some sort must be provided. Steel sleepers at present are very expensive and consequently we have to fall back upon timber. In choosing between powdered and unpowdered sleepers it must be borne in mind that there is a certain quantity of powdered lumber on the ground quite sufficient to permit of a thoroughly good experiment being made. As to the sleepers required for the balance of the Pine Creek section of the line or of the extension to Bitter Springs it is a debatable question whether we should use powdered or unpowdered sleepers. I was not aware that the Windham Creek railway, which passes through badly infested white ant country, has been built wholly of ordinary jarrah sleepers and that although they have been down six or seven years, the manager reports that he has not lost a single sleeper. I think it is a matter for the consideration of the Engineer in Chief as to whether it is necessary to continue the use of powdered sleepers. In the northern part of Queensland—the Cooktown line about 70 odd miles of railway have been laid through country which is badly infested with what ants of the worst kind, and yet all unpowdered timber sleepers have been used up on it and although some of these sleepers have been lost owing to the ravages of white ants, they have been justified so far as the loss from white ants is concerned. It must also be remembered that as the main trunk line comes south it will be getting out of the white ant country. Whether it will be wise to powder timber

like jarrah or ironbark in preference to kauri resolves itself purely into a question of cost. A difference of 1s per sleeper is equivalent to £100 per mile on a railway. If nothing is to be gained by the powdering process, obviously it is hardly worth while paying this additional money. The Engineer in Chief will determine or recommend whether unpowdered sleepers should be used on the main trunk line further south.

(Taken at Melbourne.)

WEDNESDAY, 24th NOVEMBER, 1916

Present

Mr. Ruxley, M.P., Chairman,	
Senator Keating,	Mr. Fitzgibbon,
Senator Blair,	Mr. Gregory,
Mr. Linton	Mr. Baupain

Algeron Combes, sworn and examined.
69 *To the Chairman*—I am in no official position at the present time. I was a member of the Royal Commission to inquire into the Northern Territory railways and ports. I arrived in the Territory in June, and left early in November. I have been following the route of the proposed main line from Daly Waters, north to Darwin. I do not consider that the best route, because it is hopelessly poor country. I have been twice over the proposed extension of Katherine River to Bitter Springs, once in July, and again in October. It is very doubtful if the permanent line is to go as proposed, along the telegraph route, whether this extension is essential to the completion of the scheme. It is quite likely that the better route is over to the west down through the Victoria River country. Of course, I have not seen the country, but I have heard it described in evidence. I have heard the evidence of station managers and others who knew the Victoria River country, and they speak highly of it. It is used for cattle and the southern end will possibly do for sheep—at Wave Hill and around there. The people at Wave Hill would not go to Bitter Springs because they would have to cross a wide stretch of waterless country. If Darwin is the port, the cattle would go to Katherine. I do not think that the extension to Bitter Springs would do all help to develop this part of the country. I should say that the route recommended by Messrs. Clarke and Lindsay would be the better one, continuing further south, and tapping the Wave Hill country. For twenty years Mr. Lindsay advocated the other route, but when he came to consider it again, he favoured the extension to the westward. Mr. Lindsay is not in any way interested in land there that I know of. If the railway be taken as suggested by Mr. Lindsay, it would be going away from what is known as a fairly good show for a tin mine. Tin mining is very uncertain. I have been at Marranbar. The field started in June, 1913, and I was there in October when it seemed to me that there was very good prospecting show. The deepest shaft was about 40 feet and I do not know that it is any deeper now. Tin mining is the most uncertain of all mining. If the field is what it has been said to be it might be advisable to take the Bitter Springs route right through it. It is a good thing to open up a country by a railway, but I say I know nothing of Marranbar recently all I can say is that it was a promising show when I was there. The establishment of the Govern-

ment battery will be a good thing if there is enough pro, but at the time it was proposed I thought it was a mistake—that it would be better to subsidize some of the claim holders, and sink some shafts. Under the circumstances, I should advocate the other route. I would not go near the mining district unless it turned out something very exceptional before the construction started, and that is not likely in this case. I think that the route suggested by Mr. Lindsay is a better paying proposition, because it opens up better country which is there for all time, while the mining might last only a few years. In my opinion, Wyndham is the proper port for most of the Victoria River country, and Wyndham is 200 miles nearer than Darwin. There is not much trade at present at Wyndham, and not much has been done in the way of improving the port. There is not a water supply, but there is a good rainfall, and water could be conserved. All that, of course, is a matter of money. I have seen the Admiralty chart and I think that a good harbor could be made there, there being a tidal rise and fall of 20 feet. As a matter of fact, 4,000 ton steamers go there now, and cattle are shipped. The Western Australian Government have been running a steamer of this size, and taking cattle to Fremantle. As I have said, it is the natural outlet for the Victoria River country. I have seen in the press that freezing works have been started, but I understand that they were stopping through some trouble. If freezing works were built there, they will get all the Victoria River cattle except those from, perhaps, within 100 miles of the Katherine. That, of course, would not help Lindsay's route. There would be a certain amount of traffic on that route from Wave Hill, but Wyndham, if opened up, would get most of the trade. In my opinion, the whole scheme seems to be greatly exaggerated. No doubt they are bad, particularly at Darwin, but not so bad as they have been represented. They are not nearly so bad as I have seen in Malaya where wooden sleepers are used, and I should have no hesitation in using wooden sleepers south of the Katherine. Of course, I should use metal sleepers if they were available, but if steel sleepers have gone up in price by 100 per cent, I should have no hesitation in using wooden sleepers. There are some steel sleepers on the existing line which have been in use for about twenty seven years, and inland they are standing well, but near the coast they are showing the influence of the climate. I did not see any of the wooden sleepers, though I saw some made of concrete. There has not been much of a test of the concrete sleepers at this place, because they were not put on the main line, but on the wharf line, where the only locomotive was a sort of freak shunting engine. I know of no poorer country in Australia than that on the telegraph line route, and that is a poor country by the fact that it is nearly all unoccupied. There are no other surveys that would bring this railway through better country, if the railway has to go this way the telegraph route is as good as any other. I know that there is the agreement with South Australia, but the northern part presents such a hopeless proposition that I think the agreement ought to be varied to carry out the agreement as now suggested would be of no use to South Australia, because that State would have to pay its share of a heavy loss. Beyond the Macdonnell Ranges, it does not matter, from a South Australian point of view, where the railway goes so long as it goes to a port that will develop the Territory. A railway from Katherine to Bitter Springs and Anthony Lagoon—about 550 miles

from Darwin—would strike good country, but then you are only 230 miles from a good harbor at Pilew Islands. The trade, if it goes to Darwin, will be penalized to the extent of 300 miles, which, at Queensland rates for cattle, represents about 12s or 15s per head. That might continue until the country got settled, but there would be agitation for a line to Pellev Islands, and then the other line would be left without traffic. The extension from Katherine River to Bitter Springs is, I think, a useless proposition. I have not seen the west country, but it appears to me to be the better of the two. I have been at Providence Knoll, where there is a small patch of limestone. The country does not improve for at least 40 miles from Katherine. At Bitter Springs there is a big patch of limestone, about 20 square miles or so. The rest of the country is quartzite and sandstone—Cambrian or pre-Cambrian—the poorest possible country. After Bitter Springs, we have Hodgson Downs and Nutwood Downs, which are what I should call second class pastoral country, the number of acres that the country is, per block, on Downs and Eley station, about 4,000 square miles, run about 5,000 head of cattle, and this seems right up to the capacity. Nutwood Downs, near Daly Waters, is about 800 square miles, and there are 5,000 head of cattle. These cattle, however, run all over the country. By taking up 800 square miles, to occupy the whole of the permanent water and no one else can take up land. Baulonia Downs, near McArthur River, has 400 square miles leased, and carries 2,000 head of cattle. The owner told the Northern Territory Railways and Ports Commission that he reckoned he runs over 4,000 square miles, and that this is necessary. The boundaries are not marked, and I am not sure. I spent a week riding up the Roper River Valley to Bitter Springs, and I saw patches of good country though, on the whole, it is very poor. The valley is from half a mile to 5 miles wide, with barren hills on either side. There is no possibility of irrigation or water conservation between Bitter Springs and the Katherine. There is no permanent water. The Katherine is a permanent running stream, but it is doubtful whether one could call Bitter Springs permanent water. The permanent springs are almost unfit for use, and the water-hole above is only enough for a head station. The water in the Roper gets better 25 miles down, where it is just about drinkable. Twelve miles south of Bitter Springs you might call it permanent water at the abandoned station of Eley. Beyond that in about 100 miles there are four wells, but when I was there three were dry and we had to stay two days to water forty horses. There is no possibility whatever of minerals south of Marranbar until you get to Daveport and Marranbar. According to Dr. Jensen, it is proved that there is sub-artesian water. There are seventy bores on the Barkly Tablelands, and they have never failed to strike water at 200 feet to 300 feet. The water varies in quality, and most of it might be classed as good stock water. I take it that the Northern Territory is purely a pastoral proposition for the present generation. There have been experiments with agriculture for thirty years, I suppose, without success, and much money has been spent in this way. Most of the things that have been tried would doubtless grow but commercial success is another matter. Of course, I am now speaking of the northern part of the Territory. I think there is a possibility of sheep being successful, and much evidence was given before the

Royal Commission on this point, showing that sheep would do on the Barkly Tablelands, and at the head of the Victoria River, though it is doubtful whether this could be said of country further north. When I was there sheds and to feed were being put up at the Government sheep farms at Bitter Springs, but the sheep had not arrived. I do not see much prospect for sheep doing well there. The Government may be able to keep sheep alive, but, as I say, commercial success is another matter. Dingoes are very plentiful on the Roper. The spear grass in this part of the Territory is very bad for sheep, and the remaining grass is very coarse. In the early seventies, sheep were tried on the Katherine. About 12,000 were sent from South Australia, along with 2,000 head of cattle, to a station formed near where the proposed railway is to cross the Katherine. After three or four years, however, the venture was abandoned, the animals mostly dying or disappearing. It is evident that the people who carried out the experiment know something about sheep, because, although it was a year, none were lost on the road from Adelaide. Dr. Brown, of Adelaide, is supposed to have lost about £40,000 in this experiment. Sheep have been tried about 70 miles from Darwin by the present Administration, but I do not think that any are alive now. South of the head of Victoria River, Newcastle Waters, and Anthony Lagoon, there is every possibility of sheep doing well. According to the evidence we heard, Avon Downs Station has been a sheep station for about thirty years. According to the estimate of the owners, Barkly Tablelands can carry about 2,000,000 sheep and 600,000 or 700,000 cattle on 20,000,000 acres—1 sheep to 5 acres. The highest estimate we got was 150 to the square mile. There is spinifex south of Barkly Tablelands. Cotton grows wild in the north at Port Essington, but whether it could be made commercially successful, I am doubtful. Experiments in this regard have been made at several places in the Territory, and sugar has also been tried. In the eighties, about £20,000 was spent on sugar near Darwin, but when they got 7 tons the first year, and 5 tons the next they gave it up. Cocoanuts will grow, but I hear that they do not ripen properly. Coffee has been tried, and I think the reason for its non-success is the six months drought every year. I lived on the Malay Peninsula for five years, and I know what a tropical country is. Rice has been grown in the Northern Territory to some extent; at one time about 3,000 Chinamen were there, and as all Chinese understand rice growing it is likely that it was given a fair trial, but they did very little good, and gave up the business. I saw an attempt near Pine Creek when I was there, but I do not think the soil is suitable, or that the rainfall tends to success. My experience of the Northern Territory does not present a very bright outlook from an agricultural point of view, and I do not think that that view is worth worrying about. On a small scale there may be agriculture, but I regard the Northern Territory as a pastoral, and, possibly, a mining proposition. I was all over the mining fields at Pine Creek, but everything seems to be going back. There is one good copper show, I believe, about 40 miles east of Pine Creek. There is certainly a good show of copper ore, but there are many difficulties. Nothing can be done without a railway, which would cost £250,000 and there is no wood, not even firewood for mining purposes. There is no flint, and all the ore would have to be taken to the seaboard.

60. To Mr. Sampson.—My idea is that the Territory should be developed from the coast line by railways between the hinterland and the nearest port. I have already said that I regard the proposal before us as so hopeless that the agreement ought to be varied. There are only two routes, one by Bitter Springs, and the other down to the west, and I assume they would connect somewhere south of Newcastle Waters. I would carry the line more directly south than Messrs. Clarke and Lindsay's route, say to Wavo Hill head station, and connect somewhere about Renner Springs, or further south. If the line I have spoken of were built, Darwin would get the traffic in the absence of a line to Wyndham, and that, in my opinion, would be in accordance with any scheme of development from the coast line inward. If a line were not built to Wyndham, it only means that the export of sheep and cattle from Wavo Hill would be handicapped by 200 miles extra haulage to Darwin. If the route I suggest were taken, it would serve most of the Victoria River country, but it would not avoid a railway to Wyndham when the country became more settled. As a pastoral proposition the line would serve fairly well. The deviation of the line to Marrabey all depends upon the prospects of the field. The Victoria River country will justify a railway sooner or later on its merits. I should prefer the western line with a branch. The construction of a main line on the best route, and with a view to its being a commercial undertaking, is a matter of course that involves much investigation. It depends on the traffic to Marrabey, and the possibilities of traffic south of the junction. Then we have to consider the relative cost of the different routes, what the grades will be, and the possible traffic from the Victoria River country. My impression now is that the western line is the best.

61. To Senator Keating.—Throughout the Territory, where there are creeks, there is a possibility of conserving water; but when I was speaking a little time ago of the Roper River, I had irrigation in my mind. There we have presented opportunities for the construction of dams for the conservation of water for stock and domestic purposes. On the McArthur River, near Borroloola, there might be facilities for the conservation of water for irrigation purposes. That is the only place I saw in the Territory where there is a large area of fairly level country suitable for irrigation. I had about ten years' experience of water supply and irrigation work in Victoria and New South Wales, and I think the country about the place I have mentioned would be good for agricultural experiment. It appears to be fairly good soil. There is a lot of limestone country on the McArthur River, it makes the fertile flats lower down. Owing to the presence of limestone the country is better than any I saw elsewhere. It is very hard to say what the soil there is capable of producing, but I think, for one thing, lucerne would grow. There are a few patches of it round about Borroloola, and it seems to do very well. The line I suggest would go through the country I have just mentioned, but that was too small a consideration to influence me in my preference for this route. There is a pretty fair area of this land, perhaps 40 or 50 square miles, but apart from that consideration, the route I favour is the shortest from good country to the sea board. There are patches of country elsewhere in the Territory where conservation and irrigation might be carried on. For instance, there might be some land on the Roper on to

which water could be pumped from a water-hole, but these are only isolated instances. I think the agreement should be varied to cut the railway route should be from Oodnadatta to Alice Springs, and the Macdonnell Ranges going north; then south from Pellow Islands to Anthony Lagoon, Alroy Downs, and the Frew River. I am doubtful about the Frew River, because, according to Mr. Gordon, of South Australia, it is pretty rough country. If the route I have mentioned should prove impracticable, the alternative route from Anthony Lagoon to Tennant's Creek, and thence along the overland telegraph line could be adopted. The line from Port Augusta to Pellow Islands is 250 miles shorter than from Port Augusta to Darwin, and 250 miles in a long haul like this is a big thing. I do not consider that the agreement should be varied to the extent of not connecting Oodnadatta right through the Northern Territory to the sea, but I think that the line should go from Oodnadatta to the Pellow Islands, on the Gulf. I do not think that Darwin has any more strategic value as a port than Pellow Islands or Wyndham; an enemy might land at any one of the three places, and even at Port Essington, for that matter, which is about 200 miles from Darwin, and has a better harbor. Admiral Creswell gave evidence before the Commission, and laid stress on the comparative importance of Darwin as a naval station. As a strategic point for Australia in relation to the waterway between the Archipelago and the Pacific, Admiral Creswell considered Darwin most important; but, on the other hand, General Gordon, who was then Chief of Staff, refused to name any point as being of great importance, the fact that from Darwin it would be more easy for Australia to harass a hostile fleet passing from Europe to the Pacific than it would be from any other point. I regard Darwin and Essington as practically of equal value from this point of view. If all the railways that have been mentioned are built, the question of fuel becomes a big one. I think that coal at Darwin now is about £2 10s a ton, and the working expenses of the Darwin line are 8s to 10s per train mile, as against about 3s. 8d. in Queensland. According to a geological report, there is coal in western Queensland, south of the Gulf, for hundreds of miles, and this might be developed and made use of. I do not think, however, that any steps have been taken to develop this deposit. I have not seen any of the coal, and I do not know its value. According to Dr. Jensen, there is no oil or coal in the Territory of any commercial value. As I have said, my view is that for a generation at least the Northern Territory is purely a pastoral proposition, and for its development railways must be constructed from the nearest ports to the hinterland. The exports will be cattle, wool, and sheep, and these require to reach the nearest possible ports. I do not entertain any very high hope of tropical production in the Territory, and I think that the non-success in this regard is due to the rainfall—six months deluge and six months drought. In the real tropics the rainfall is the same all the year round. I have had experience of railway construction in the Malay Peninsula and Siam for five years. My work was that of deciding when, where, and how to build railways. The conditions in the peninsula and Java are not anything like those of the Northern Territory. Nominally, the Northern Territory is in the tropics, but actually it does not seem to be so, for the vegetation is altogether different, and the land so much poorer with the six months' drought

In the real tropics the rainfall is very evenly distributed throughout the year, there is no winter and no summer—no season. The Northern Territory is a tremendous proposition, though apart from that the railway question appears to me simple enough. It is a cattle and sheep proposition. The chief things wanted are water conservation and bores, with the opening up of stock routes. I know that for the last twenty or twenty-five years there has been great railway construction development in Africa—in the Sahara and other places which some years ago were regarded as absolutely unproductive. I have not, however, gone into the question more than reading about it in the technical journals. I do not see how any information in regard to this water development elsewhere could help us so far as our railways are concerned. Information of the kind might, however, guide us in making better use of the coastal country.

62. To Mr. Gregory.—I should say that the construction of the proposed strategic railways of south eastern Australia is of more importance at the present time than Territory railways. The pastoral development of the country will, I think, come from western Queensland and western New South Wales. The railway which would give the greatest return would be one from Pellow Islands to the Barkly Tablelands. There should be a railway from Wyndham into the Victoria River country, and one from Oodnadatta to the Macdonnell Ranges. It would pay people to transport cattle at Queensland rates 500 or 700 miles to a port, but it might not pay at Northern Territory rates. I should say the cattle trade could stand that. Of course, it is a matter of the class of railways. The present line from Darwin to Pine Creek is not capable of carrying any really heavy traffic; it is not strong enough, but needs to be rebuilt at a cost of £400,000 or £500,000, with heavier rails and strengthened bridges. Until that is done the line cannot carry proper train loads. I do not think that 41-lb. rails are heavy enough, there ought to be 60 lb to 70 lb rails. I have heard that in Western Australia new areas have been opened up by railways with 41-lb. rails, on which heavy loads are carried on easy grades. The grade is 1 in 60 between Darwin and Pine Creek. It all depends, however, on the amount and nature of the traffic, and cattle is a difficult traffic to handle in the Northern Territory. It lasts for only five or six months in the year, and the question is what to do with the men afterwards. In a State like Queensland, with a big railway system, the men could be transferred from one place to another, but in the Northern Territory it appears to me that they would have to sit down and wait for the next season. As to light railways for opening up the country, I would point out that the Darwin and Pine Creek we have a region of very heavy rainfall, even at Bitter Springs it is about 35 inches. The saving in the first cost between the heavier and the lighter rails is not very much, and I do not think that we could build railways for £5,000 per mile with the present price of labour in the north. South of the Victoria River you will get grades of 1 in 100. The fencing works established by the Western Australian Government at Wyndham will undoubtedly get most of the cattle. At one time, the great idea was to drive cattle, but railing is coming more and more into favour. Driving is done on a big scale very cheaply, from 1s to 1s 6d per head per 100 miles. Cattle can be driven from Arltunga River to Bourke, 1,400 miles, for £1 a head. I have never gone into the question of utilizing

the tide in the north west in the generation of electricity for power purposes. I believe that some one now is experimenting with the generating of electrical power by means of the sun's rays at the Macdonnell Ranges. I did not see any mining timber in the locality of Marranboy. Along the river, and in the water holes, there is what is called paper bark, which I take to be a species of tree, 2 or 3 feet in diameter, which might be useful. It appears to be sound, but the great percentage of the timber in the Territory is useless stunted stuff, and nearly all hollow, not fit even for good firewood. As the tin mining field is developed, and they get deeper down the expenses will increase, and water will become a very heavy consideration.

63 To Senator Story.—I prefer the deviation to the west in order to touch the Victoria Downs country. As a strategic railway, it is necessary to have a line north to a certain point, say, Newcastle Waters, Fellow Islands, Wyndham, or Darwin, possibly it would be advisable to have a railway to three places. I do not think it is correct that, with a line from Darwin to Port Augusta, the trip to England could be shortened very considerably. I think it is practically the same distance from Colombo to Darwin as to Fremantle. If the idea was to go overland through India or Russia, the people would not go to Darwin, but Queensland. I have not been in the Macdonnell Ranges country, but from evidence given to the Commission, there seems to be considerable mining and pastoral possibilities there. A railway for 300 miles from Oodnadatta to the Macdonnell Ranges would probably cost £2,000,000, and it would be run at a loss, though it would be justifiable. It would open up a lot of country, and there is no question as to the route, Port Augusta is the natural outlet for all that part of Central Australia. Sooner or later a railway will have to be built southward to the South Australian border, but when I reported, I was not looking very far ahead. The building of a line from the Macdonnell Ranges would take four or five years, and to fill up the gap would take about the same time. The only justification for taking the line to Bitter Springs is, in my opinion, that it would develop the Marranboy tin field. The country traversed is of the poorest possible kind, and not capable of development. The evidence of experienced pastoralists shows that the best of the country would not carry more than four head of cattle to the square mile. The labour of the 150 men or so who might be thrown out of employment if the proposed railway is not proceeded with might be utilized on the Marranboy tin fields, if they were miners, but I think there is plenty of other work for them in the way of making roads and tracks, and works for water conservation. In Darwin, there is an L shape pier capable of handling about 20 tons an hour and what is required is a new causeway on which the trains can be run alongside the ships. This work would cost probably £80,000 or £100,000. I have heard it said that people are likely to settle about Bitter Springs, but in the last two or three years the Administration has surveyed and thrown open 200,000 or 300,000 acres for agricultural settlement and there are only ten settlers. Why not put the men there? It is only 70 miles or so from Darwin, while Bitter Springs is 265 miles away from the sea board. The land subdivided is quite good, and possibly, better than at Bitter Springs. I certainly do not think there is any necessity to build a railway which would be a long proposition simply for the sake of keep

ing 150 men at work, there is plenty of work to be found for them some day it would be a good thing to have a line from Darwin to junction with the McArthur River railway for strategic purposes, but that is a long way ahead. If we spend money in that way now, there will be no money to build huts really required for settlement purposes. It is possible, of course, that in the present unsettled state of the world, a strategic railway may be needed within, say, the next eight or ten years. Assuming that there are freezing works at Wyndham, there is evidence that there is a good, well watered stock route to that place from Victoria Downs, and it is quite likely that, in view of the extra distance and freight to Darwin, such a stock route would be used. The same thing, however, would follow if you put the freezing works at Fellow Islands. The difference in the distance to the sea board is even greater. If the line were continued along the telegraph line, and a spur made to the Victoria Downs country it would add considerably to the distance, so much so that it would probably kill the traffic to Darwin, and send it to Wyndham. As a railway would go, the Wave Hill country is almost as near Port McArthur as it is to Wyndham. I propose a line right across east to west joining the two Pellow Islands would very likely get the traffic, even if a railway were built to Wyndham. The distance from Wave Hill to Pellow Islands and to Darwin is practically the same. The plan prepared by H. Y. L. Brown shows the possibility of coal in the Territory. Bores were put down in Anson Bay two or three years ago, but nothing was found. The land spoken of by Brown towards the Barkly Tablelands has not been bored, but it has been examined by Jensen. I am pessimistic about the prospects of tropical agriculture in the north, but I should think that tobacco is the thing that would grow there. For some reason or other the growth of tobacco seems to have been a failure in spite of the favorable climate. I do not know any of the particulars of the reasons why tobacco culture was abandoned in the Northern Territory, but I still think there are possibilities. For thirty years or more experiments of all kinds have been tried, but none successfully. There is no doubt that maize would grow there in parts.

64 To Mr. Kenton.—If the trade were good enough, steamers would go to Pellow Islands. I did not examine that port as much as I should have liked, but Mr. Jensen sent a fortnight there, and is as well qualified to express a judgment as I am, and he considers that it would carry a railway. There would be no great length of piering required. The chief difficulty is taking the railway out to the islands, and bridging the arms of the sea. Where it is proposed to make a harbor there is a depth of 24 feet at low tide within 100 feet of the shore. Pellow Islands would make quite as good a harbor as would Darwin. The latter has had about £80,000 or £100,000 spent on it and probably requires an other similar expenditure, and if the money were spent on Pellow Islands we should have a harbor quite as good. A good deep sea harbor there would not only serve the Barkly Tablelands but also the Normanton district, in Queensland. I do not know the Queensland port but we had evidence that it is hopeless to get a good harbor on the Queensland side. My contention is that the through route I propose will save 250 miles of haulage from sea to sea and thus, taking the cost of the line at £5,000 per mile, would mean

a saving of £1,500,000. I should say that Wyndham is about 200 miles from the western border of the Territory at Turkey Creek. As to whether, before Wyndham becomes a port for the Victoria River country, it would be necessary to have a line of railway from Wyndham to the border, I should say that Wyndham would still be likely to get the trade without a railway. This would depend on all sorts of things—markets, seasons, and so forth. The evidence is that there is a good stock route right through at every season of the year. I laid out about 400 miles of railway in the Malay Peninsula which would form part of the scheme of an overland route to Great Britain. I think it will be a long time before that comes to pass. Such a railway would greatly reduce the mail times, but a lot of railways will have to be built before that is an accomplished fact. I understand that Veeley Brothers have a chain of country from the head of the Victoria River to near the Katherine, and I should say they are bound to send their stock that way. If they sent it to Bitter Springs they would have to cross a patch of absolutely waterless country. The deviation proposed by Clarke and Lindsay would take the line away from the Marranboy tin fields. I think that oil might be used successfully as fuel in the Northern Territory, but that is a large question. Darwin and the northern part of Australia is very well situated so far as the supply of oil is concerned, because there is Borneo, and possibly New Guinea to draw on, and, in time, oil may be used in preference to coal. If oil could be obtained in large quantities, it would be a better proposition than taking coal from the south. My Commission was recalled before it had finished its investigations, and it will be seen from my report, that I do not recommend a railway straight out for the reason that I have not seen sufficient of the country. The Commission was appointed for two years, but with the changes of Government it ended in thirteen months. Had I had another nine months, I should have seen the southern part of the Macdonnell Ranges country, and also the Victoria River country. As I have said before, I regard the proposed extension from Katherine to Bitter Springs as a hopeless proposition.

65 To Mr. Lindsay.—By profession, I am a civil engineer. I am a native of Tasmania, and at sixteen years of age I joined my father, who was a civil engineer in the Wimmera. I was three years with my father on water supply and irrigation work. I was then some seven years in New South Wales, engaged in water supply and mining works, including copper smelting. In 1898 I joined the Victorian Railways, and for four years was engaged in all sorts of railway work. In 1902 I was appointed by the Colonial Office as Chief Surveyor of the Malay States railways, this included part of Siam. I was there four and a half years and on my return to Victoria was appointed superintending surveyor in charge of railway surveys in Victoria. This meant investigating and reporting on all railway propositions, making surveys, and so on. In 1913 nearly seven years afterwards I resigned that position to join the Northern Territory Commission, my first official work was reporting on the proposed strategic railways of south eastern Australia. Apart from the Northern Territory schemes I have in the last fifteen years investigated and reported on about £25,000,000 worth of proposed railways and have designed and made surveys for about 1,000 miles of railways costing £5,000,000 or £6,000,000. I am now occasionally engaged in private practice. I was in the Nor

thern Territory from June to November, 1913, and my knowledge of the country is confined to that period, with the exception that some years ago I was in Darwin. I have, however, supplemented my experience by a careful study of the question, and have examined witnesses from all over the Territory, so that I think I have now a fairly good general knowledge of its requirements. There is no practical need for the building of a north south railway as a commercial enterprise, though it might be justified for strategic purposes. Commercially, the line would fail for want of traffic, but on this point it would be better to take the railway in sections. For 300 miles from Oodnadatta to Macdonnell Ranges, there are possibilities, and the line might, in the future, pay working expenses. The next 300 miles to Tennant's Creek is very poor, excepting in patches, as I understand from reading the evidence. The following 400 miles to Pine Creek is also wretched country. Unless for strategic purposes, I would suggest a go easy policy for railway construction through the centre of Australia. I believe in a line to Macdonnell Ranges, a line to Yellow Islands and from Anthony Lagoon to Cameroonal. If it is decided to follow the telegraph line, then the line from Katherine to Bitter Springs is a section. If the Marranboy tin field is sufficiently good, it is probable that the line should be taken across there instead of missing the field by 12 miles. The only immediate justification for a proposed extension is as a link in the north south railway. The country proposed to be traversed has been known ever since 1872, and it is still unoccupied. At one time, Bitter Springs formed part of the Elsey cattle station, but it was abandoned. Presuming that the section between Katherine River and Bitter Springs is determined on, I do not think that a branch line between Katherine River and Pine Creek would be preferable to a line to Wyndham, which I regard as the natural outlet for most of the Victoria River country. The northern portion would be better served by Darwin, because of the distance only. I would favour some arrangement with the Western Australian Government to connect up the Victoria River district with Wyndham, because to develop that part of the Northern Territory is the proper thing.

66 To Mr. Sampson.—I do not know the Daly River, but I know the Katherine, and I have been at the Alligator and the Roper. I think that the soil in the lower reaches of the northern river system is quite unsuitable for irrigation, it is too stiff. In the Malay Peninsula, there is practically nothing else but rice growing and tin mining. I should say that 90 per cent of the rice there is grown in swamps in fairly stiff soil, between the hills. North of the Katherine there are a few square miles of nice looking limestone country, which might be irrigated by pumping water out of the river. No doubt a great amount of water could be conserved on the Katherine, if required but I know of no large area to justify the building of great reservoirs, and so forth. I am very doubtful if the dairying industry could be carried on in any of the coastal country, because, with insect and other pests, the cattle do not do well. The flies in the wet season are something frightful. The cattle there seem to live, but I am doubtful whether dairying would succeed. Amongst the other products grown in the Malay Peninsula is rubber, which I think, in some respects, so far as the climate and soil are concerned might be produced in the Northern Territory. However, as I have said before, the trouble

is the six months' drought. I do not think that the prospects in regard to irrigation colonies are too bright. The only place I thought worth trying was, possibly, on the McArthur River, where the soil is better, because it comes from limestone country higher up. There are also good sites for dams and storage. Along the Adelaide River, there are fairly extensive flats which are periodically inundated, and ought to be suitable for rice culture. I have had no evidence on that point, though I have heard a good deal about it. I think there might be settlement on a lot of these rivers, but that, of course, means a small steamer proposition, and not one of rail ways. I see no chance of closer settlement in the country where there is six months' drought every year. Irrigation is an expensive business, with many drawbacks, and I do not think we need worry about it for a time. A comprehensive report on the subject might be useful, but I do not think the prospects warrant it. I have heard of no areas sufficiently large to justify such a policy. River valleys like the Alligator, the Roper, and so forth, are very much flooded in the wet season being frequently many feet under water. The salt water in the Roper extends about 80 miles up.

(Taken at Adelaide)

WEDNESDAY, 27 FEBRUARY, 1916

Present

Mr. RILEY, Chairman,	
Senator Keating	Mr. Gregory
Senator Story	Mr. Sampson
Mr. Toulson	Mr. Laird Smith
Mr. Tullyson	

The Hon. John Laws, M.L.C., Director, Bagot, Stokes and Laws Ltd., King William street, Adelaide, sworn and examined.

67 To the Chairman—I have a good knowledge of the Northern Territory. I went overland from Adelaide in 1871, right through to Port Darwin, and, in 1873, through there to Port Essington, so that I have gone across the continent of Australia from one side to the other. I have been over that part of the country involved in the extension of the railway now being considered from Katherine to Bitter Springs. There is no doubt that the railway will have to go very near to Bitter Springs if the intention is to take it through from Katherine to Oodnadatta. The country to the west of the track would probably be drier, and not subject to the inundation that occurs from Katherine to Bitter Springs, where you get into low lying land. It is one of the sources of the Roper River. I should like to explain that I am interested in pastoral country in the Northern Territory, and also in South Australia, along the proposed rail way from Oodnadatta to Pine Creek, but I wish the Committee to understand that I am not advocating anything for my own special benefit. My evidence is for the benefit of the whole community. A railway along the proposed route would undoubtedly assist in the sending of cattle for treatment at Darwin. The travelling of fat great stock down to Pine Creek would reduce their condition very considerably, because cattle in that country are not like those farther south. They do not thrive on the feed as they go north. The feed is very long and rank, and not succulent,

so that cattle reduce very rapidly. A railway would assist very much indeed in that direction. Pastoralists would undoubtedly avail themselves of a railway—I refer to a railway from Bitter Springs north to Darwin. A railway is already sanctioned from Pine Creek to Katherine, and the continuation to Bitter Springs would be a very great convenience. The extensive freezing works which have been erected would be able to treat all the cattle that the pastoralists would be able to send from the neighbourhood. It is not sheep country, but cattle and horse country, and much better for cattle than for horses. The alternative route is a safer route, because there would not be so much inundation. The blue-grass swamp country is very much subject to inundation in wet weather. The alternative route would be a little longer, but a little cheaper—I am referring to the western side. The western deviation is down the Wave Hill district. I would advocate the straight line down to Oodnadatta, though it would be a little way out of a straight line. From what I remember of the country, it would be much easier to take the western line. There would probably be 25 miles difference between the two lines. The western deviation would just escape Newcastle Waters, and go west of Lake Woods, which, when full, is 150 miles round. I think it would be a great many years before the line would be a paying proposition, but it would help materially in opening the country on either side. The only rich tin field is below Katherine, but that is quite a recent discovery, and I could not express an opinion about it. A report made about twelve years ago said there was tin there, but the question was whether it could be worked at a profit. Efforts have been made to raise sheep there with very unsatisfactory results. I brought 10,000 from Ayon Downs, and took them round to Newcastle Waters. The grass, however, was very rank, and did not do for them at all, and some of them got cancer. There is a great deal of country on either side of the line that might be taken up if there were greater facilities for communication. The cost of railways is very great indeed, it cost me £45 a ton to get supplies there. I do not think there is anything more I could add, beyond a belief that if there was a railway a great many stations would be formed on either side of the line, so the people could get better provisions and material. This, of course, would help to develop the country. We had to send our loading 550 miles by camel, so that you may understand the cost. Barkly Tableland is much better. I have not been on it, but I understand that you can get water at about 350 to 400 feet. That is right across what is known as Barkly Tableland, which is very good country indeed. The extension of the railway will, to a certain extent, go towards Barkly Tableland, but the people there could get to Burketown easier than to Darwin. Their stock nearly all goes into Queensland.

68 To Senator Story—In my opinion, the Bitter Springs route is the better, though more difficult to construct. A short line to Port McArthur would help the Barkly Tableland very much indeed, but I should have thought it better for the people there to send to Burketown. A shorter distance to the seaboard would, of course, save a great deal in carrying, and the Barkly Tableland is the best country we have. The eastern route, following the telegraph line, is the better. It follows John McDouall Stuart's route,

and be discovered nearly all permanent waters. You could easily go right on to the Macdonnell ranges from Oodnadatta. The country in between is so inferior that it is scarcely worth considering a railway, unless you go right on to the ranges. There we have a good rainfall of 12 inches, it increases from 4 inches as you near the ranges. The rainfall in between is too poor to warrant a railway, unless the railway goes right on. At Charlotte the rainfall is only 4½ inches a year, but as you go north you get a good rainfall, as at Alice Springs. There is a patch of sandy desert country which is marked as an arid basin. There is no telling, however, how far it goes. The only proof is a bore about 12 miles from the South Australian border. I should go right through the country with a railway if I wanted to make a success starting from either end. I would not spend one penny in the Northern Territory unless I could get coloured labour, which, I am sure, could work the Northern Territory profitably. I had six years of hard work there, but if a man has a defect in his constitution, and gets a fever, he has to clear out, or he very soon dies. You cannot get a European to work there like a coloured man.

69 To the Chairman—The Chinaman is as good a man as we can get for mining work there. We had white men from Ballarat for the gold mines, at £4 10s and found, but they did not stay long before they cleared out. We got a number of Chinamen from Singapore.

70 To Senator Story—I suppose that there is no possibility of that kind of labour coming on, but I think that the Territory is hopeless without coloured labour. I am sorry to say this, because I would sooner see white labour. There are a few aboriginals there, but they are getting fewer every day. They are not good for manual labour, but some of that part they are only good for stock work, which, of course, is all you need for stations. South of that, however, you want to cultivate land as well as raise stock. So far as I know, there are not many minerals south of the Katherine, except at Macdonnell Ranges. There have only been a few specks of gold discovered between the Katherine and the Macdonnell Ranges, but there are gold and precious stones, with wolfram and mica. I recommend that it would be better to follow the suggested route on to Bitter Springs.

71 To Mr. Tullyson—I know the Victoria River district by repute. It is excellent country, and a great deal might be utilized if there was water for irrigation in the future. It is rich, dark alluvial soil, but, of course, I speak only from hearsay. I think the country would be better served by connexion with Windham. The district covered by the western scheme is, so far as I know, poor, dry country, until you pass the waters. It is from the Katherine River crossing that the divergence of route occurs. I do not think that the extension of the line from Katherine River to Bitter Springs would in any way prejudice the north-south route. The more easterly route that I suggest would, I think, traverse better country and give a better return. There is some good country on the other side of the Roper, to the north of Bitter Springs. The railway to Bitter Springs would tap some good country. The extension to Katherine River is, I think, practically ineffective, unless it be continued to Bitter Springs. There is a strip of brid

country, with one small exception. In order to make the extension worth while, it ought to go on to Oodnadatta. A line built out in a westerly direction would have to depend entirely on pastoral occupation. If the line be continued to Bitter Springs, I think the pastoralists will make use of it for transporting cattle. As I say, the line would have to depend on the pastoral traffic. A good many people from Barkly Tableland would use the line. The country there is very fairly stocked with both sheep and cattle, and there would be more traffic as the line was extended southwards to Newcastle Waters. If the line is to be constructed at all, it is better to go from Katherine River to Bitter Springs.

72 To Mr. Tullyson—It would, in my opinion, be profitable to send sheep from the Barkly Tableland to the freezing works in the north if the line were constructed. The Bitter Springs country is subject to inundation, but higher up there are only one or two creeks to cross, after that you are practically free from flood country. With freezing works in Darwin, it would be advisable to continue the line from the north for the development of the Territory. As to the rainfall in the Macdonnell Ranges, the Northern Territory is partly tropical and partly sub-tropical, but the rains, as a rule, occur from November to March. I think that the railway will assist very greatly in the defence of Australia. I have never suggested the abandonment of the Territory, but I would not like to take it on unless I could use coloured labour. I am not aware that in South Africa coloured men objected to go below a certain depth. As to the Chinese labour having "picked the eyes" out of the mines, I do not think, as an old miner, that they are very much different from Europeans in that respect. Whenever there was a good patch, I tried to get it, and the Chinamen would do the same. When the railway was first constructed there, Mr. Charles Miller told me that he could never have carried out the work without Chinese labour—he could not get Europeans to do it. That was not owing to the isolation of the country, but because the climate is against Europeans working there. I am not versed in what has been done at the Panama Canal in the way of enabling white labour to be utilized, except from what I have seen in the papers. It is not a matter of wages, because, as I have already said, we paid £4 10s and keep, but the white men would not work there, simply because the climate was against them. Stock men have a different life from that of the miner, it is not such a task to ride a horse as to use the pick and shovel. Then, further, the stock men do not ride much in the heat of the day, but in the early morning, or the cool of the evening. Apart from the coloured labour question, I would, as a matter of development in the Territory, suggest a railway policy similar to that now under consideration—I would follow the telegraph line right through.

73 To Mr. Fenton—It is nearly forty years since I was in the Territory. I know that considerable alterations have been made in regard to the employment of labour in tropical countries. A Britisher in India manages very much better than he did forty years ago, but there are the work and personal services are done by the natives. There are, of course, exceptions to every rule, and a man might work in the Northern Territory for four or five years without harm. I am glad to hear that on the railway construction

at the present time there are 200 white men engaged, and that the medical testimony as to their health is exceptionally good. I suggest, however, that if there have been 200 men engaged there for eighteen months or two years, they may not have been the same men all the time. A good strong Britisher is a valuable man, if he takes care of himself. The Victoria district may be all right, but past that is some of the worst country we have. It is dense scrub country, which no one has thought to take up. I speak of westward of Newcastle Waters, where the country is all wretched. By taking the westerly route, you miss all the country round about Bitter Springs, where the land is very decent indeed. On the eastern side of the telegraph line is better country. I reckon a better class of country would be served by the more easterly route. To make Port McArthur the port would be "robbing Peter to pay Paul." It would be of some advantage, but freezing works would have to be established, while there are already freezing works arranged for at Darwin. From my knowledge of the country, and speaking generally, I regard the route I have suggested as the better one, and I differ from the report of the Royal Commission. Having regard to the development of the Territory as a whole, I believe it would be better to construct a railway from north to south, as well as from south to north simultaneously.

75. To Mr. Sampson.—I do not know if the Burketown meat preserving works have gone in for freezing; a fair number of cattle are recovered there. But, of course, the large companies have gone inland. I do not know the country of the westerly swing at all. The southern portion of the deviation on the western side is poor country, and it is poor country as it tapers back to the telegraph line. The southern portion of the eastern deviation, as it is coming back, is much better, because there is water there. I should say, however, that it is not good pastoral country, because it is too dry. All the country east of the telegraph line is taken up, but that is not so on the west. The country on the eastern side is all good grazing country. My stations there run from Newcastle Waters up towards Daly Waters, and below Powell's Creek—about 6,000 square miles.

76. To Mr. Gregory.—I said a little while ago that in the southern part there was a large area which was supposed to be artesian bearing, and I presume that, to a great extent, there has been a systematic survey of the artesian belt; if not, it is only supposition. The only man who explored the fringe of the country is Captain Barclay, and the only bore is about 12 miles over the border from South Australia. This country carries a good many stock in the dry season. There has been no trial north of that. I think it would be a great advantage to have the whole of the artesian area geologically mapped out. The basin seems to run between the Australian Bight and the Gulf of Carpentaria. I have always advocated that there should be a special effort to locate the water supply. It all depends on circumstances, as to what railway distance cattle can be profitably freighted—what the cattle are worth, and what the freights are. From Oodnadatta it costs 30s. per head to get the cattle down to Adelaide. We could pay the same rate north to the freezing works, which would pay us very well at the present prices. We could

fatten stock for the freezing works, and I should be very glad to pay 30s. per head. I should not like to express an opinion as to whether, with cattle raising, the railway could be made to pay. A railway would assist very materially in developing the country, but I could not say whether that development would be sufficient to make the railway pay. As to mining development, it all depends upon the magnitude of the minerals. If you have anything like a Kalgoorlie or a Coolgardie, the railway would pay; but, so far as we know, there are not sufficient minerals to make any material difference to the railway. If there were a mineral belt on the eastern side, it would support the views I have expressed. To construct a railway from Fellow Island to Barkly Tableland would be what I have called "robbing Peter to pay Paul" in view of the fact that large freezing works have been built at Darwin and at Wyndham. No doubt other works would be of advantage to the settlers in the immediate district, in the way of cheaper rations and quicker communication, but only one locality would be benefited. If coloured labour were introduced, I should be content with a system of indenture. I may say that I would rather watch a Chinese coolie work than I would work in a mine myself, though I have done a good deal of it in my time. I have not known a very successful closer settlement up there. I think, however, that the Government works on the Victoria River were unprofitable.

(Taken at Adelaide.)

THURSDAY, 3rd FEBRUARY, 1910.

Present:

Mr. KIRRY, Chairman;

Senator Keating	Mr. Gregory
Senator Storey	Mr. Sampson
Mr. Denton	Mr. Laird Smith
Mr. Finlayson	

William Wallace Andrews, Assistant Engineer of Railway Construction, South Australia, sworn and examined.

76. To the Chairman.—I have been in my present position for three years, and prior to that I was in charge of the Palmerston and Pine Creek railway for about eighteen years. I went there on the construction work of the second half of the railway. During the twenty-one years I was in the Territory I was resident engineer. I stayed on in charge of the line until 1909. I am fairly familiar with this country, but I have never been south of the Katherine. I have, however, been on the Victoria River, which is pretty well west of Bitter Springs, and I know that the Western Downs is very fine pastoral country. It would mean lengthening the main line if the section under consideration were to swing out. So long, however, as the deviation is not too great, it is wise, from a traffic point of view, to serve the best country you can. I cannot speak from personal knowledge of the eastern line, but from what I have heard I should think the Victoria River country is better north developing than on the Downs. So long as good country is tapped, a few miles lengthening does

not make much difference. It is only from hearsay that I have any idea of the engineering difficulties around there. I know there are no large rivers, and I never heard of any serious floods beyond the Katherine.—I mean serious floods from a railway point of view. I believe there are often large swamps, but nothing to cause washaways. I am not in a position to say whether, as a general policy, it is advantageous to continue the line as proposed by South Australia.

77. To Senator Storey.—The line from Darwin to Pine Creek was constructed by contract. As to the climate, we have had some exceptionally good white men working there. When I left, there were two or three men who had been employed in the contractor's time, over twenty-one years ago. They were employed on the maintenance of the line. I feel sure they did not do as good work as the men in the south, but they did as good work as anybody there could. There, however, were exceptional men—good, sound men. The whole of the line, with the exception of the bridges, is laid with steel sleepers. On the bridges there are wooden sleepers. At first we used karri for these sleepers, but they did not last very long, and since then we have been using New South Wales iron-bark. We use it on the bridges, because there the white ants could not get at them. Timber sleepers are no good where the white ants are. The only timber we found to resist the white ants in the ballast was the eypress pine, which, however, is too scarce and expensive to use.

78. To Mr. Gregory.—We did not use the Queensland ironwood.

79. To Senator Storey.—In my opinion, there is not much to choose between the western and eastern deviation from a construction point of view. I can see the Victoria Downs country capable of furnishing as much traffic for the railway as the Barkly Tableland could; but I am only going on hearsay as to the eastern part of the country. I know the western part personally, and the Victoria Downs is first-class country. As to the mineral prospects north or south of the Katherine, I know that the mines north never did very much. I have no knowledge of the Mirramboy tin-fields, but Mr. Todd was in operation there. I know nothing of the country south of the Katherine, or of the country south of the Macdonnell Ranges, because the furthest I have been north is Oodnadatta. I think a good deal of labour could be saved by taking the line from the south, though I do not anticipate there would be much saving in regard to material. You get your rails delivered at Darwin at very little more than at Port Augusta—you have 260 miles to go from the north, and 150 miles from the south. The extra cost would be, I think, in labour. That is, you can keep up your supplies of labour better from the south than from the north, and you get acclimatized as you go north. The maintenance of the Palmerston to Pine Creek line would take only thirty-five or forty men altogether. The line itself was built with coloured labour, but we had European gangers. In regard to the latter, there was trouble, because they were always coming and going, it was the usual thing—too much drink. Drink is not the only cause of men breaking down, because some constitutions cannot stand the climate. At the time I was there concrete sleepers were only in the experimental stage. I have not used such

sleepers, though, of course, I followed Mr. Joseph Timmins' experiments. I think these concrete sleepers would only pay to use at present on lines with very heavy traffic, where heavy construction is required. They would cost a lot more than the others to handle, on account of their weight. I do not think these sleepers would save much on current maintenance, but they might live much longer, and the cost of renewals would not be so great. The road would, however, in my opinion, require just the same attention. I may say that of the steel sleepers used on the Palmerston-Pine Creek line, the greater portion are still in, and many have not had to be touched in the way of packing or straightening for years at a time. The traffic was very light, but still they held the road so well that they had not to be touched. I have heard that they are now using a different pattern of steel sleeper, though not to replace the old sleepers, but to use with the heavier rail. I do not think that the new pattern is more suitable for the light rail. I like the old pattern for light traffic. It is a stamped steel sleeper, which holds the ballast in it, and it can be used on curves. It is hard to describe the peculiar shape, but it holds the ballast in such a way that it does not shift at all. The use of concrete sleepers entirely depends on the type of sleeper. In some experiments there has been a lot of trouble with the concrete disintegrating under vibration; but it is claimed that that difficulty has in some of the recent sleepers been got over. Of this, however, I cannot speak from personal experience. None of these sleepers, so far as I know, have been used in Australia on railways, though some have been used on the tramways laid down by Mr. Timmins.

80. To Senator Keating.—I could not say whether these sleepers are of Italian design, and I do not know how long they have been down.

81. To Mr. Gregory.—The steel sleepers rusted somewhat in the Palmerston station yard, where they were exposed to the salt air, otherwise they lasted splendidly. As to wooden sleepers, the white ants are very bad. We tried jarrah, and the white ants attacked it, and we also tried New South Wales ironbark. We have not tried Queensland ironwood; and I dare say it was a question of costs. We wrote to Queensland at the time we were trying the timbers, and I do not recollect the Queensland wood being recommended to us. The ironwood grows in the Northern Territory, but the trees are so far apart that it would not pay to get it. I have not followed up the experiments with concrete sleepers on the tramways. There is a far greater hammering action on railway lines than with the lighter cars of the tramways. If it could be ascertained that these concrete sleepers were suitable for heavy train-loads, it would be worth while inquiring whether they could not be manufactured on the rivers and creeks of the Northern Territory. It is a question of whether you can get the gravel on or near the spot, because every mile of cartage adds to the cost. I am speaking of cartage by horses and drays, as against rail cartage. I would not recommend wooden sleepers in that portion of the country. I have not used porcellaneous sleepers. I do not know of any having proved defective. There has been a very heavy increase in the cost of steel sleepers—from 10s. to 12s. each. I think that, at such a price, concrete would be well worth trying,

but I am not prepared to give an off-hand opinion. I have seen so much damage done by white ants to terrah that, personally, I would not touch it. Ironbark would, I believe, in many cases last much longer. The little white ants down south are a very different proposition from the Territory white ants, which are nearly as big as bull-ants. As to the proposed deviation to east or west, a new mineral discovery would decidedly affect my recommendations. The fact that there was a tin area 40 miles square and 14 miles from Bitter Springs, would most certainly affect my recommendations. I do not pretend to be able to make a recommendation beyond saying that it would be worth lengthening the line a little to get traffic by tapping the best country. I should hardly think that you could get sufficient traffic merely from the pastoral resources to warrant the construction of the line. But the point is that we do not know what will follow the pastoral development, which is a good thing to start with. There were practically no efforts at closer settlement at Darwin in my time; the most has been done since the Commonwealth took the Territory over. I know of no cases where closer settlement has been successful. I do not think you will ever develop agriculture with white labour—that the growing of rice, cotton, and so forth, could be successfully carried on by a white population.

82. *To Mr. Finlayson*.—I think that the establishment of freezing works at Darwin would lead to a very large stocking up of the country, and this would bring a certain amount of traffic to the line. I do not think the line would be used for stock, except from the Katherine to Darwin, because there is good travelling country to the Katherine. From Darwin to the Katherine the country is very poor, and I do not think you will carry much stock on the railway south of the Katherine. The extension of the line south of the Katherine would not result in any great direct stock traffic; but indirectly it might. The transport of cattle would be the main source of revenue, and the mineral traffic would assist. I am a great believer in the mineral resources of the Territory. It needs the conjunction of all traffic to make a railway pay. Primarily the line should go direct towards the mineral districts and good cattle districts, providing the lengthening of the line is not too great. If the proposed line is to be part of the main line, the lengthening should not be very great. I approve of the idea that the extension of the line merely as a local proposition would not be worth much; but it is the fact that it is a link in the chain of the main line that gives it its value. The extension of the line for local purposes would be difficult of justification, but it is a different matter altogether when it is part of the main line. I regard the Victoria River district as good pastoral country. From what I have heard, I should think that this pastoral country would be better opened out by Wyndham. So far as I know, there are no difficulties in construction. An extension of the line towards Bitter Springs would be worth favorable consideration if there were good pastoral country there which could be developed once railway connexion was given.

83. *To Mr. Sampson*.—The present carrying capacity would, I suppose, give an indication of possible future developments; but my opinion is really of very little value on such a matter. I am not in a position to express an opinion as to

whether the Victoria Downs will develop an agricultural population. If I were charged with the duty of developing the Territory, I should go for a through line, and not for lines running from suitable ports inland, as is the case in the various States. My idea is to have a trunk line, because we want a backbone for any system of railways, and not isolated branches. In planning a railway system, I think you want a trunk line with branches to it. The Territory would, to a great extent, be isolated if there were these local lines. The development of the Territory will, I think, depend principally on the export trade. If there were a system of railways running from the coast inland, with freezing works at the various ports, I think you would find it difficult to induce people to go to the Territory. The importance of the ports would depend on the amount of traffic for the trains. But, first of all, you have to get your people there. There are not the same inducements for people to go to the Territory as to other places in the Commonwealth. The great advantage, to my mind, of a trunk line over a system of lines from the coast is the closer communication with the other States it would give to the Territory, and this would tend to induce people to settle there. I think that at first the population to the Territory will come from the south, and not from immigration. I have not prepared a list of reasons for preferring a trunk line, but I have always held the opinion that a main trunk line is essential to the development of the country. I think that, by a system of water conservation, irrigation colonies might be formed in the northern portion of the Territory—that is when you have the labour to make the agricultural settlement. The idea is, I think, practicable. The trouble is, that the rainfall is only for three or four months in the year. Conservation dams could be made; but without the labour that would not be one of the best means of solving the question of settlement. With the necessary labour, it is practicable to store water and establish irrigation settlements, and by that means have closer settlement areas. Many of the rivers are perennial. I have, in a general way, followed the experiments that have been going on in regard to the cultivation of rice, cotton, and so forth. As to dairying areas, there is, of course, the tick trouble anywhere near Darwin. I suppose that the tick is possible of eradication.

84. *To the Chairman*.—It only adds about one-eighth to the cost of the sleepers to lay down a 3-ft. 6-in. gauge on sleepers for a 4-ft. 8½-in. gauge, with a view to subsequent alteration, and I would not think it worth while to depart from the present plan. The bridges, I think, are to be built for the broader gauge, for which it is wiser to prepare.

85. *To Mr. Gregory*.—I do not think much has been lost in interest and so forth; but it is a matter of figures and probabilities. There is no danger of the long sleepers breaking if they are properly packed. The packing must be done under the rails, and that is a matter for the engineer. If the longer sleepers crack in the centre, that is a matter of seeing that the packing is properly done.

86. *To Mr. Fenton*.—I think it would be wise to proceed from Oodnadatta north, as well as from the north southwards.

Lawrence Allen Wells, Deputy Commissioner of Federal Taxation, South Australia, sworn and examined.

87. *To the Chairman*.—I have had some experience of the Northern Territory, having spent a good many years there. My first experience was on a survey of the border of Queensland and the Northern Territory from 1883 to 1889, and afterwards I was engaged in exploration work. My last long experience was in the Victoria River district, where I made a survey from 1905 to 1908. Altogether I have seen a great deal of the Territory. I am not in favour of continuing the line right down from Oodnadatta, because I do not think that is the best line, as a paying proposition. I favour parts of the line. The part now being investigated would, I think, form part of the main line down as far as Daly Waters or Newcastle Waters. I should think that the line from Katherine River to Bitter Springs would be continued now. I have no knowledge of the Territory about Bitter Springs, but I have been all about the Katherine. I do not think that the country below Katherine to Bitter Springs is any good. If it were attempted to travel fast stock through that country they would be found to get low in condition. There are not many cattle about that locality. I think that a railway would cause some development, provided there were meat works at Darwin. I think it would benefit the country from Newcastle Waters and south-east of that. I do not think the westerly deviation would be quite justified. I do not see any great benefit to be obtained, because the cattle there now could easily be taken into the Katherine, where they are taken now. They often have to go that way to get to the south, because the other way is not open except in good seasons. I know that part of the country well, as my survey went through it. There is only a small patch of good grazing country at Willeroo. If the railway took a swing over there, it would be only to the benefit of that part, and the benefit would not be much. It would be all right to go on to Bitter Springs. In the Territory the tropical rains are about Christmas, and last about two months, and the cattle build up quickly, but will not stand driving over that class of country. It is scrub country, with long grass that will not feed. There is no sheep country for 150 or 200 miles south of Darwin; but below that there are very good areas in the Ord River country, at Wave Hill and Victoria River Downs, and Stuart's Creek, and there is a large patch of undeveloped country south-east of Wave Hill. On the Fitzroy River country sheep stations have been running for years. On Ord River a small flock is kept, and they get the same prices for wool as if down south. From Camoosial to Anthony Lagoon is good sheep country. I do not think there is any possibility of closer settlement or small farming. There was an experiment in farming tried; but there was nothing in wheat, because they do not get the rain at the time it is wanted. The rain starts about Christmas, and is all over after the end of March. That applies down to Wave Hill. There are white ants, and I brought away a piece of sheet lead in which they had eaten seven or eight holes in order to get at wood, and they had also done destruction at the superintendent's quarters at the Cable Company's place. The house there is built on concrete blocks. I know that the white

ants will eat green paw-paw. It would be found in the morning that the white ants had made a start on the pack saddles; and they are said to have damaged a set of billiard balls at Port Darwin, after eating through the box—these were composition balls.

88. *To Senator Story*.—I have a knowledge of the Territory from Oodnadatta north, but not much beyond the Charlotte. I know all the western portion. About Musgrave there is sufficient country for a few holdings. About the Charlotte, and right up to Alice Springs, is good pastoral country, and it is all held now. Some of it is good sheep country. I think the line should be extended from the Katherine south, as part of the main line. The other line should be completed in some way or other; but I think it should go from Bitter Springs to Camoosial. There is good country all the way, and most of it is sheep country. The greatest population is on the east coast of Australia, and this would give direct communication. It would be a good scheme if a line were taken from Newcastle Waters west. That would avoid crossing any rivers or creeks, and it would tap all the sheep country; it would be just north of the desert. It would be of some advantage to the Tennant gold-field, and it would be a much shorter distance to Port McArthur than to Darwin. There would not be much chance of getting water between Newcastle Waters and that part, except by well sinking or boring. I favour the east route, because it would benefit the Barkly Tableland. I think that the westward deviation would be too far away to benefit the Tennant gold-field, and that it would be costly, and subject to washaways and so forth. It is very rough country. When I said before the Commission that there were many miles of good country in the Victoria River district, I meant Wave Hill and the southern half of the Victoria River Downs, and the belt of country lying south-east of Wave Hill. The people occupying the land could send their stock to the Katherine; but there are no sheep yet, because there is no railway. I think a line should be surveyed running westward from Daly Waters or Newcastle Waters; it would be an easily-constructed line. When sheep get into the country I think the line will pay; but it will not pay only with cattle, because people would rather drive cattle than put them on the railway. The belt of the country between the Katherine and Bitter Springs was discovered after I left there; but there is tin all through that country. I think that mineral development, added to the pastoral traffic, will go considerably towards paying the cost of the railway—that the mineral development will mean much, particularly in the Macdonnell Ranges. I would continue the Oodnadatta railway to the Macdonnell Ranges, but I would eventually bear off to the east. I would continue the Oodnadatta line to the ranges, and then continue the line north. My only knowledge of minerals in the Macdonnell Ranges, or northern Australia is only from reading. As to the climate, my opinion is that a white man can work where a blackfellow can, if you pay him to do it. I do not say it is nice work. My experience is that I took five men there from Adelaide, and we worked the whole of the first summer. We had to climb mountains and clear large gum trees, and the men could there do as much work as any—as much or more than any coloured men. In

81. To Mr. Sempron.—I think that the swing over towards the Barkly Tableland should be at about Nevershire Waters. The construction to that point would depend on the immediate gathering ground. I think that the bulk, or, at all events, half of it, would lie there.

93. To Mr. Fenton—I agree with the statement that there are about 10,000 miles suitable for dairy

Even now it might be wise to consider this, and allow the Darwin line to end at Katherine for the present.

A line from the Western Australian border to Port McArthur would be about 560 miles in length, and there would be no engineering difficulties from the border to Newcastle Waters.

96. *To the Chairman.*—The Tramway Trust have constructed 100 miles of tramway in Adelaide, and we have used wooden jarrah sleepers throughout the construction until the last two years. We found that jarrah resists the ravages of the white ants very well, and in very few instances have we had to renew them. But the price of jarrah sleepers has been steadily increasing. When we first started we paid 3s. 11d. each, while the last lot we obtained cost 4s. 7d. for 7 ft. 6 in. x 9 in. x $\frac{3}{4}$ in., and this difference means £70 per mile, which, of course, is a very serious item when there are many miles. Two years ago a

proposition was submitted to us by Mr Joseph Furness to use his patent reinforced concrete sleepers. We went very thoroughly into the matter, and laid several miles of track with them. These sleepers cost much more than the jarrah sleepers, but owing to their size and bulk, instead of laying them 2 ft 6 in apart, as with the jarrah sleepers, we put them 3 ft 3 in apart. This reduced the number per mile from 2,100 to 1,600. We have laid altogether 9,351 concrete sleepers, and these have been on the track, in some cases, nearly two years. On one route the cars have passed over the track 54,000 times, on another route 77,000 times, and on a third about 18,000 times since the sleepers were laid, and we have had no trouble whatever with them. On no occasion have we had to open up the road and look at the sleepers or deal with loose kyo. The road has been quite rigid, and the maintenance cost is less than with wooden sleepers. We find the concrete sleepers more rigid than wooden sleepers, and another feature is that they have an infinite life, practically speaking, as compared with the more or less definite life of the wooden sleepers. This, of course, is a matter of great importance in railway and tramway construction, though I speak only from a tramway point of view. I may say that there has been a small portion of the railway track in South Australia laid with concrete sleepers, and also on one of the suburban lines in New South Wales, and I think they have turned out very satisfactory. I shall be glad, if the Committee desire, to open up the track and expose some of the sleepers which have been down two years under the traffic I have mentioned. The concrete sleepers cost about £20 or £200 per mile extra, even allowing for the extra spacing. The cost of the concrete sleeper is about 10s 6d, and of the wooden sleeper about 5s 6d, the latter, of course, including adzing and so forth. As to handling the concrete sleeper weighs on the average 3 cwt., and the jarrah sleeper 1 cwt 1 qr 2 lbs, and both are the same length, namely, 7 ft 9 in. The concrete sleeper, however, is 6 inches thick, while the jarrah is 4½. If I were to start to put down a new tramway track I should prefer a concrete sleeper. Our gauge is 4 ft 8½ in, and the maintenance cost with concrete is much lower than with wooden sleepers. I have had no personal experience of concrete sleepers on railway work, and I should like to experiment before expressing an opinion, because railway construction is very different from tramway construction, so different that you really can not make a comparison. The maximum load we get on a pair of wheels is about 8 tons, but we do not travel at such high rates of speed as on the railways. Our maximum speed is 28 miles an hour, while on a railway it may be 55 miles. We have not yet made our own concrete sleepers, but we have arranged with Mr Furness to take over the plant in order to do so. I do not anticipate, however, that the sleepers will prove much cheaper.

97 To Senator Story.—The making of the concrete sleeper is a simple process. Very fine concrete has to be puddled into boxes and the great thing is to make sure that it is properly puddled without any blow holes. There would be no difficulty in manufacturing such sleepers in the Northern Territory, provided the material were hauled. The labour is unskilled, and all you require is cement, gravel and sand and the reinforcement.

98 To Mr. Panton.—In the case of the concrete sleeper the chair is embedded in the concrete, and attached to the reinforcement, and forms part of the structure. The rail has to sit on the chair, and is held in place by either bolts. In railway work the rails would be all exposed to the weather, whereas in the case of tramways they are not. In no case have we found these fastenings work loose. They are long and tapering wedges.

99 To Mr. Gregory.—That is the only danger I would fear as an engineer, and it is a matter of trial.

100 To the Chairman.—The maintenance men would see that the wedges were kept right.

101 To Mr. Panton.—I should say that the rigidity would make a difference in the resilience. A certain amount of resilience is necessary for comfortable travelling.

102 To Senator Keating.—Resilience is more necessary on railways than on tramways, because of the higher speeds and longer and heavier is my opinion. I do not see why concrete sleepers for railway purposes would minimize the resilience beyond what is really safe, of course, I am speaking without actual experience, but that is my opinion. I do not see why concrete sleepers should not be made on the spot in the Northern Territory. It is one of the advantages of the concrete sleeper that, with a good supply of gravel and sand, all you have to obtain from elsewhere is the reinforcing material, chairs, and so forth. It is a practical proposition.

103 To Mr. Finlayson.—We gave them an average of 20 days to dry, and this should not be reduced in a dry climate. It is a decided advantage not to allow them to dry too quickly. I should strongly recommend that a trial be given of these sleepers, because the question of the wood supply is a very serious one. In a country infected with white ants you want something to resist the pests. No timber is ant proof, but jarrah is the best of the lot.

(Taken at Adelaide)

FRIDAY, 4th FEBRUARY, 1916

Present

Mr RILEY, Chairman,

Senator Keating,

Senator Story,

Mr Panton,

Mr Finlayson,

Mr Gregory,

Mr Sampson,

Mr Laird Smith

Henry Yorko Lyell Brown, sworn and examined

104 To the Chairman.—I am out of business now, having resigned about four years ago from the position of Government geologist, which I had held for about twenty-nine years. I have been up to the Territory on five or six occasions, and have spent a good deal of time there. I think there are great possibilities in mining, particularly gold, though there are other minerals like lead, copper, and tin. Gold, however, is the chief thing. As yet they have not been deeper, speaking geologically, than 200 feet, and I think they ought to go deeper. The most important places from a mining point of view are Pine Creek and Union. There are several other gold-bearing districts, but those are the principal. I have been through that part of the country where it is proposed to build this railway extension, but there are no mining possibilities just there,

Primary rock is favorable to minerals, and this is mostly flat tableland. The nearest mining country is about 15 miles after you leave the Katherine, at Maudslayi's Creek, after that it is all plains and tableland. I do not think that the section of railway now before the Committee would do much in the way of mining. I have a knowledge of tin mining, but the Marraboy field is a discovery since my time. I think that the area likely to be tin-bearing is limited. I think that the route along the telegraph line, in a general direction, would best develop the mining industry of the Territory. A main trunk line would be the best, and the shorter it is the better. The section now proposed will form part of the main line. I should say that the railway will lead to stations being taken up. There are a few stations further south, but only second class, and sheep do not go in that part.

105 To Senator Story.—In my opinion the mineral resources of the Territory are well worth developing. I think that mining, together with the pastoral industry, when you get further south, will prove the main sources of revenue. The pastoral industry is not much good in the tropical parts, where you have to contend with red water, and so forth. I do not think that the devotion to the west would go far enough. So far as I know there is no mineral country there, but it is good soil on bank plain. A line running due west would tap the Wave Hill station.

106 To the Chairman.—The tin field does not go very far, and is very patchy. In my opinion no minerals will be found immediately south of the Marraboy. I have here a geological map which I prepared and which may be of some use to the Committee.

107 To Senator Story.—No valuable minerals have been found near Tennant's Creek, but there are indications of minerals. There are great lodes of very good iron ore there. After that we do not get any minerals until we reach the Macdonnell Ranges. There is nothing very important in regard to minerals at Macdonnell Ranges so far as has been found, but there are a good many gold claims worked there. They found about and sink a few feet, but they do not continue, owing to want of capital. I have reported several times on this mineral country, and as to the possibility of large reefs or lodes being developed that is a matter of further prospecting. The indications are all right—quartz reefs, and so forth—and then have had 4 or 5 ozs. to the ton. All the working is by private parties. It is very expensive working on account of the cost of stores. They get their stores from Oodnadatta, and it is very costly, and prevents economical prospecting. It is first-class country generally in the Macdonnell Ranges, wonderful grass country, and I think, fit for agriculture. It is fairly well watered, and at Alice Springs there is abundance of water, while in the rainy season the creeks are running. It would be possible to conserve water for irrigation, and I should say the country would be suitable for dairying, though it is very far from the market up to now. The climate is very good, hot in the day and cold at night. For three or four months in the year you get no nearly every night. Alice Springs is about 1,800 feet above sea level and there are hills 2,000 to 3,000 feet above. The country south of the Macdonnell Ranges is mostly pastoral, and there are no stations there, but I do not know of any sheep. I think that the continuation of the line north from Oodnadatta would be beneficial in

the way of opening up that country, and would have a good effect in reducing the deficit on the line running to Oodnadatta. White men do not work very hard in the Northern Territory, and I do not think they can keep their health. The mosquitoes are so bad that north of the Katherine, you have to get into a net, and the Katherine itself is bad enough. I would place the mosquito country north of Bitter Springs, south of which there is a better climate and fewer mosquitoes. The men after working all day, have to creep into a mosquito net at night, and do not get any sleep until very late, and are not fit next day. Men engaged on the railway work may be all right for a time. The first time I was there for eight months I had no fever, but the next time I was down for a month. I think that when one goes the first time one can stand it, but not so well subsequently. I do not favour the development of the country by short railways running into various parts, as, for instance, Port McArthur and Wyndham, a main trunk line is the best, though these short lines would be very useful to the stations. I do not think the Northern Territory will ever be properly developed until we get a main line through from south to north. Such a line will make a tremendous difference in the way of opening the country. At present the journey is such a long way round that few people will go.

108 To Senator Keating.—The geological map of which I spoke was prepared in 1898, and does not show the Marraboy tin field. No systematic and complete geological survey in the Northern Territory has been made for the Government of South Australia, though I made several trips there for special purposes, and saw different mines. On one trip I examined the west as far as the McArthur River, that was since the map was published. I have also made a trip to the west. It would not be fair to myself or the Territory to regard this map as the result of a systematic survey, it is only the first map, and subject to considerable modifications, though, in a general way, it is all right. Much has been done since the map was made, and so geological reconnoissances, made as discoveries were reported, nothing anterior to discoveries. The only thing that has been done in the way of systematic prospecting is boring for coal and gold under Government supervision within the last ten years. I recommended Port Keats for boring for coal. The longest time I was in the Territory was about eight months, in connection with prospecting and discovery. I went to Darwin and Tanami. When I speak of the Territory as offering possibilities for successful mining enterprise I give an impression consequent on my several visits. As to which railway route would best serve the mineral interests of the country from Pine Creek southwards, I should say the first mineral-bearing country is due south, and down the telegraph line, which, I think should be followed. To the west there is some really good agricultural country towards Wave Hill. It is not far from the Katherine, and it extends into Western Australia. There is not such good country on the eastern side.

109 To Mr. Gregory.—There is no later geological map than the one I have here, but there are several reports on various districts, kept, I believe, in Melbourne. The mineral belt I spoke of is south of Powell Creek, there is only a little gold found there. I do not know how far the area extends north and south. The little gold found is the only indication, but it is a lovely mineral country, though I have no idea of an auriferous area. In the north there is not a very

large mineral area, but gold has been worked to a certain depth, and the chances are that it continues to a greater depth. That country was geologically examined when we were boring for gold, and we brought it up at 700 feet. I assume, from my knowledge of the country, that these holes go to a depth, indeed, they have been proved by the diamond drill to do so, but shafts ought to be sunk. The diamond drill is used to find if the gold goes down to a depth. The shafts are usually very patchy. As to the diamond drill being like working in the dark, it enables you to find whether or not a lode extends down, but in order not to miss the best way is to sink a shaft. The country also contains other metals, there is a wolfram mine near Pine Creek. I think there is plenty of good mining timber it is not very large or heavy, but I think there is plenty for ordinary purposes. For big mining work some would have to be imported. Mining is very expensive, and I think wages are the main outlay. I could not say that I would be content to open up a mine with white labour, because it is too expensive, seeing that the men get £4 10s a week. Under ordinary circumstances I could not carry on mining with white labour. At Tanami the men are £5 a week. Very little work was being done when I was there. I did not see a report a few years ago that there were 200,000 tons of rich ore in sight at Tanami, and I do not think it can be correct. I think that the alluvial gold in the north has been pretty well worked out by the Chinese. Under the South Australian Government Chinese were allowed to obtain miners' rights though they could not hold licenses. I think enough having been done at Pine Creek for gold, and the next thing is to sink a shaft. I read the other day that there have been some very good results there. Much greater good would be done by assisting people to sink shafts than by diamond drilling. There is plenty of water to be obtained, on the gold fields it can be got within 100 feet. I say they get too much water at certain times of the year, and very little at another time. I think that, as a rule water lasts all the year round in the water holes. The artesian area over the northern portion of the Territory has been mapped out, and is shown on the map I produce. On the "Tablelands" an effort was made to obtain artesian water, but there is no chance according to the geological indications. There is no possibility of obtaining artesian water further north than between parallels 23 and 24. At present I think the pastoral resources are of greater value than the mineral resources.

110 To the Chairman.—There is any amount of ironstone gravel, and sand, and the proposed railway will go right over it. It is very level country, and there is any amount of sand which, I think, would be suitable for making concrete.

111 To Mr. Sampson.—I do not think that the British companies who spent large sums of money in prospecting and then abandoned the gold fields went down far enough. They spent all their money on surface buildings, electric lights, and so on. They never ascertained whether the lode went deeper down. The diamond drill has proved that there is gold below 200 feet, and it cannot be known whether it is payable unless shafts are sunk. These English companies often made their money on the shares, and never intended to work the mines. I do not think there was any mining reason why these companies did not go deeper, every line was in favour of the reefs going down. One reason why I favour the telegraph line route for the railway is that all the water holes are there and you generally find people round the water holes. It is level country with no en-

noing difficulties, though there are some very deep creeks. There is no other reason why the telegraph line should be the best route. There are some mountains north of the Macdonnell Ranges called the Davenport Ranges, and some gold has been found there. It is pretty good country for grazing.

112 To Mr. Laird Smith.—I believe that one of the English companies built a railway of their own, and, therefore, it was not the cost of carrying that prevented development. In the case of the tin fields the railway would tend to cheapen the cost of the carriage of stores. The Chinese did not develop the mines on lines satisfactory to me, but I think they are first-class miners for working underground. There could not be better miners when under white control. Copeckup lines would be a means of development, but I think the main line is the principal thing.

113 To Mr. Fenton.—In the case of one mine at 350 feet there was so much water that they had to stop, there was a creek within 50 yards of the mine, and I think the water ran from there. Generally speaking, I think that the water in the mine was an indication of a fair subterranean flow, but a good deal comes from the surface. You can get water almost anywhere in the Northern Territory, fresh, with very little salt, and it would not do any injury to vegetables. I do not think that the subterranean supplies would be required for the purposes of settlement because there is sufficient on the surface. If the people made dams in the rainy season they could conserve enough for all their requirements. I do not favour the western swing, it would be better to have a branch line running into the trunk line, and it would have to be carried on eventually to Western Australia. Climatically, and otherwise, white people generally can get on all right.

John Alexander Douglas Brown, sworn and examined.

114 To the Chairman.—I was in the Territory twenty-one years, and I represented it in the South Australian Parliament at the time of the transfer to the Commonwealth. I think that the only way in which the Territory can be developed is by a main trunk line going along the telegraph line as near as possible. It is not to be expected that people will go so far inland without some means of communication. My experience of the Territory extends only from the coast to Katherine River, south of that I can only speak from hearsay. I have not been around Bitter Springs. I would advocate a direct line with branch lines where necessary. Of course, it is too much to expect that the line can be built right off at once, and I have always advocated that if it were brought 150 miles further than Pine Creek, making about 200 miles, it would be sufficient for the time, which would enable people to take their cattle to the freezing works at Darwin. I also have advocated extending the line from Oodnadatta to the Macdonnell Ranges. This would be a means of enabling people in the pastoral areas to properly develop the country. I cannot say that the Territory has made any progress in the past twenty-five years. I have not been there since its transfer to the Commonwealth. The railway has been extended for 50 miles, but I advocated 150 miles. With other 60 miles the people in the Victoria River country and Wave Hill would be enabled to take their cattle in. It would never do to travel there overland to Darwin, because they would lose in weight three times what the freight

would come to. The most convenient point for trucking would be the first nearest Victoria River. It is better holding country at Wyndham, but it could go to much cheaper by truck than by road. Mr. Laurio has always told me on no account to drive cattle into coast country, because the loss in weight would take twelve months to recover. By rail they are taken straight to the freezing works and have no time to lose in weight. From a defence point of view I think the direct line would be the proper one, it would be a mistal on policy to have a railway running like a corkerew. I favour the direct line, because, in a cattle country, a few tails driving is neither here nor there, so long as they do not get to the coast country. I was in the Territory when Chinese were introduced, and I worked with them and over them. I would not go so far as to say that the Territory cannot be developed except by coloured labour, it would depend a good deal on the work on which they were employed. If it came to mining and the mine would warrant it, I would prefer white labour. I never felt any ill effects from working on the surface or below. I would not think of working a mine with cheap coloured labour if the mine would support white labour. I do not think it impossible for white men to live there. As I say, I was there to only one year, and I know people born and reared there, and who have been there for forty years. These people are getting old, but they are healthy and strong, and they have children and grandchildren. It all depends on the constitution. Women possibly suffer more than the men. I had fever for a short time, but I do not think it did me any harm. White men can do railway construction and stand the climate, especially as they get farther away from the western coast. People talk of cheap coloured labour, but it takes two coloured men to do the work of one white man. The Chinese appreciate the "Governments stroke" very well, or, perhaps, I ought to say, they have a special "stroke" of their own, not working for themselves. The agricultural development attempted by the Administration, under the existing conditions, was bound to be a failure for several reasons. It was trying to fight against nature, and the existing conditions, with regard to land tenure, and so forth, it was only wasted energy. I expressed my opinion freely at the time, and I have never changed it. In 1904 of 1,000 people at Daly River at the end of twelve months, I said he would not have ten, and that has turned out to be quite true. There is no local market, and in shipping produce he had to contend with coloured labour markets and with Queensland and New South Wales, where the wages are lower. If he had had coloured labour I do not think he could have made a success. Everything that can be grown in the Territory can be grown almost to better advantage in other places. Queensland is a better country than the Northern Territory. I have seen cotton, both wild and cultivated grown in the Territory as well as in any other part of the world, but it cannot be grown with white labour. White labour, of course, grows it, but you cannot compete with the cheap labour elsewhere. Cotton has to be gradually harvested, and cheap labour is necessary. I shall believe in cotton harvesting by machinery when I see it—when I see a machine which can go between the rows and discriminate ripe pods from green.

115 To Senator Story.—I do not know that the extension of the line of so much as 300 miles in Oodnadatta is required to strike the ranges

Anyhow, the present railway finishes in a point, and is rather an ornament. If the railway were extended from Oodnadatta to the ranges there would be a possibility of reducing the loss on the present line. At the present time there is great loss, a dead loss, and the loss would be reduced, even if the line were not made into a paying concern. As to any deviation the tin fields at Murrumbidgee have got to be provided whereas the Victoria River pastoral country has already been proved. I think the rail ought to follow the telegraph line as near as possible.

116 To Mr. Kinslay.—In view of the fact that there is a mineral field and excellent pastoral country around Bitter Springs I will favour the route I have indicated. I think the Victoria River people could, without much trouble and expense, drive their cattle to the railway line. I am my knowledge of the Katherine River district I think it is a suitable centre for the Victoria River people to entrain cattle to Darwin.

117 To Mr. Sampson.—I have no personal knowledge of south of the Katherine, and I cannot say which swing of the railway would bring most revenue. So far as it has developed to the present the Victoria River country will bring the most traffic. I do not know that the pastoral country on the eastern swing, or the western swing is any better than that in between.

118 To Mr. Laird Smith.—I do not think the pastoral people would trouble a great deal to entrain their cattle when they had country through which to drive them. I would like to see the railway go right through it, and I gathered that the Government was in duty bound to construct it direct from north to south without any deviation to the land. Mr. Mitchell, K.C., of Victoria, said that the Commonwealth could not go out of the Territory, but they are getting over the difficulty by attempting to go to "unconquered" and connect. There is no time stipulated in which the work shall be completed, as I think there should have been, but I look on it as a solemn agreement with South Australia that the line shall go from north to south to South Australia. Without the railway the country is useless, because it would be impossible to get people to go to these practically inaccessible places and try to develop the country. In the event of the rush war being conducted from north to south I should not have the slightest objection to encroaching branches from the Queensland border into the Barkly Tablelands. It was always my idea that a rich line would be constructed not only from Queensland but from New South Wales. The Barkly Tablelands and the New South Wales trade would naturally gravitate to Queensland ports. If the white ants are as bad on the south as they are on the north it would be a waste of money to use wooden sleepers. No timber will withstand white ants, though cypress wood will do so for many years. There are steel sleepers still standing after thirty years in the Territory. It is hard to say whether white labour had been imported at first for the lines in the Northern Territory, there would now have been settlements. No man would live in the Northern Territory for preference, there must be an inducement to keep him there. I have always thought it a crying shame that men should build 10,000 square miles of country in one week. Wave Hill is 8,000 square miles also, and that is not the proper way to develop the country. I do not suppose that on the 10,000 square miles held by the Victoria River Company, there are fifty men but

people employed. The universal opinion is that the Commonwealth will honour the agreement made.

119. To Mr. Gregory.—Without going over the country I could not give an opinion as to whether it would be profitable to build a railway representing £150,000 in interest unless there was a reasonable amount of population, but the country is at present useless without the railway. Whether such a railway as I recommend would be profitable without a much larger population in the Territory is a peculiar one, but we have to do something with the Territory, which, without a railway, can never be developed. I should say that, eventually, the line will pay, and much sooner than will the east-west railway. Without a railway there will be no population. It is possible that some big mining development may attract population, otherwise I do not see what is to bring it. There can be no closer settlement in the growing of cotton, rice, and similar products with white labour, and I do not see why these should not be grown with coloured labour, tropical products can be grown only with coloured labour. I have always objected to the less-than principle. I do not think the clearing and developing of the land would cost much with coloured labour, and without such labour I do not see that there are any possibilities in the tropical portions of the Territory. To protect the Territory, you would want an immense population if trouble came and without a railway I do not see how we are going to defend it. I have heard that there is a large area of country that will grow tropical produce within a reasonable distance of Darwin. I do not think that sugar will ever be grown in the Northern Territory, owing to the shortage of rainfall. Cotton, rice, and so forth grow remarkably well, but wheat growing is impossible. I do not think that the country has ever been prospected for minerals to any extent—not 20 miles on either side of the line. There are large bolts of auriferous country north of the Katherine, but I am doubtful whether they have been sufficiently proved. I have been at Kalgoorlie, and I do not look on the Territory, from a mining point of view, as country as good as Western Australia, but the place has never been prospected any distance from a railway line. I do not think there is any possibility of coal. There are areas of tin within 4 or 5 miles of the line. There is any amount of tin-bearing country—there are two or three districts I could mention—and I have seen specimens better than from any other place. In order to develop these positions the Government would have to subsidize them. It would be as well to have those holders of large areas with money, so long as they complied with certain conditions—if there were those conditions. But I think that ten different persons owning 1,000 square miles each, would prove a greater source of revenue than one person owning 10,000 square miles. The land was taken up without classification or survey, and those who took it up picked the eyes out of the country. Mr. W. H. Stevens, the manager of several stations has said that the Victoria River country compares favorably with any in Australia.

120. To Senator Keating.—As to the cultivation of tropical products I know there have been some big failures in sugar growing, but most of it has been experimental. I have seen tropical products growing very well in the Belmural Gardens. As to the objections expressed by Mr. Combes, I honestly think that there are better countries than the Northern Territory for tropical agriculture, and that is because of the soil.

Another cause is the heavy downfall of rain for a few months, followed by too long a spell of dry weather. Sugar, when it is coming to maturity, wants rain, of which it does not get sufficient. If the pastoral industry is to be the main thing we cannot look for an extensive population, but mining may cause an increase, and the agricultural question will settle itself automatically. That, I think, will be the evolution of the Territory—pastoral industry first, followed possibly by mining development, and agricultural settlement afterwards. I hold that view irrespective of the question of the use or non-use of coloured labour. It is not a factor which would affect my view. The difficulties presented in the way of obtaining a reasonably dense population would be equally the difficulties of any other Power. There are, however, people in some parts of the East so densely packed that they would be glad to take any country. I should hardly like to express an opinion as to whether the fact of our not settling the Territory with a fairly dense population would not afford an opening for some other Power to settle it with a population fairly dense. Any extensive coloured settlement from the East would be a great and serious trouble, but I do not think there is any great danger of it.

121. To Mr. Fenton.—I think one of the best possible things would be a fine mineral discovery, especially of gold. Gold discoveries have been reported towards the west, but nothing of any value. I have always heard of the western side of the railway as being very much more barren than the eastern side, but I know nothing of the mineral prospects. A lot of money has been made out of the buffalo hide trade, but the animals do not increase very fast. They are located at Port Eslington principally, and would not contribute to the traffic of the railway to any extent. I would not like to express an opinion whether it would pay to keep herds of buffaloes. Goats thrive very well in Darwin. I do not think that the Territory from a mineral point of view has been tested as it should have been, and the expenditure of a considerable sum of money in this direction would be advisable, as it is the only way in which the country can be proved. There have been considerable alluvial deposits there, and workings. The contention of mining men that where you find alluvial it must be shed from reefs somewhere near has, I think, been burst up in many instances. There has been no reef development to any extent in the Territory. I saw the other day that a bore had been sunk several hundred feet and had got 3 oz stone.

(Taken at Melbourne)

MONDAY, 7th FEBRUARY, 1916

Present

Mr. RILEY, Chairman,

Sensor Keating,	Mr. F. Iulayson,
Sensor Lynch,	Mr. Gregory,
Sensor Story,	Mr. Sampson,
Mr. Fenton,	Mr. David Smith

Norma Garrett Bell, Engineer in Chief and Acting Commissioner, Commonwealth Railways, recalled and further examined.

122. To the Chairman.—I do not know of beds of gravel or sand suitable for the making of concrete sleepers along the route of the proposed extension to Bitter Springs, but we have had some

difficulty in getting this material along the route from Plus Creek and the Katherine. The officer who surveyed this route is in Melbourne, and he should be able to give evidence on the subject. I have been told that there is sand and gravel along the existing line, but I do not know personally where this material can be obtained. As to the use of concrete sleepers for railway construction, experiments have been made during a number of years in Russia, Italy, the Argentine, and other countries, and many designs have been tried, but, according to the information with which I have been supplied, concrete sleepers have not proved a success.

123. To Senator Story.—The concrete sleepers are reinforced. I read recently in an American journal that so far no efficient concrete sleeper has been designed.

124. To the Chairman.—I have seen the Timms patent, and have had letters from Mr. Timms, so that I am acquainted with his design. A recent work on permanent-way construction contains photographs of concrete sleepers taken out of a road in America after they had been for some time in use, and a statement appears in it of the number of sleepers experimented with, and the results of the experiments. The sleepers referred to did not prove a success, and were broken by the load that they had to carry. We have not gone further than the Katherine in the construction of earthworks along the route of the Northern Territory railway, and work has been delayed because most of the men who were engaged in it have been called upon to repair damages caused by floods. It will be a few weeks before constructional work can be resumed. We have all the rails needed to finish the line as far as the Katherine, but have none for any construction beyond that point. At present it takes time to obtain rails. Steel sleepers cannot be obtained at reasonable prices, indeed I doubt if they can be got at all. Rails cannot be imported just now. They cost about £10 15s a ton in the Old Country, with another £5 a ton for freight, and sea carriage is very uncertain. I am willing to experiment with concrete sleepers, though I have already tried them. I have not experimented with Mr. Timms' design. The concrete sleepers that I have used were designed to take bolts right through them. The Timms fastening is a better one, and allows the rails to rest on iron plates. No doubt a concrete sleeper could be designed which would suit both the 3 ft 6 in and the 4 ft 8½ in gauge. The use of concrete for sleepers is not a new idea, it is one on which experiments have been made for many years. The first cost of concrete sleepers would be greater than that of wooden sleepers, but were they otherwise satisfactory the difference in cost might be compensated for by their longer life. Concrete sleepers are costly to make and expensive to carry, and, of course, increase the difficulties of maintenance, because they are so heavy. There are 392 men working on the line to the Katherine, and they will be available for the extension to Bitter Springs if the construction of that extension is not delayed too long.

125. To Mr. Gregory.—The railway is being made with earthworks and bridges suitable for the 4 ft 8½ in gauge. That is being done in compliance with the Act. Moreover, sleepers to carry rails laid on the 4 ft 8½ in gauge are being used. It is estimated in Western Australia that a jarrah sleeper lasts about fifteen years. If the 6 in gauge is to be used on the proposed line for 10, 12, or 15 years, it would be cheaper to lay down in the first instance a 7-ft. sleeper in

stead of an 8 ft 6 in sleeper. The traffic that is to be expected does not warrant expensive construction, the justification for the 4 ft 8½ in gauge being the connexion with the southern lines. In my opinion, it is a mistake to use rails of less than 60 lbs. in weight. In Queensland, on several occasions, they have used a lighter rail, but they have recently reverted to an old determination not to use any rails lighter than 60 lbs. Such a rail allows the use of heavy engines and large train loads. The revenue from the Northern Territory railway has largely increased during the past few months, an increase due greatly to the traffic caused by the extensions southwards, but we expect considerable cattle traffic when the meat works at Darwin start. The extension of the line to Bitter Springs and the opening of the meat works at Darwin should increase the traffic on the line, which I think will then pay working expenses, though not interest on construction. As to the use of sleepers suitable for rails laid on the 3 ft 6 in gauge, the Crown Solicitor recently gave the opinion that there is nothing in the Act to prevent their use, and if there is no likelihood of the conversion of the line from the 3 ft 6 in to the 4 ft 8½ in gauge within a reasonable period, money could be saved by using the 7-ft sleeper. The price of pulverised sleepers at Darwin is 8s 4d, and of unpulverised sleepers 7s 6d each. The unpulverised sleepers have not been in use long enough to enable me to come to a decision as to their merit. Some 600,000 have been laid on the east-west line, but they have not been down more than a few months. I do not propose to order more for the Northern Territory yet, and have been trying to get sleepers of Queensland timbers—blood wood, for instance—which I know to be suitable, though I have not succeeded so far. We got 8 ft 6 in sleepers for the east-west line at 4s 3d alongside the railway. I am aware that concrete sleepers have been tried by the Adelaide Tramway Company and by the South Australian and New South Wales railway authorities, though I have not tried them in the Northern Territory. About 100 concrete sleepers are laid in a railway near Sydney. These experiments are a better test of concrete sleepers than an experiment on the Northern Territory line would be, because the traffic is heavier, and I think that a trial of six months should be sufficient to determine whether the sleepers are suitable. I should have no objection to trying concrete sleepers in the Northern Territory, but the traffic on the Northern Territory is so light that the test would not be a severe one. However, I am following the experiments elsewhere, and am in touch with the Engineers for Existing Lines in the two States concerned. I know what the sleepers cost. I have obtained photos of the roads in which they are laid, and I hear from time to time about their conditions. I shall cause an inquiry to be made as to the occurrence of sand and gravel in suitable localities in the Northern Territory.

126. To Mr. David Smith.—The objections to the use of concrete sleepers are these. The sleepers are costly to make and to handle and are difficult to deal with when in a road. Under heavy traffic they are said to shake to pieces. In big centres, where there are plenty of maintenance men available, the weight of the sleepers is not of much consequence, but it is a serious matter in places where there are only four or five men for the maintenance of, perhaps, 20 miles of road. It is not clear to me that a saving of ballast can be effected by using concrete sleepers. Ballast is used below sleepers to distribute over the formation the weight of the train, while top ballast is

used to true the line and to deaden the noise of running. Bottom ballast would be required for any sleepers, whether wooden, concrete, or steel. Top ballast might be unnecessary for ballast concrete sleepers in line, because of their weight, but such sleepers would be very noisy to run over if they were not ballasted. Steel sleepers make a quiet road if they are properly boxed up. Such sleepers are used on the road from Normanton to Crocody, and that is a very quiet line to run over. Concrete sleepers on a safe bed, properly boxed up, might be a little hard, but would be quiet. I take as much interest as any one can in the experiment that is going on, because I am desirous of obtaining the cheapest and best sleeper that it is possible to get. All sleepers are packed below the rails only, the centre being left slack. That is done to prevent rocking. There would be no saving by using concrete sleepers in doing without ballast underneath. In wet country a great depth of ballast must be used to distribute the weight of the train over the formation, but in dry country less ballast is used. Whatever kind of sleeper you use you require the same number to support the rails. I do not think that any great advantage would be gained by using chain gripping the flanges of the rail for about 6 inches instead of dogs, which grip only about 1 inch. There is no danger in running over concrete sleepers; they are perfectly safe. I have read that private railway companies expect to recoup themselves for expenditure on railways by increasing the value of the land through which the lines run. In Queensland owners of land in benefited areas were required to guarantee an earning of 3 or 4 per cent, but I understand that the law under which that provision was enforced has been recently repealed. The measure seemed to me a fair one.

127. To Mr. Finlayson.—I could not tell you offhand what saving would be made by using sleepers for the 3-ft. 6-in. gauge. In any case the bridges should be made strong enough to carry a line laid on the 4-ft. 8½-in. gauge, but the widening of cuttings for that gauge could be left until it was proposed to convert from the narrow gauge. The Northern Territory line will have to be continued a very long distance outwards before it meets a broad-gauge line, but if the railway were extended from Oodnadatta northwards it might be wise to construct on a 4-ft. 8½-in. gauge. Small savings could be effected by adopting the 3-ft. 6-in. gauge in the Northern Territory, but economy is to be brought about rather by the method of construction—that is, by making the grades steep and the curves sharp, and by building low-level bridges. There will not be much earthwork on the country below the Katherine and Bitter Springs. The grades and curves and bridges on the line from Pine Creek to Katherine River have been designed for the 3-ft. 6-in. gauge, and on that section a saving could have been made by providing for a line on the 3-ft. 6-in. gauge; but not much saving is to be made in that way between the Katherine River and Bitter Springs, because the country to be traversed is level. It is estimated that the excavation work will amount to only 45,000 yards. About half the cost of the line will be in rails and sleepers. For labour we are paying 14s. a day as the minimum rate. I do not think that the difference between the cost of wooden and concrete sleepers would be less relatively in the Northern Territory than in South Australia. In comparing concrete and wooden sleepers in the light of tests made in tramways it must be remembered that tramway conditions are very different

from railway conditions. Tram rails have a great lateral strength, and are made deep to maintain the roadway between the top of the sleepers and the top of the rail. The rails used in a railroad have to be supported at more frequent intervals than those used in a tramway, as otherwise the cost of maintenance is increased. In Queensland at one time the centres of the sleepers were 3 feet apart, but the cost of maintenance was so great that a distance of 2 feet between the centres came to be adopted. I have no prejudice against concrete sleepers, and would be only too glad to use them if I thought that they could be used economically. But all the information I have on the subject is against them. The American opinion is against them, and I have heard that where they have been used in Italy they have been taken up. The concrete sleeper would, of course, resist the attacks of white ants, and would probably last longer in a railroad where the traffic was light than where the traffic was heavy.

128. To Mr. Fenton.—The Committee will see the results of American experiments with concrete sleepers set out in a book entitled *Sellow on Steel Rails*, which I produce. In Queensland the number of sleepers used to the mile with a 3-ft. 6-in. gauge was 2,640; on the east-west railway I am using 2,240 sleepers to the mile, and in the Northern Territory railway there will be about 2,080 sleepers to the mile. If Mr. Hobler has estimated the saving by using wooden sleepers suitable for a 3-ft. 6-in. track at £180 a mile, the total saving on the section under consideration would be between £7,000 and £8,000. As to the route which the line follows from the Katherine to Bitter Springs, I understand the intention was to keep as near to the Barkly Tablelands as possible. If what is called the "western swing" were adopted, the Roper River country would not be opened up. I have been told that Wyndham is the port of the Victoria River district. I do not think that the Bitter Springs route was adopted simply to give access to the tin-fields at Marranboy. So far as I know, the reports favour the Bitter Springs route rather than the western route. I shall ascertain what reports there are in the office concerning both routes.

129. To Mr. Sampson.—I agree with Mr. Francis that if there is no prospect of the 4-ft. 8½-in. gauge being adopted for the next fifteen or twenty years sleepers suitable for the 3-ft. 6-in. gauge should be used, that is, supposing wooden sleepers to be adopted. If steel or concrete sleepers were laid down, they should be made sufficiently long to carry the 4-ft. 8½-in. gauge. I am not aware that concrete sleepers have been used in the Northern Territory railroad, but I shall ascertain, in view of what Mr. Evans has said, when such sleepers were put down and what their condition is. I have not travelled through the country south of the Katherine. The line from the Katherine to Bitter Springs will pay its working expenses, and will increase traffic on the line already made. The ment works at Darwin are nearly finished, and it is expected that during the season they will treat 320 head of stock a day, which will mean a considerable cattle traffic on the line. For the five months ending 15th November last the loss on the present line was £276, and for the corresponding period of the previous year it was £2,355. Practically the line from Darwin to Pine Creek is now paying working expenses. That is due largely to the increase in traffic caused by the conveyance of material for the extension of the line; but I expect that there will be an increase of traffic from the grazing areas which will be served. It has been estimated that the traffic

estimated over the section under consideration will amount to £12,000 per annum, and the expenditure to about £11,600. I can furnish the details from which these estimates were made. What is called the westerly swing has not been surveyed, and, to the best of my knowledge, has not been estimated at any great extent.

130. To Senator Lynch.—Although the westerly swing was recommended by a Royal Commission, I understand that the Victoria River district, where there is a patch of good cattle country, has its outlet at Wyndham, and would be best served by a railway to that port. If the western swing were adopted, the Roper River country would not be served. In comparing the cost of this line with the cost of light lines made by the State—such, for instance, as that from Dalby to Moll, in Queensland, which cost £1,611 per mile, and that from Jericho to Blackall, in the same State, which cost £2,137 per mile—it must be remembered that the first was really only a tramway. Second-hand rails were used, and the track was unballasted, the sleepers being buried in the black soil plain. On the line from Jericho to Blackall, 40-lb. rails were used, but most of the country is very easy. When that line was being made, labour was receiving only 8s. a day, whereas in the Northern Territory now 14s. a day is paid. Then there is the difference in the cost of material to be taken into account. The line from Charleville to Cunnamulla cost £1,760 a mile, but it goes through fairly level country. Our estimate for the section under consideration is based on the actual cost of the section under construction. It is difficult to compare the cost of one line of railway with that of another, because conditions vary so much. We used to land rails for between 26 and 27 tons in length in Charleville-Cunnamulla line was under construction, but rails at Darwin cost something like £10 a ton. My estimate of the earnings of the proposed line does not provide for a sinking fund nor for the payment of interest, neither does it allow for any writing off in respect of locomotives or rolling stock. The steel sleepers that were laid in the Normanton to Crocody road in 1899 are still quite good. Some of them had to be taken out of the salt pans near Normanton, where the tide used to flow on to the line, but elsewhere they are unaffected by exposure. There is very little traffic on that line, and the maximum speed is not more than 20 miles an hour; but in Germany and America there are many miles of lines laid on steel sleepers. Some of the best roads in America are laid with steel sleepers. The line from Darwin to Pine Creek is carried on steel sleepers which, I am told, are still in good order, although it is about thirty years since they were laid down. We made inquiries of the Broken Hill Company whether it could manufacture steel sleepers in Newcastle, and found that it could make them quite well, but that it is booked up with rail orders. Steel sleepers are easily made. They are obtainable in several designs. There is no other place in Australia where they could be made. We imported those that we got for the first portion of the Katherine River extension from America. I would prefer steel to wood for the line under consideration could steel sleepers be got as cheaply as wooden sleepers. It is questionable whether, having regard to their long life, steel sleepers would not prove the cheaper in the long run. As to the paviouring process, the Engineer for Existing Lines in Western Australia tells me that pavioured jarrah sleepers are still in the road on the Marble Bar line, and that they look all right. I have some under observation, but they

have not been laid sufficiently long to enable me to come to a determination as to the value of the process. I have not heard of any sleepers on the east-west line being attacked by white ants.

131. To Senator Keating.—I have not heard that trains cannot be run at sixty miles an hour over steel sleepers. On the contrary, I have seen numerous illustrations of high-class roads in America in which steel sleepers have been used on which trains must be run at at least that rate of speed. When a curve is very sharp, the gauge has to be widened out, and wooden sleepers would have to be used if steel sleepers allowed no provision to be made for such widening; but the clips are often made so that they can be varied a little, to give the necessary clearance on curves. I do not think that Mr. Evans' statement that in the Argentine steel sleepers were replaced with wooden sleepers to enable trains to be run at 60 miles an hour is correct. The steel sleepers replaced with wooden sleepers may have been defective in design. High speeds are run on steel sleepers. It is not likely that trains will be run at 60 miles an hour in the Northern Territory for some time to come. The cost steel that Mr. Trevellick Smith is going to produce will be of no use for the manufacture of sleepers. For the manufacture of sleepers steel plates have to be rolled and stamped out. The plant required is quite entirely from casting plants. Except in places like the salt pans, steel sleepers last a very long time. The sleepers in the Northern Territory line may show some signs of corrosion where they have come into contact with the sea air, but elsewhere there has been no corrosion at all. I have no prejudices against concrete sleepers, and for years have been trying to secure one of suitable design. It might be possible to make such sleepers advantageously at places along a route where sand, gravel, and water were available. Concrete sleepers might prove more serviceable in the Northern Territory line, where the traffic is light, than they have proved under heavy service conditions. I think it would have been well to lay a mile of track with concrete sleepers by way of a test instead of putting down only 100 of these sleepers, as was done in New South Wales. If we can get the rails, perhaps six months will elapse between the authorization of the proposed extension to Bitter Springs and the laying of the sleepers, in which time a thorough inquiry could be made concerning the advisability of using concrete sleepers. The Department is not in any way committed to the use of wooden sleepers, though the estimate of cost provides for wooden sleepers. There will be very few loops. A sum of £14,000 has been put down for station accommodation, including water supply, pumps, cattle yards, and stations. There will not be any raised platforms.

132. To Mr. Sampson.—Mr. Williams, the surveyor, can give you evidence regarding the other route that has been suggested for the continuation of the line south from the Katherine.

133. To Mr. Gregory.—The traffic on the Northern Territory line will vary considerably. There will be a cattle season, which may extend over from three to six months. It is quite possible that in parts of the year only one train a week may be run. Where there is a great deal of traffic and the road is well ballasted white ants do not cause much damage to sleepers, but where there is little traffic, and earth is banked round the sleepers the white ants often do great harm. They do more harm to sleepers in the sidings than to sleepers in the main roads. In the dry country the white ants are not likely to be bad. On the east-west

line there are very few white ants. I have seen traces of them, but not many. The country is too dry for them.

With the consent of the Committee the following addition is made to Mr. Hesketh's evidence:—
[Extract from "Preservation of Structural Timber," by Howard P. Wells.]

As a result of its studies, the American Railway Engineering Association Committee concludes "that no form of reinforced concrete tie has been made which is suitable for heavy and high speed traffic, but believes a properly reinforced concrete tie, with proper fastenings, may be found economical in places where speed is slow and where conditions are especially adverse to the life of wood or metal." The Lake Shore and Michigan Southern Railroad placed some "Buhrer concrete ties" in its track at Sandusky, Ohio, in 1904. There were no renewals up to 1908, and all the ties appeared in good condition. The more common experience, however, seems to be like that of the Pennsylvania Railroad, which placed 500 of these ties in 1903. The concrete broke and crumbled, and by December, 1905, all had been removed on account of breaking. Kimball, Pectol, McKee, Chenoweth, Keeler, Hickey, and Alfred concrete ties tested by various railroads all cracked or crumbled in two or three years, although in a few cases longer lives were secured.

(Taken at Melbourne.)

TUESDAY, 8TH FEBRUARY, 1916.

Present:

Mr. Rixley, Chairman;

Senator Kostling,	Mr. Finlayson,
Senator Lynch,	Mr. Edwards,
Senator Story,	Mr. Laird Smith.
Mr. Fonton,	

Thomas Ernest Day, Chief Surveyor for the Northern Territory, sworn and examined.

134. To the Chairman.—I have been Chief Surveyor in the Territory for about four years, and I have been all over the section of the proposed extension, which, in my opinion, will have to be part of the main trunk line. The line need not necessarily run direct to Bitter Springs. There is some good country on the west side of the telegraph line. The country has been mapped out, and I think the Commission could get a plan showing that the country right through is all good pastoral. One of our surveyors travelled through with horses from Victoria River to Newcastle Waters, and by the time they got to the line they were in good condition notwithstanding that they had no more than one drink a day. The journey took about six weeks, and they picked up wonderfully. There are no permanent waters. It is good country provided there is water, which no doubt could be conserved. There are very few cattle there at present, and the trouble is that there is no water. There is a lot of beautiful sharp river sand suitable for concrete work, but I could not say as to the limestone right through. There is plenty of ironstone gravel which would make good concrete, though, perhaps, it is hardly sharp enough. If there is plenty of rock you could make all the necessary sharp stuff. Concrete would stand the white ants as well as steel. I am sure that, with the exception of cement, you could get the raw material to make the concrete sleepers. The material is handy to the track—all over the country. You could not go wrong for gravel and sand, which are of principal things, except, of course, cement. There would be engineering difficulties on the western side. Just where the western swing is shown there are a lot of rough hills, but a more direct route would

practically miss these. There are no serious difficulties. It would not be so difficult on the eastern side as on the western side by a long way. At the same time the western side is not a serious problem. There are no mountains in the Territory, the highest hill being only about 600 feet. They are nearly all broken ranges, and I have no doubt they could be practically avoided. Even through the ranges flats might be found. I think very little of the neighbourhood of the line as a white man's working country. I speak from practical experience, and I know that the white manual labourer does not stand the climate. I took forty survey hands there about three years ago of tip-top physique, and in our first season they had all had enough. Our experience right through is that the white man cannot stand hard work in the wet season. I know that there are white men working on the railway, but the point is do they give you anything like the value in work they would in the same time in the south? I deservy, however, that a white man will do as much as a coloured man there, but we have to pay the white man much more, and we get a less percentage of work then in the south. White labour is very costly. As to settlement, this country is going to be occupied for a long time by the pastoralists. We have only about half-a-dozen settlers, although we have surveyed some thousands of acres for agricultural settlement. What are the settlers going to grow and live on? The products could only be tropical, and they would have to come with Java sheep labour, which is only complete with steam from Darwin. I do not believe in agriculture at the top end of the Territory at all. I am not prepared to speak as an expert in regard to mining. The gold return was £1,600,000, to say nothing of what the Chinamen have been taking out for years, and which would bring the actual result up to about £2,000,000. There seemed to be no lodes, but simply little pockets.

135. To Senator Story.—I cannot speak of this as a geologist, but, as a "man in the street," I believe that there is a possibility of gold being discovered by deeper sinking. I have heard from geologists that they are pretty sure of this. The trouble is water, but that, I take it, is a matter of machinery. Once they are down to a depth of about 100 feet, they are practically flooded, and it is beautiful fresh water. There is some good pastoral country, as a recent survey shows, but there are no permanent waters. The stock at Victoria River get out on this country in the wet season and come back; there are then plenty of water holes. I would not say there would be less engineering difficulties in going straight through than by adopting either swing, but there are less engineering difficulties on the east than on the west. I would not say that the eastern swing would be easier than the middle route; they are probably about the same. I would not like to express a decided opinion without a further examination. It would be just as feasible to travel stock to the railway on the eastern route, but, of course, the fact has to be faced that on all stock routes they have to provide water. There are seasons when there are no permanent waters and they cannot get across. We have not as yet had any sinking. There are plenty of places for dams and so forth, where the water could be conserved, and there must be quite 40 inches of rain there. There is some good holding ground, and some of the billabongs could be cleaned out and dams put across the ends. They could conserve any amount of water for travelling stock. I have been right through from Oodnadatta to Darwin and all

over the Macdonnell Ranges. I should say that the climate is good enough for white men to work at Newcastle Waters—say, between the 17th and 18th parallel. It is hot, but there is no humidity, and the nights are cool. Further south we get a hotter dry heat, and even in the Macdonnell Ranges the nights are beautifully cool in summer. As to the kind of land for pastoral or agricultural purposes, going north from Oodnadatta, I may say first that at a place called Anacora there is a bore, the only artesian bore in the Territory. This has an output of 750,000 gallons of fresh water. That is about 80 miles from Oodnadatta as the crow flies, or 4 miles inside the Territory boundary. Between that and Oodnadatta is a country subject to drought, but good pastoral country with saltbush plains, mulga, and all kinds of edible bushes. As you know, this is at present all occupied. There have been very severe droughts in the last few years. From there I went to Arltunga, at the extreme eastern end of the Macdonnell Ranges. The country between that and sandhill country, covered with all sorts of edible bushes, with saltbush flats between; then comes stony tableland country with good saltbush and cotton bush. When you strike the ranges there are very rough tiers, with beautiful saltbush plains between. A well is being put down 40 miles to the north by McDill Brothers, taken in conjunction with the well at Anacora. This fact is interesting, because it shows that good water can be got in this country anywhere. They sunk the well 71 feet, and obtained enough water for 500 head of stock. All through that country there are similar localities with similar indications until you strike the ranges. These facts, in my opinion, speak well for the prospects of obtaining good water. I may mention that all this has been set forth in a report to the Department since I went there. In the Macdonnell Ranges there are plenty of permanent waterholes, and every facility for the most extensive conservation by shutting in some of the enormous gaps. It is splendid sheep country. They were in the pink of condition near Arltunga, although there had been practically no rain for eighteen months. It is a great point with that part of the country that, so long as the stock can get water the feed seems to hang on; and there is nothing in the world like saltbush mutton or beef. None of the country would be fit for agriculture except in the ranges, and then only with irrigation, for the purposes of which there are possibilities of conserving plenty of water. Professor Spencer spoke very wisely of the possibilities of this water conservation; and every visitor is struck with the small engineering difficulties of retaining large volumes in the gaps. The ranges extend for 400 miles with beautiful level flats in between where anything could be grown. At Alice Springs wonderful wheat has been grown, and water can be got anywhere there at 20 feet. Going north from the ranges for, say, 120 miles—I am speaking of the eastern side of the telegraph line—we get to the Davenport Ranges. All the country between is of second-class or third-class pastoral. I am not calling any of the land first class, because it is isolated, and it would not be wise to do so; but it is good land given facilities for transport and water. I would call a big lot first class if a railway ran through and the waters were proved. I may say that I am quite an enthusiast on this district, which I thoroughly examined in a visit of eight months. I saw it in a bad season, and was struck with its possibilities. I do not think there would be any deficiency of water, although we have not tried south of the

ranges, except at one place—Mount Skinner—where in a bit of limy country we got a supply for our own use at about 10 feet. I do not speak of north of the Davenport Ranges, because I have not been there. I have been 150 miles on the west side of the telegraph line, and I have examined the Burt Plains, which is some of the best pastoral country we have. In order to tap and develop the pastoral country in the northern part of the Territory I should say that the railway ought to be run so far south as Newcastle Waters, assuming it is not possible to run it right through. If the line were extended from Oodnadatta as far as Alice Springs, you would open the best of the country at that end. I know there is a considerable loss on the working of the Oodnadatta line, but then, of course, Oodnadatta is not in the best of the country. It always seemed to me a pity that the line should have stopped at Oodnadatta, which is really nowhere. It is stony, desert-looking country, but wonderfully recuperative with a good season. If the line were extended from Oodnadatta to the ranges I unhesitatingly say that the present loss would decrease, if the line did not become actually payable. My reason for saying that is that the ranges are well adapted for sheep, but at the present time the wool cannot be got away. It is probably the best country in Australia for horses, and it has been proved exceptionally fine for cattle. The cattle, however, lose 25 of their value by the time they get to Oodnadatta when travelling in anything like a bad season, and this, I take it, would more than cover the cost of trucking. As to running a line west of the telegraph to open up the Victoria River district, I hardly think it would be of any value, because we have to face the fact that freezing works are going up at Wyndham. The commerce, I take it, of the tablelands goes to Queensland and that of south of Newcastle Waters goes to Adelaide. Commercial administration is really what we have to look at, and I do not think a line there now would be of any value. The proposed through line should run either on the eastern swing or by the more direct route from Katherine—a direct line.

136. To Mr. Laird Smith.—If I were asked to submit a railway scheme for the development of the Territory I should come down the centre, because I believe that even on the western side, in the unexplored country, we should get a lot of good pastoral lands. A line coming near the centre would be a main drain for both sides. I do not think there would be much difference in the cost between the eastern and the central route. I cannot say definitely what the distance is between the western swing and the eastern swing from the centre route, but I do not think it can be more than 40 or 50 miles. I think that the present holdings along the railway are too large, and that with smaller holders there would be more produce. There are no fences or paddocks; the cattle are allowed to roam, though, of course, they are confined to their waters. When a man takes up a bit of country he provides permanent water, and, as the cattle naturally come where it is, it acts as a fence. The practice is to let the cattle wander and remain away, the owners having the knowledge that they will come back in the dry season. The tick is bad down as far, I should say, as Tennants Creek, from that south it is not known. In the ranges there is no tick. When the railway is constructed to Bitter Springs I think the owners will always drive their cattle to Katherine River. If the people from Roper River can get into Bitter Springs they will prefer to drive to the Katherine. North of the Katherine, however,

driving is bad, especially from there to Pine Creek, and there is no doubt that the cattle sent to the freezing works in the north would have to be trucked. In the wet season the country would be too boggy, and in the dry season there is no nourishment, and the cattle would not be fit for killing. As to whether a large quantity of the rolling-stock would be idle at times, or whether there would be a continuous supply of cattle, is a point on which I should not like to express an opinion. I do not think, however, that the supply of cattle would be continuous at the rail head. There would be, say, only six months in the year when cattle would be supplied at the rail head. They do not fatten all the year round. I should say there would be a supply of cattle for the freezers for six months.

137. *To Senator Lynch.*—I favour either the eastern swing or the middle route. On the western swing there is a large area of settlement whereas there is only a small one on the eastern swing, but I suggest the eastern because there is there good country that might be taken up. It has not been taken up by now owing to the want of water, and people are just realising the fact that there is water there. For years on the Barkly Tablelands there was no stock, and they are just now finding out that a good sub-artesian supply can be got at 300 or 400 feet. I favour the eastern swing on the prospect of water on Barkly Tablelands, and the fact that the country south will eventually be taken up. The country on the western side is taken up by only two or three pastoralists. Dorrill Australian Estates Company has, speaking from memory, about 12,000 square miles, and then we have Wave Hill, Gordon Downs, Willeroo, Delamere, and two or three smaller holdings on the borders of Western Australia which really belong to Vesty Brothers. What I mean is that the whole of the land I have mentioned, from Wave Hill inclusive, has been purchased or leased by Vesty Brothers. Then there is Bradshaw's run which has very little cattle on it. In all this territory in the neighbourhood of the western swing the outlet is Wyndham, the natural port, and they could drive the cattle there for most of the year, because the country is the better for driving. I favour the eastern swing because it will eventually serve a much larger area of settlement. As to the size of holdings which would be reasonable I should say about 500 square miles, whereas, as I have said, there is one 12,000 square miles. I have had no experience of white labour in Queensland, and I do not know of men having worked there continuously in the open for a number of years. I can only speak from my own experience, and I know that white men do not do the same work as in the more temperate parts of the Commonwealth. I think that if the development of the Territory is to take place by means of tropical agriculture there will have to be cheap labour of some kind, white or otherwise. If we have to compete with Java, or any eastern country, we must have some kind of cheap labour. If they could find machinery to do the work well and good, but failing that we must have cheap labour. I am speaking purely from the point of view. I am a lover of the idea of a white Australia, and was enthusiastic on the subject myself when I went to the Northern Territory, but my four years' experience there convinced me that, under present conditions, another view must be taken. Of course, whether in generations to come different acclimatised races of white people spring up, is another matter. I think there would be much better hope from the

importation of some of the southern European races. There are some Maltese there now on the railway, and they are shaping remarkably well. The Territory is in about the same latitude as Mexico, but, in the latter country, there is a mixed race. Another point is that there is trouble to get white women to go to the Territory; and no country can be a wife man's country without white women. I think I shall be borne out by any body who has lived in the Territory when I say that white women cannot live there if they have to do their own housework, and that sort of thing. We hear of women who have resided there and have done well in the past, but they had their Chinese cooks and servants, which help cannot be obtained now. I tried to get a Chinese cook for my own wife, and £10 a month was the lowest wage.

138. *To Mr. Fenton.*—The part of the country to which I am referring may be taken as north of the 17th parallel, or you could probably go further north and draw a line from the bottom of the Gulf of Carpentaria. The climate gets better as one goes inland; the coastal country is the worst. There may be fine men at Port McArthur or at Darwin—there are exceptions to every rule. I do not say that I have a preference for the central route as against the eastern route, but I should not favour the western swing at all. If it were a question of the two swings I should favour the eastern, but I think it possible that a route could be got through the middle which would be as beneficial, perhaps, as the eastern swing. Above the tablelands we are going to get development on the Roper River, and so on. I have been through the Marranboy tin-fields. I am not a mining man, but, according to geologists and mining people, it is going to be a great place. I know of no one who has reported on the eastern swing, but I suppose the Engineer in Chief has done so. I have made no report on any railway route.

Victor Albert Williams, Surveyor, Commonwealth Railways, sworn and examined.

139. *To the Chairman.*—I have been in the Northern Territory engaged on railway location and survey work. During the last two years I have been travelling backwards and forwards over the route of the proposed extension making surveys. Our instructions were to proceed to Bitter Springs, where I understand, the Administrator proposed to build a township. The Government Experimental Farm is at Bitter Springs, and I understand that that is the reason this point was made for, with a view to a township in the future. We had no discretion whatever on this point; we received our instructions to survey to that place and did so. The country along the route of survey is very poor indeed, with no pastoral settlement. There is the Government farm of which I spoke, and I understand there are there about 2,000 sheep, 60 or 60 cattle, and probably 30 or 40 horses. The railway will terminate at a point about 1½ miles west of Bitter Springs. There is no permanent water on the route; but at King River I think that, by boring, you would strike a very decent supply. That river is about 25½ miles from the Katherine on the survey route. That is the only stream that will have to be bridged over—the only bridge of any size; of course, there are some small culverts, and so forth. It will be very hard to conserve water on the route, but on the upper Katherine it is possible. I am doubtful whether the country is high enough to gravitate water. The Katherine is 350 feet above sea level, and Bitter Springs is

just 450. As to concrete work there is a limited supply of sharp sand in the bed of the Roper creek. There are numerous ironstone ridges running into the Roper, and I think there would be a fair supply of ironstone gravel to be got. There would, however, have to be investigations as to the thickness of the seams, and so forth. I believe that you will be able to locate gravel beds for the purpose of manufacturing concrete sleepers. There are good surface indications, but we have not made tests in the way of trial shafts beyond small cuttings. The earthworks, however, are very small on this section of the line. The white ants were a great trouble to us, and I lost a number of field books. They crawled up the legs of the table and ate right through the books, and we lost a quantity of clothing, boots, and so on. In my opinion, it would be unprofitable to use any timber work on this section of the line. Steel or concrete sleepers will be necessary, unless you can preserve or protect the wood, and I know of no way of doing that. As to mineral development there is the Marranboy tin-field, but, so far, there are surface indications only, with very little developmental work. I understand that the battery was put up on the strength of a geological report by Dr. Jensen and others. This railway would cheapen the cost of material at Marranboy, seeing that the line will run within a little over 12 miles of the field. I made a trial survey over to Marranboy, and they have the full particulars in the Commonwealth office.

140. *To Mr. Simpson.*—We have proposed no other route except the one we surveyed. From the Katherine to Newcastle Waters the country to the west is very dry—a waterless area. That is the route along the centre. I have explored the country for about 10 or 12 miles from the line, and to the westward it seems to continue dry—poor, miserable, and uncultivable. I know nothing of the Victoria River district. I know the country on the western side is very low and swampy, and subject to inundations from the overflow of the Katherine; that is why we kept slightly to the east. We did not follow the telegraph line, but we were not far from it, crossing it once. There is a small cattle run extending from the Katherine to King River, about 26½ miles, containing some 1,000 head. These could easily be trucked from the Katherine. There is no more cattle raising until we reach Bitter Springs. They come from the stations lower down the Roper—Elsie station. I was there during the muster, and from the report there appeared to be, roughly, 4,000 head. Then, lower down the Roper—about 90 or 100 miles—there is the Hodgson Downs station, where, I understand, they muster over 7,000. They drive their cattle up to Darwin via the Katherine. I have not heard of them shipping them from the Roper. South of the Roper I travelled to just above Daly Waters. The country appears to improve towards the Barkly Tablelands. The grass is shorter and smaller, and our horses did better on it than on the grass further back towards the Katherine. A fair number of stock runs on the country further south. Our instructions were to take the best route from the Katherine to Bitter Springs, and to keep as direct a line as possible, avoiding, of course, all engineering difficulties. I gather an extension of the line to the mine would think very little traffic; it would be necessary to carry the line further towards the Barkly Tablelands. I should say it would have to be extended to Anthony Lagoon, over 200 miles from Bitter Springs, before it would secure any amount of traffic. That would increase the distance. If the eastern or western

swing were to terminate at Newcastle Waters or Powell's Creek the Barkly Tablelands could be served quite well by a branch line.

141. *To Mr. Finlayson.*—I was in the Territory three years, and during that time was on the routes surveyed. I did the first section from Pine Creek to Katherine. I know the country between Darwin and Bitter Springs fairly well. The country between Katherine and Bitter Springs is very much better for pastoral purposes than the country from Katherine to Darwin; it improves as you go south. The prospects of railway traffic, as well as of the development of the country, improve the further south the line is taken. As to the reasons which inspired our instructions to touch Bitter Springs, I think that the water supply was one of the chief. Bitter Springs is the head water of the Roper, and at that point there is a good water supply. At intervals along the Roper Creek, just after the wet season, there are clay-holes full of water. This is a good road followed by teamsters. You must go towards the tablelands to get traffic of any consequence. I consider that this section is a proper link in the chain connecting Darwin with either the tablelands or South Australia, and I believe the line is probably the best route for a north-south railway. So far as I know the country to the west of the surveyed route is not as good as that to the east. It is badly watered for one thing, and the country is very poor. Bitter Springs is the head water of the Roper, and at that point there were engineering difficulties that prevented the line being taken nearer to the tin-fields, it would be a costly line to take from Katherine by Marranboy. That was not exactly the reason the line did not take in the tin-fields. At the time the survey was authorised there had been no known of Marranboy, which was in its infancy. When I went from Bitter Springs, south to Daly Waters, I was engaged on a trial survey between those two places, roughly, 100 miles. That survey is still going on, and I am going back to the work very soon. There is an inquiry of some sort as to railway construction south of Bitter Springs, towards Daly Waters. We felled trees in clearing work, and found that most of them had been hollowed out by white ants, and were nothing but shells. The timber was practically useless. The ants crawled up outside the legs of the table to which I have referred. They had built a small tunnel of mud and worked their way up through that, always at night time. I have had no experience, and would not like to say what would be the life of a wooden sleeper on a railway line in that country.

142. *To Senator Lynch.*—I prefer the eastern swing, because I consider it will tap much better country, and that the traffic would be much greater with a better water supply. I have not been on the western side further than 10 or 12 miles off the survey line, and I do not know what that side is like from personal experience. I have, however, held conversations with land surveyors who travelled through it recently, and from them I have got the information. According to the map before me large pastoral areas are taken up on the western swing, while on the east there are only small scattered settlements. The eastern swing contains the better country; but on the western side I understand the pastoralists could ship their cattle from the Victoria River at Darwin and Wyndham. That is one reason I take it why land has been taken up there. I am not aware that the first section of the line from Katherine to Bitter Springs would really determine the route. We shall lose distance certainly

in getting back on the western swing, but not very much. From Bitter Springs we take a westerly swing, tapping Willeroo, and then work south. I realize that my suggestion would mean a zigzag line, with a sharp angle at Bitter Springs, should it be decided to adopt the westerly swing and go to Victoria River, but still I think it could be done if the success warranted. I have noticed no traffic whatever from Barkly Tablelands district to Darwin. White men cannot live in that country. Carriage rates are excessive. They pay £37 a ton at Daly Waters to cart flour from Pine Creek. Private enterprise cannot stand that. So far as I know, Barkly Tablelands hold out a fair prospect as a pastoral proposition, I have not been there, but have heard reports. My evidence is that, so far as this 65 miles strip, it will not serve any settlement on the route at all, only one station—a Government station—and the Maranboy tin field will benefit. So far as I can see the prospects of this line carrying any stock are very poor.

(Taken at Melbourne)

FRIDAY, 10th MARCH, 1916

Present

Mr RILEY, Chairman;

Senator Keating,	Mr Finlayson,
Senator Lynch,	Mr Gregory,
Senator Story,	Mr Sampson
Mr Fenton,	

David Lindsay, surveyor, sworn and examined

143 To the chairman.—I was a member of the Royal Commission for Northern Territory Railways and Ports which inquired into matters connected with the Northern Territory, and in that capacity I have considered the question of the construction of railways in that Territory. I know the section of the railway which it is proposed to construct between the Katherine River and Bitter Springs. I have always opposed the continuation of that line in any particular direction until an examination of the country has been made by experts. The route between the Katherine River and Bitter Springs traverses very poor country most of the way. It follows the valley of the river, and a lot of the country there is subject to inundation. I cannot understand the object of taking a railway there. I have studied this question, not merely as a member of the Commission, but for the past thirty years. I spent twelve years in the Territory, six years on the north coast. The route that I desired the Government to examine before undertaking the construction of the line to the Katherine River, would have traversed country from Pine Creek in a southerly direction and would have crossed the Daly River, so that it would have served, not only the Roper valley, but the Victoria River valley. It was because the Government had decided to build the line to the Katherine that Mr Clarke and I recommended a westerly swing from the Katherine, with a view to tapping the Victoria River country, thus making a line which would be only a few miles longer than would be the direct line, or than the line would be if an easterly swing were adopted. We desired to visit that country as a Commission, in order to ascertain whether it would be better to adopt the westerly

swing or the easterly swing, or to build a straight line. But the Government stopped our work, and, so far, nobody has examined the country which would be traversed by a line with a westerly swing, or by a straight line between the westerly and the easterly swing. I understand that the central line would traverse a level plateau, upon which a railway could be built very quickly. Of course, in a land of heavy rainfall, it is necessary to keep as near to the water-shed as possible. A line with a westerly swing would probably cost more to construct, but it would get into better country sooner—into basalt country. Thus, we were afforded no opportunity of examining the country, and our recommendations were based on sworn evidence, because we were recalled before our labours were finished. The idea underlying the suggested westerly swing was to tap the great western part of the Territory—the Victoria River country. This westerly swing would probably touch Willeroo, and would tap the good country in the Victoria River district. I have visited the Maranboy tin field, but there are very few tin mining fields that would justify a deviation from a main trunk line. The westerly swing which I have suggested would take the line much farther away from the Maranboy tin field. I do not know what has happened to that field since I was there, but, at that time, there was nothing visible to influence any man's judgment as to the route which a railway should follow. Even if the Government erected a plant there to enable the ore to be treated, my opinion in this connexion would be unaffected, because a mining field has a life—shorter or longer, according to the value of its mines—whereas the Commonwealth wishes to construct a railway which will serve the Territory from the stand point of permanent settlement. I favour the westerly swing of the line, because I think it would be a more paying proposition, inasmuch as it would tap the country which carries the most cattle. Sixty or 70 miles of the route from the Victoria River to the Katherine River is very poor country to travel stock over. It is true that the pastoralists travel stock over it now, but there is all the difference between travelling a few hundred head of cattle which are intended for local consumption, and travelling thousands of cattle which are intended for export.

144 To Senator Story.—Theoretically, a direct line between the two suggested swings would prove a benefit to the Victoria River country. There are two belts—the eastern and western—both of which are capable of settlement and development, and if it be possible to run a line direct between the two, that would be very much better than swinging a line to either, and neglecting the other. I have no personal knowledge of the country that is included in the loop between the two swings, but I believe it is a level plateau upon which a railway could be built very cheaply. It was practically unexplored country until twelve months ago. It is country with a rainfall of from 25 to 30 inches annually, and I have not seen any country with that rainfall in which water could not be obtained at a greater or less depth. A main depot for cattle might be found to enable cattle to travel from Bitter Springs on the east and Victoria River on the west, that would be a necessity. I do not think it is advisable to deviate from the direct line for the sake of developing a mining field, unless it is a very extensive one and I am of opinion that the Maranboy field is not an extensive one. I would rather run a

2 ft. line from the main line to the tin fields. I have no knowledge of the country immediately south of the Victoria River district. I do not agree with the suggestion that a spur line should be run due west from about Newcastle Waters, with a view to opening up that country. It is too far from a port for dairying purposes, and I would regard such a line, in the absence of any knowledge upon which to base a conclusion, as merely a theoretical one. If there were a large number of sheep in that country, that fact would be a justification for the construction of the line. But, until the trunk line has been built, a branch line such as that suggested is scarcely worthy of consideration. I think that a railway into Port McArthur is probably the most important developmental railway that it is possible to construct in the Territory. That port is the natural outlet for 200,000 square miles of country, and has no other port to compete with it. It is distant, in a direct line from Port Darwin, about 650 miles. The next nearest port capable of being utilized as a place of shipment for frozen cattle is Townsville, on the east coast of Queensland. There is no other port in the Gulf of Carpentaria which can be made fit for shipping cattle or frozen meat. The main purpose which will be served by a trunk line to Darwin is, that through communication will be established with the other States, and, as a result, it will be possible to conduct business more expeditiously and cheaply, and as a strategic railway it is very difficult to say whether it is more important to continue the existing line south from the Katherine River, or north from Oodnadatta, assuming that both projects cannot be undertaken simultaneously. The southern line would connect the southern half of the Territory, and, in the absence of that line, that portion of the Territory cannot be profitably occupied, although it is capable of profitable occupation. The northern portion of the Territory, even after a railway has been constructed to the Katherine, will be capable of occupation only to a limited extent. But it is absolutely necessary to have the railway at both ends, in order to obtain the full advantage of the country. In other words, both railways are vital to the development of the Territory. I certainly think that a straight line between the suggested easterly and westerly swings is preferable to either of those swings, assuming that the country is suitable and that there are no engineering difficulties. The difference between the mileage would be very small. If we desired to serve two villages with a railway, obviously, a line that was equally distant from both would be preferable to a line running into either one of them.

145 To Senator Lynch.—When Mr Clarke and I suggested the westerly swing, we knew that there was a suitable place at Willeroo where cattle could be shipped, we were aware that powerful influence was being exerted to get the line constructed to Bitter Springs, and consequently we had to make some recommendation. That recommendation we carefully safeguarded by pointing out that we had not been afforded an opportunity to examine the country. I do not think that the building of a railway with an easterly swing would assist to develop the northern river country. But if the McArthur River line were built, that would develop the best pastoral country in the Territory. There is a deep sea port available at Pelly Islands. It is a natural port. If such a line were constructed first, its cost would be saved in the less expenditure that would be involved in carrying material for the remainder of the transcontinental line. When the Pine Creek

to Katherine River railway was projected, I wanted expert examination made of the country, to enable us to form an opinion as to whether the line should trend in a westerly direction. There is a lot of good country on the Daly and Flora Rivers, and such a line would bring that country into the closest possible connexion with Darwin. I think that an expert examination of it is practicable within a reasonable time. An expert could get over it at the rate of 20 miles a day. He could travel on horseback. The tributaries of the Victoria River have their rise in the country of basaltic formation, which is coloured yellow on the map. The Territory to the eastward of the telegraph line is sandstone and lime. Much of the country around Bitter Springs is of a poor character, and I do not know what purpose a railway there will serve, except as part of the main trunk line. Of late years, Bitter Springs has been brought into prominence by the Administrator, when, I think, has formed an absolutely erroneous idea of it. Bitter Springs has no qualifications as a main centre upon that trunk line. The western swing of the railway would suffice to develop the Victoria River downs country for many years. If a railway were constructed to Port McArthur, the produce from the Roper River country would go to that port and the necessity for building a line to tap the Roper River country would thus be obviated. The westerly swing of the line would inflict no injury on the Barkly Tableland, because it would touch that tableland about Newcastle Waters, just as near as would the easterly swing. The Victoria River country is too far from a port of shipment to be practicable for dairying. On the Flora River, however, it would be practicable. I still adhere to the opinion which I previously expressed in favour of the westerly swing of the line.

146 To Mr Sampson.—I favour the construction of a line from the McArthur River to the Barkly Tableland, to the neighbourhood of Anthony Lagoon. I have been over that particular country, and, as a matter of fact, I surveyed all the boundaries of the runs on the Barkly Tableland in 1886. If a line were constructed from the Katherine River to pick up the overland telegraph line at Powell's Creek or Newcastle Waters, it would be perfectly practicable to run a branch line out to tap the Barkly Tableland. The country that would be traversed is absolutely level. The country between Port McArthur and the Barkly Tableland is capable of great development, and that fact in itself would be a partial justification for the construction of a line of railway there. I may mention that, on the McArthur River, a very big irrigation scheme could be carried out. If the westerly swing of the main line were adopted and the railway were continued south from Newcastle Waters towards Oodnadatta, whether it would serve any considerable portion of the Barkly Tableland would resolve itself into a question of the freights to be paid on the stock, and of Townsville competing against Port McArthur. Assuming that the freights to be paid on the railway were on the Queensland basis, it would be possible to take stock from a great portion of the Barkly Tableland. The distance is not too great to enable stock to be driven, but the route passes over too much poor country. In regard to the westerly swing of the line which I advocate, I base my opinion chiefly on the evidence which the Commission elicited from those who were settled within the Victoria River and Willeroo country. That evidence indicated that the country would be developed, and

its carrying capacity trebled or quadrupled if that swing of the railway were adopted. If the railway followed the telegraph route, there would not be much development in the matter of cattle raising, because the country is not capable of carrying a large number of stock. A fair amount of revenue might be obtained from that source, but I do not suppose that the traffic on the section of the line between Bitter Springs and Daly Waters would pay. At the head of the Roper River, however, there ought to be considerable settlement. Around Bitter Springs, there is an area which, by means of drainage and irrigation, might possibly be made suitable for closer settlement. There is a strip of country on the tableland to the west that looks as if it would grow wheat. There are one or two problems there which have not been settled, and which ought to have been. There are many million acres of loamy soil, which grows grass 3 feet high, and which is as good as a wheat field. That is on the route between the Katherine River and Daly Waters. The rainfall varies from 16 to 30 inches. Under similar conditions of climate and rainfall, in India, wheat is grown. Some time ago, South Australia made an experiment with Indian wheat and it grew well. But, despite all that I have urged upon the Government of the day, they have not proceeded with one experiment which would have settled this problem. It is one of the most important experiments which ought to have been undertaken in the Northern Territory. I know of no country in the world that is easier to develop by means of irrigation than our parts of the Territory. There are five rivers there, great level plains, and, in many cases, the land 3 miles back from the river is lower than are the river banks. Thus, no great elevation of water is required. But irrigation cannot be taken under present labour conditions. It is impossible to pay the wages that are demanded, and to compete successfully with countries which are within a stone's throw of our own continent, where agricultural work is carried on by coloured labor, whose standard of living is so much beneath ours. But the soil, climate, water, and storage capacity are there.

147 To Mr Gregory.—The Railways and Ports Commission, of which I was a member, took evidence in Melbourne and Adelaide before it proceeded to the Territory. That evidence, however, was not sufficient in itself to justify the Commission in recommending any particular railways. The Commission spent about seven months in the Territory itself. Its members were prevented by the Government of the day from examining the country which is capable of development under present conditions by means of railways. Six months longer were required to enable them to complete their investigations. The north and south transcontinental line is the only line which can develop the Territory. But, of course, Queensland can be developed, and a small portion of the Territory, by constructing a line into the Territory from Queensland. I am of the opinion that a railway from the Follies Island in a south westerly direction, would provide a more immediate return. They would be an extension south of 60 or 70 miles of the existing line from the Katherine River, or than would a line from Oodnadatta north. But it is scarcely a fair thing to put one proposition against the others. The three lines are absolutely necessary, because the country is so vast. It must be recollected that the Territory is 960 miles long, and that a line which would develop 100 miles of it would

not touch the other 800 miles. A line from the north would require to come as far as Alice Springs, a distance of about 300 miles. The necessity for a railway from the McDonnell Ranges southwards lies in the fact that one has to cross a dry belt of Australia which has a rainfall of only 6 inches, which is in South Australia. It is only in occasional years that stock can travel through that country. Even in good years, a large number cannot be travelled. Ultimately, I believe that a railway north of Oodnadatta would become a payable proposition—possibly within the next twenty years. At the same time, I do not think it would prove a paying line within the next ten years. That country will carry sheep, and these will pay when cattle will not. I would not take any notice of sensational returns from the Marriarraby field, unless the crushing ran into thousands of tons. I am so extremely sceptical in regard to extensive payable tin fields existing in Australia, that I am naturally sceptical about Marriarraby. I would rather see the transcontinental line built with a westerly swing than with an easterly swing, in the absence of an inspection of the country. I believe that such an undertaking would secure a great deal more traffic in the immediate future. There is no settlement on the Roper River worth speaking of.

148 To Senator Lynch.—It might be many years before there is any settlement there.

149 To Mr Gregory.—The Victoria River is the natural outlet for the Wave Hill district. But, so far as we can gather from evidence which has been adduced, the mouth of that river is too uncertain to enable it to be made a port. Darwin is a better port than Wyndham, because the latter is a place which is not fit for white men to live in. It lies at the foot of a high mountain range, and is shut off from the breeze, whereas Darwin stands on a level plateau, and is open to all the winds that blow. Moreover, it has a fine harbor, and is right on the sea. I am very doubtful about the possibilities of dairying in the Northern Territory. The hot season of the year is too long, and I do not think it is possible to make butter successfully in the wet season.

150 To Senator Lynch.—I have already said that I favour a central route being followed, provided that there are no engineering difficulties—and I believe that there are none. The object of the westerly swing of the transcontinental line which I have recommended, is to tap the settlement on the Victoria River.

(Taken by Sectional Committee at Darwin.)

SATURDAY, 15 JULY, 1916

Present

Senator STONE, in the Chair,

Mr SAMMONS, Mr LAIRD SMITH

PATRICK CHILL, Protector of Aborigines and Superintendent of Central Aboriginal Station, sworn and examined.

151 In Senator STONE.—I have been 34 years in the Northern Territory and have ridden over nearly all of it. On one occasion I rode from Townsville to Wave Hill, on another from Townsville to Jacot close to Burketown. Then from Normanton to Corolla Downs, on the Barkly Tableland from Townsville to Borooloola and round to Cape Don, &c. I was manager of Delamere Station for five years, and temporarily

managed Corolla Downs Station for fifteen months, while I was head stockman and horse breaker for five years.

In regard to the railway your Committee has under consideration, my opinion is that it should be built to follow as direct a line as possible so as to serve the country both to the west and to the east. If you swing to the west you will tap the Victoria River Downs country, which is certainly good cattle country, but not better than exists on the east, given plenty of water about Farram, on the tableland country, and from there to the head of the Roper, and from Hodgson Downs to the head of the Limmen River, is all good cattle country. There are permanent waters there, but not sufficient to run large bodies of cattle. To go to the westward in the vicinity of Willeroo you could not get along the line proposed, because of the difficult nature of the country, without great cost in cuttings. It is very swampy at times near Lake Woods, and you would require miles of bridges. After leaving the Katherine River, and before reaching the King, there is a stretch of country over which it would be far more difficult to construct a line than to the east, on account of the number of small water courses to be bridged. By keeping out further and going in a direct line to Willeroo, there are no great engineering difficulties, although the country is rough. But when you leave Willeroo to approach Delamere there is a large range of broken country—called a "jump-up"—about 200 feet high, that you would have to get down. After leaving this "jump-up" country, crossing to Killarney, there is a little more broken country until you get close to Battle Creek, where there is another "jump-up." I have not been beyond Killarney Station in the direction that is marked on the map, but I believe it is desert or hedge wood country between the head of the Armstrong and Lake Woods.

152 To Mr SAMMONS.—I would not run a railway in a direct line from Katherine to Daly Waters as you would be missing all the present settlement. Possibly, later on, when water is available, land along that route would carry a lot of stock. I would prefer to run a line direct from Katherine to Daly Waters, thence on to Newcastle Waters, rather than direct from Katherine to Renner Springs. By going to Renner Springs you run the line into low, flat country before you reach Anthony's Lagoon. By turning off at North Newcastle and keeping Abner Range on the west the low lying country at the source of the creeks known as Mon Moona and Bundara would be missed. All the large creeks on the tableland running west drain into low, flat, coolieish country, forming large tracts of swampy country, and for months in the year are uncrossable. Lake Sylvester and Lake De Burg being places where the waters from Corolla and Brunette empty their waters. Lake Woods is filled by the Newcastle Water Creek and other smaller creeks. All these depressions would have to be bridged for miles to get a line across. By turning or bearing to the south east at the end of Abner Range, near North Newcastle, and running direct to Eva Downs, nearly all the lake lake country could be avoided.

153 To Mr SMITH.—The good tableland country begins just north north east of Newcastle Waters and runs right to Cameroonal. It is good cattle and sheep country, and now they are making available good artesian water there. I have not been along the telegraph line below Powell's Creek, but I know the whole of the country on the eastern side from Newcastle Waters to the Queensland border in a south easterly direction. All the tableland country is beautiful country, with

patches of what is called desert country, but that is the best of the land for stock purposes in the dry season. All the country known as the Barkly Tableland is the best country in the Territory.

154 To Mr SAMMONS.—It is better than the Victoria River Downs, but it has not the water.

155 To Mr SMITH.—If a direct route were adopted for the line, I think, as a cattle man, that the owners would certainly truck their stock rather than drive them, excepting at certain periods of the year, when cattle would improve while driving.

156 To Mr SMITH.—I am of opinion that the territory we have referred to would be used purely for producing cattle and sheep, that is, for pastoral purposes. Later on, agriculture might develop if a market were available. I think a direct route would be quite practicable and free from hardships, excepting perhaps in the vicinity of the King River and the western branch of the Birdum or the Ilcey Creek, where bridges would be necessary. In driving from lots of the western country it is practicable to drive to the railway line, but at times it would be necessary to go round by the Flora River, and in that case you would not return to the Katherine, but pick up the line near Poo Creek, because the waters on the other route are too small for large bodies of cattle. As to whether it would pay the squatter to truck his cattle rather than drive them, it must be remembered that the farther you swing the line to the west the better it will be for the people there, but, wherever you go, bores will have to be put down to enable the cattle to reach the railways at all times. Up to August or September they could probably get in, but under difficulties.

157 To Mr SAMMONS.—I have not been through the central country, but I believe it is good pastoral country. The carrying capacity all depends upon the water available. With water supply for the stock, the country to the east of the telegraph line should carry 125 cattle to the square mile. Taking it on the whole I would say that with water it should average 100 head of horned cattle to the square mile—I do not mean sheep. From Katherine to Willeroo is about 72 miles in a direct line, but about 90 miles by the west route. If it were a question of constructing a line from Katherine to Bitter Springs, or Katherine to Willeroo, I should look at the matter from a pastoral point of view, go to Willeroo, but you are only just touching the cattle country at Willeroo, and they would still have to drive from Delamere, Victoria River Downs Station, and Wave Hill to Willeroo. It depends upon the owner and the price of stock as to whether it would be better then to send the cattle to the railway or to Wyndham. As a general rule, owners would take advantage of a railway provided the rates were not extortionate and the trucks were suitable. I do not think that a country like Willeroo or to Bitter Springs would pay at the present time. Willeroo, however, would pay better in the near future although the country about the head of the Roper will probably soon stock up. From a cultivation standpoint, you stop a good way short of the agricultural districts at Willeroo. When you get to Bitter Springs you get on to country which may quickly support small agriculturists—men with a few cattle, and who will grow grain of some sort. I think you have a greater difficulty in constructing your line to Willeroo. In my opinion, the minimum area of the best land on the Barkly Tableland that a man could make a good living from, assuming the country was fenced and ample water, and railway facilities for getting his

stock to market, would be about 25 square miles. The tableland would do well with horses, but you would get a quicker return from sheep. It is good, healthy stock country right through, but there are patches in which the minimum would have to be raised. At the present time under existing conditions, I would not attempt to start on the tableland without 300 or 400 square miles.

168. *To Senator Story.*—There are no natural waters along the direct route; horses must be sunk. When I spoke of running to Willeroo, I was speaking from a pastoral point of view and was disregarding the possibilities of the Marranboy Tin Field. There are also big possibilities near the head of the Roper, while the country to the east and north-east of Daly Waters carries 10,000 to 12,000 cattle. Another station carries 6,000 or 7,000. All that country will carry stock, and it has big possibilities with the flats for agriculture when they can find somebody to consume their produce at a reasonable rate. The Victoria River district is carrying four or five times as much stock as the country to the east of the telegraph line, therefore a line swinging to the west would earn a bigger present revenue if they drove their stock to the railway. On the other hand, the country to the east is capable of carrying large bodies of stock, but they are not there yet. If the western route were adopted, there would be greater engineering difficulties, I should say, in the vicinity of Katherine and in the vicinity of Roper Springs, than on the eastern route, and, in my opinion, would cost half as much again as the easterly route. If the railway is intended as a trunk line, the shortest route should be followed, and adjoining districts that are worth developing tapped by branch lines. If you want the biggest revenue from cattle, carry the line towards Wilgeroi, but if you want a general development from other industries, follow approximately the telegraph line. If the Government were to put down a line of bores from about Battle Creek to Katherine, that stock route would serve the Victoria River district very well, and the eastern swing would be preferable from a developmental point of view. We have iron-wood in the Territory suitable for sleepers, but the cost of obtaining them would probably be too great. We have some belts of timber, but I believe it is nearly all piped by white ants. I do not think there is any possibility of getting local timber suitable for sleepers, and, even so, it would have to be treated with some arsenical preparation to make it capable of resisting white ants. If it is considered probable that concrete sleepers may be adopted, there is plenty of sand available in the Katherine River, and there are sand hills and sand ridges, but it may have to be screened. There are lots of ironstone outcrops that would provide good material for concrete also.

(Taken by Sectional Committee at Marranboy.)

MONDAY, 24th JULY, 1916.

Present:

Senator STONY, in the Chair;

Mr. Sampson, | Mr. Laird Smith.
William Thomas Pearce, Miner, and Manager for
Walter Bell and Co., sworn and examined.

169. *To Senator Story.*—The Marranboy Tin Field, in my opinion, is an extensive one, and a permanent one. As far as is known, it is 8

miles in length by about 2 miles wide, but I believe the same class of mineral-bearing country extends for a distance of perhaps 9 or 10 miles and crosses the King River. In regard to the production likely to be expected from the field, I may mention that I know of one property where a shaft 8 feet by 8 feet has been sunk in the lode, and at a depth of 70 feet there are no walls in sight—either foot wall or hanging wall. We know by prospecting that at the bottom of the shaft the lode is at least 10 feet wide, but how much wider it is we do not know. We were going to put in levels at 70 feet, but the stone became so hard that we consider it needs machinery to break it. The stone at the bottom of the shaft at the present time carries about 8 per cent. tin. We have a No. 2 shaft, 8 feet by 6 feet, on the same property. It is down 50 feet, and we are working on the foot wall side of the lode, but we do not yet know the extent of it. There are other claims adjoining that are producing good ore for a distance of about a mile and three-quarters. I think they are on the same line of lode and should prove equally as good as the one being developed. I cannot say that the lode of which I have seen a 10 foot width extends for a mile and three-quarters—it cannot be said that it extends farther than we can see it, but it is the same lode apparently, as it runs right through the same kind of stone as we have. The centre of the field has yet to be prospected. The lode dips under the flats and comes up on to ridges again. I believe there is one claim in the centre of the field that has produced some good stone. At the north end there are several claims, but, being some distance from the battery, they did not seem to receive the same attention as those at the south end. On this field there are two parallel lodes about half-a-mile apart, which are producing ore. One claim is on a lode apparently about 4 feet wide. I believe that if we can get some capital into the place things will improve considerably. There is no question of the grade of the stone, but we are handicapped by the cost of transport. The rate for carting concentrates from Marranboy to Darwin is £20 a ton, while from Marranboy to the railway costs £12 10s. per ton. Assuming that the proposed railway came within 12 miles of Marranboy, cartage would be reduced to approximately £3 per ton. There is no question but that this field could support a fairly large population if we could get capital. We have several miners only too willing to bring up their wives and families, but they cannot see their way far enough ahead. If the railway came within 12 miles of Marranboy it would induce population to come here, as we would then have cheaper transport. This is a good district in which to settle, as there is some good country on the Deswick especially, not more than three and three-quarter miles from the battery to its nearest point.

160. *To Mr. Laird Smith.*—It is good agricultural country, and would grow almost anything; English potatoes grow there at the present time.

161. *To Senator Story.*—The climate is excellent, and until just recently it has had the reputation of being a very healthy place. Men work here quite as well as in other places, and even the hottest part of the year produces no excessive fatigue. Rates of wages are abnormal; miners get 25 per week, but the field is rich enough to pay that rate if we get cheaper transport. At the present time there are just a few claims that show a profit, but some others barely pay 25 per week. The majority of the claims would be payable in positions if the railway came within 10 or 12

miles, provided they worked them by other means than the present drill and hammer; and for that we want capital.

162. *To Mr. Laird Smith.*—The agricultural country I spoke of lies east of the battery and would be within 18 miles of the proposed railway. The railway would be the only outlet for that property. If the railway be constructed on the route proposed, it will tend to induce capital to be invested in this field. I think the field may be considered as being beyond the prospecting stage, although there are still parts to be prospected. The lodes in my shafts are continuous; there are two lodes running parallel to each other, and although I have not had an opinion from anyone of experience, I am inclined to believe that they come together at a lower depth. The richest stone at our 70 feet level carries about 8 per cent. tin, and our smallest percentage was obtained from a bulk crushing from the two shafts, when, from a parcel of 300 tons, we got a percentage of 4.6. I estimate we could send to the battery about 60 tons of stone per week from one property, and about 20 tons per week from the other. Calculating from what we have in sight, I should say we could reasonably expect to carry on for six months in the year. That is, we could send 80 tons of stone per week, carrying a minimum of 3 per cent. tin, for a period of six months. Under present conditions, with cartage at £20 per ton, our shaft, after paying all charges, showed 60 per cent. profit. It cost us £6 a foot to sink the shaft, excluding explosives, and it was payable from the beginning, is still payable, and will be payable as we go down. There is no drumming going on on this field as far as I am aware. In the event of a railway being constructed on the proposed route, a branch line to Marranboy would be costly, as it would have to cross a lot of bad country. There are certain flats where, after a heavy storm, you can walk away the earthworks of a light line. I think the mining is sufficiently developed, however, to warrant the Government taking into consideration the question of railway communication; then there is also that agricultural land I mentioned. We have had no trouble so far in getting rid of our concentrates, but we may find trouble in the wet season. For four months in the year we cannot get anything away from here, but a railway would overcome all that.

163. *To Mr. Sampson.*—The £20 per ton cartage to Darwin means free on board. The £12 10s. per ton is for cartage to the railway at Pine Creek. There are three members in my party, and the claims are taken up individually and comprise the "Star of the East," the "Osman," and the "Montana," the total area being thirty acres. I cannot tell you the quantity of stone in sight, but we value our holding at £20,000 to £30,000, basing our calculations on 4 per cent. stone. I am of opinion that the investigations we have made justify us in assuming that the field is a permanent one.

164. *To Mr. Laird Smith.*—The battery erected has proved of great assistance in the development of the field, and in my opinion we could not have done anything without it. It would be hard to find a better battery anywhere, and the treatment of the ore has been perfectly satisfactory.

165. *To Senator Story.*—In my opinion, the tin-bearing stone may go down 1,000 feet, and a mine of that extent would almost keep the battery going itself. We are prepared, at any time, to guarantee to keep the battery going one shift from our own mine.

Louis Norman Stutterd, Mining Engineer and Battery Manager, Marranboy, sworn and examined.

166. *To Senator Story.*—I have been on the Marranboy field two years and have been over most of it. I am of opinion that the tin-bearing country comprises an area about 6 miles by 3 miles. I could not tell you the exact number of claims from which I have crushed stone, but we are on the 29th parcel now. The line of lode Meares, Pearce and party are on is on the extreme west, and from there the lode runs for about a mile and a half of proved ore values. The value of stone sent in varies very much. Pearce and party put through two parcels, one went 16.1 per cent., while the second parcel of 300 tons went 4.35 per cent. I should say the first parcel was picked out of a rich loader. I do not say it was richer lead. Some of the stone on that line of lode has gone less than 4 per cent. Dunn and Ford's went 1.64 per cent., and Sullivan's went a little over 3 per cent., perhaps 34 per cent. From the parallel line of lode farther south we have received some good values. The Bull Syndicate went, I think, 10 per cent., or 10.1 per cent.; the prospectors Schamber and Richardson sent stone which went about 134 per cent. There is a distance of about three-quarters of a mile between the two parallel lines of lode, but, as a mining engineer, I think they are too far apart to come together at a depth. As to how deep the lode extends, there has not been sufficient work done to say, but the indications are good. Outside of the two lines of lode, and further to the north, there are the claims of Barton and Fischer and of Teese and Elliot. Barton and Fischer sent in seven tons that went over 40 per cent.—that was picked stone. Then they sent in a parcel of 403 tons of second-grade ore that went about 84 per cent. They are on what we call the middle field, and I think are down about 20 feet. Teese and Elliot have a parcel of about 85 tons in now awaiting treatment. I should say, by the look of the stone, it should go about 7 or 8 per cent. As a mining engineer, I should say that if railway facilities were provided the indications are good that this field may develop and carry a fairly large population. Of course, the deepest shaft on the field is 70 feet, but it is good at the bottom. Further south Hutchinson's is 60 feet deep, and that went 5 per cent. for 100 tons. As to the lowest percentage that would pay under existing conditions of breaking the stone, taking all things into consideration, I should say that 4 per cent. ore would leave a fair profit. But it all depends upon skill and method. Machine drills would have to be used, as the present hammer and drill work is too slow and costly.

167. *To Mr. Laird Smith.*—I have had a wide experience in Tasmania. The field is altogether different from anything I saw over there. We there have big ledes and low-grade materials, but there is nothing different about the treatment. We can get over any difficulty of any metallic iron. We have no need for pumping machinery here. Conditions being as sufficient; getting rid of water would not be costly. In regard to the receipt of sufficient material to keep the battery going I may say that we have had one stop. We ran out of stone, and had to stop for ten and a half shifts. We have very great difficulty in obtaining efficient men for the battery. At the present time we are more than keeping pace with

the amount of stone produced. Taking the averages since the battery started, I have brought it down from 2,000 tons—to about 800 tons now. I do say there have at times been rich hand-picked parcels sent in to enable the mine owners to carry on. If a railway existed within a reasonable distance 1 per cent tin would give a handsome profit. I know the "Great Boulder" in Tasmania, the ore is of a very refractory nature. I have found nothing of that sort here. As manager of the battery it would be a great advantage to me to run three shifts if the ore were in sight, but I could not recommend the adoption of the three shifts unless I could see a probability of keeping the battery fully employed. It is important that I should have good men on the battery tables, and to get them I have to import them, but you cannot get good men to work for a month or six weeks and then put them off until more work is available. I do not class the tin here as very fine tin, and it is not as difficult to treat as the "Reunion Bell" in Tasmania. I find the battery as up to date as any of which I have had experience. During nine months in the year there is plenty of water for three shifts, but for the other three months it is doubtful if there would be enough for even one shift. At the present time it is hardly necessary to conserve water, as the mill has been idle for the past six months on two shifts. The capital for development is in most cases coming out of the mines. Some people have put in perhaps £100 or £200, but I think most of them have got it back again. It is hard to say whether the development carried on up to the present would justify the construction of a railway to with a twelve miles of Marzabon, although it would be a great assistance to the field if it were put through. I know of no mine in Tasmania that would stand the same charges and pay 50 per cent.

168 *To Mr. Simpson*—If we were working the battery three shifts we could put through about 45 to 50 tons of stone per day, and that would yield on an average 4.9 per cent tin. The dolly out has been 65 tons of tin running one shift. The freight to Pine Creek is £12 10s per ton, and I believe if you take a truck you get it through from there to Darwin for 12s 6d per ton. Wharfage is about 10s per ton, and covers, as far as I know cost of loading and unloading. I do not know what arrangements are made at Pine Creek, or whether any additional cost is incurred in putting it on the trucks. There is no difficulty in carting from the mines to the battery—we can cart at all times. I should say the shafts should be sunk to 200 feet before we could be satisfied as to the permanency of the lode. With the railway constructed from Pine Creek to the Katherine the teamsters would probably charge £5 to £6 a ton to cart to the latter place, that line would save £9 to £7 a ton. If the railway were brought within twelve miles of Marzabon the ore should be run across for 35s, or at any rate £2 a ton. To deal economically with the ore on this field it is necessary to sink main shafts, install compressors rock drills, and winding plants. A decent compressing plant would cost about £200, but in the case of Pearce and party, for instance, that would be sufficient for their three leases, although they would need three winding plants. It would mean very heavy expenditure to install up to date machinery.

169 *To Mr. Laird Smith*—I should require a man mun of 500 tons of stone a week to warrant putting on three shifts but if that amount were put in for three months in the year I would certainly recommend three shifts. To bring

in 250 tons a week would mean tram lines. I think tram lines would be better than an aerial system. We burn wood for the boilers, for which we pay £1 a cord, which is equal to about £2 10s a ton. There is no possibility of getting coal here on account of the freight. Timber for mining purposes is obtainable, although not plentiful. In my opinion only the main levels and shafts would need timbering.

170 *To Senator Story*—It would be a great advantage to the field to have a railway, it would help in lots of ways. If there is rich pastoral country on the eastern side of the telegraph line I should certainly advocate the extension of the railway in this direction. I know nothing of the pastoral country further south.

(Taken by Sectional Committee at Darwin.)

MONDAY, 7TH AUGUST, 1916

Present

Senator Storr in the Chair,

Mr Sampson | Mr Laird Smith.

Robert Charles Waters, Assistant Surveyor, Commonwealth Railways, sworn and examined.

171 *To Senator Story*—I have been engaged since September last on the survey of the section of the proposed railway from Warlock Ponds to Daly Waters. I know nothing of any part of the Northern Territory excepting that along the route of the proposed line. It is 27½ miles from Darwin to Warlock Ponds. For 20 miles south of Warlock Ponds there are medium earthworks required, but from there to Birdum Creek (31½ miles) they would be very light. A bridge about 160 feet long is required across the Birdum. From Birdum Creek to Daly Waters (35½ miles) it is practically flat country, which would require little earthworks, and presents few engineering difficulties. The country between Warlock Ponds and Birdum Creek is mostly red sandy loam, and from Birdum Creek to Daly Waters it is mostly gileghoi country with patches of red sandy loam. Gileghoi is the name given to holey yellow clayey water washed country. Only one bridge will be required between Warlock Ponds and Daly Waters. From Katherine to Warlock Ponds is 75 miles by the railway route, and I have no knowledge of the bridges required. I have no knowledge of the Willeroo country which would be traversed by the proposed western swing. The principal herbage on the red sandy loam country is cane grass, which is good feed in the rainy season, but towards October and November there is not much left of it, and what is left is very dry, and I know it is very poor feed for horses and probably it is no better for cattle. There is no Mitchell grass, but a little *Flinders* grass grows along the banks of the creeks. Warlock Ponds is supposed to be permanent water, but I get very low at times, last year it was very low. Water required for locomotive purposes must be obtained from bores. The bore put down at No 2 Well is reported to be yielding 30,000 gallons a day.

172 *To Mr. Sampson*—There do not appear to be any materials suitable for concrete making on the other side of Warlock Ponds—there is no sand there at all. I have had no experience of concrete sleepers. I think the country between Warlock Ponds and Daly Waters is worthless from an agricultural point of view, as I do not know anything that would grow there. The red sandy loam might grow cereals if you could get the rain at the proper time, but between Warlock Ponds and Daly

Waters the rain is not certain even in the wet season. One season they had only 3 inches, while last year I estimated they had there 40 inches. I know the lowest record at Daly Waters is about 9 inches, but I am speaking of No 2 Well. The water obtained from the wells is good drinking water and good stock water, and the supply is said to be satisfactory, although I do not know whether it has been tested properly.

173 *To Mr. Laird Smith*—I know of no engineering difficulties on the section of the route I surveyed. There are no heavy grades or sharp curves. It is flat country and first class country through which to run a railway. There is plenty of ballast and plenty of earth for forming close to the proposed route, and the country there is not liable to inundation.

Nicholas Waters, Inspector of Police, Darwin, sworn and examined.

174 *To Senator Story*—I have been 34 years in the Northern Territory, and have travelled a great deal over it. I have been also along the telegraph line as far as Powell's Creek. It is very mixed country, some good, some bad. I should say it is fair pastoral country, and is rather better a few miles away from the telegraph line on each side than it is on the actual route of the telegraph line. Water is not very plentiful—there are wells here and there but I believe there has been a good deal of trouble with them, and I understand that bores are being put down now. I think bores are cheaper and more satisfactory, as sometimes the wells cave in, or the buckets get broken away, or they become contaminated. I think the line to the south should go in the most direct route and follow as far as practicable the telegraph line, if it is the intention of the Government to construct a main trunk line. In my opinion the best scheme you could have would be to first construct the main trunk line on as direct a route as possible, then make a branch line to tap the Victoria River Downs country, which at present carries two thirds of the stock in the whole Territory, then construct another line to go into Camooweal. I think the country between Katherine and Willeroo is slightly better than the country between Katherine and Daly Waters. From an engineering point of view I believe the country on the Bitter Springs side is more level and presents fewer difficulties. I think it is an urgent necessity to have a through railway, otherwise the country will never go ahead in point of development. I think the construction of the line should be pushed on from the northern end, as there is a considerable amount of plant here, later on it should also be constructed from the south, as you would then be working in a better climate and with cheaper labour.

175 *To Mr. Sampson*—I am of opinion that the Barkly Tableland country should be served from Darwin. Of course the Queensland Government would join in, they already intend to stand to continue their line to Camooweal. It should then be with the people themselves whether they prefer going to Darwin or to Townsville. There appears to be a fair depth of water for any proposed harbor at the Tallow Islands, but I do not know about engineering difficulties. I think it is much more important to have a line from Camooweal than a line to Borroloola. It would not be much use constructing a line towards Willeroo unless you continued it into the Victoria River Downs country. I prefer a direct route to the south. Between the proposed easterly and westerly swings is a stretch of very dry country on

which stock would not do any good excepting for five or six months perhaps in the wet season, then they would have to be shifted. If I were asked to disregard a main trunk line and were asked whether it was better to construct 60 or 70 or 80 miles of railway from the Katherine, I should certainly say construct it to Willeroo rather than to Bitter Springs or the Ilkeby, because the Victoria River Downs country and south of Tennant is the bulk of the Territory, provided it has water. I understand that Dr. Chiewing has reported that water has been found amongst the sand hills at shallow depths. I have visited the Victor a River Downs, Delamere, and Willeroo, and have been along the Victoria River to the Ilkeby. At some little distance from the coast the good country starts. Below Victoria River Downs the country improves—the grass gets better, it is shorter and not so rank, and makes excellent hay. From Willeroo south to Victoria River Downs the country improves, but I am afraid it has no agricultural future as it would cost too much to get the produce to market. I do not know the Barkly Tablelands, except by reading reports and talking to people who have been there. The country from Katherine to Daly Waters may have a future from a stock standpoint, though it is a little patchy. If a man had a bore there, and plenty of water, and 200 square miles of country, he could get a decent living out of it. From Newcastle Waters to Powell's Creek the country is flat, it does not flood a little beyond Newcastle Waters, but it is soft country, which gets very boggy during the wet season—so much so that you cannot run a dray or buggy across it. You must keep to the track otherwise your horses would disappear.

(Taken by Sectional Committee at Darwin.)

WEDNESDAY, 9TH AUGUST, 1916

Present

Senator Storr, in the Chair,

Mr Sampson | Mr Laird Smith.

John Anderson Gilruth, Administrator, Northern Territory, sworn and examined.

176 *To Senator Story*—I am aware of the objects of the visit of the Sectional Committee to the Northern Territory, and at the request of your Committee have prepared the following statement.

The linking up of the northern part of the Territory, directly or indirectly, with the great centres of population by means of a railway will prove of the utmost advantage to the Territory, and cannot be gainsaid. But that any system of railway within the Territory proper will prove a payable undertaking in the near future is more than doubtful. I presume that it may be taken for granted that the chief reason for the construction of a railway between Darwin and the south is connected with the defence of Australia and the danger especially of enemy occupation of the Darwin and Bynoe harbors—I believe the principle of the only effective natural harbors for large ships on the northern littoral of Australia. At all events but for these harbors and defence generally one sees little valid argument for the north-south Transcontinental railway in the near future. Being therefore primarily a strategic rail, way obviously it should connect with the most densely populated portion of Australia by the most direct route reasonably consistent with the development of the intervening country. By far the

greater portion of the population of Australia is to be found within that area, which may be bounded on the north-west by an imaginary line from Rockhampton to Adelaide, the most populous centre being Sydney. It has been already agreed, however, that connexion should be made with the south-western corner of that area. There only remains to be considered therefore the best route to subserve the economic development of the intervening country and the possibility of linking up, by one or more branches, with other railway systems to add to its effectiveness from a strategic stand-point. The present section of railway construction is approaching Katherine, from which point it is undecided apparently what exact route shall be followed. Evidently there are under consideration—(1) a direct southerly route, presumably in a bee line to Oodnadatta, or at least Alice Springs; (2) a somewhat indefinite westerly swing via Willeroo to meet the telegraph line again at Renner Springs; and (3) that via Mataranka, approximately following the overland telegraph line. No. 1 does not seem to have anything to commend it. Between Katherine and Macdonnell Ranges it would not pass a single permanent water. Until Sturt's Plain, north-west and west of Newcastle Waters, is reached the country on the whole is indifferent pastoral country, even were it well watered. A central line would tap only one corner of good country and have to run through a lot of rough country south of the Katherine.

177. To Mr. Sampson.—From Wanggalla to the south in a straight line you would pass through inferior to medium pastoral country.

178. To Senator Sturt.—South of the 18th or 19th parallel the country generally that would be traversed would be sandy and spinifex (tridax) covered, of comparatively little pastoral value, till the Burt Plain, north of Macdonnell Ranges, is reached. The Burt Plain extends north-easterly to almost the 31st parallel in the vicinity of the Davenport Ranges, but there is no permanent water. We might obtain water from wells, but the indications we have gathered from Anthony's Well incline me to the belief that the supply would not be great. No. 2.—The Railways Commission plan No. 4 shows the westerly swing recommended as proceeding direct from Katherine to Willeroo Station in a south-west direction, thence in a south-east direction to Renner Springs, the distance being estimated at 320 miles as against 310 by the partially surveyed route on the east near the telegraph line. But doubtless a survey would raise the mileage of the Willeroo swing very considerably, that is their estimated mileage owing to the broken country necessitating deviations from the straight line. South and south-west of the Katherine the country is broken and rocky, and of small pastoral value. The King River, one and a half to two chains wide, with high banks and many creeks, would require to be bridged. A line would therefore probably run nearer and almost parallel to the Katherine River till almost north of Willeroo. It would seem that the time for such a route to be adopted, if at all, was prior to the extension beyond Pine Creek or even Brock's Creek being commenced. The general trend of the direct railway to Oodnadatta or even Alice Springs would be east of south, and a sudden deviation back to westward from the Katherine crossing would require a strong reason for its adoption. The Willeroo deviation would only supply the northern extremity of the vast Victoria River district, traversing the Willeroo and part of Dela-

more Stations alone. After leaving the last country west of Delamere the line would traverse north and west of Sturt's Plain at least, the class of country I briefly describe later. The map shows that such a swing would not assist very greatly the southern portion of the Victoria River country, at all events hardly sufficient to encourage wool-growing instead of cattle raising. From Willeroo southwards the country is poor along the route to the main stations. To avoid that bad country it has recently been decided to provide a new stock route. This will start from the Katherine River at a distance of from five to ten miles from the telegraph station, then run in an irregular manner until it crosses the King, then go to Wanggalla; then up the Dry River until it reaches a point about two east of Delamere; thence it will strike the heads of several creeks until it reaches Wave Hill Station. That stock route will be provided with bores and will give a better route for cattle. No. 3.—As to my reasons for "advocating construction to Mataranka at present time," which are requested, I think there must be some misunderstanding. I have advocated the construction of a line to Mataranka, but I have never advocated Mataranka as a terminus. I recommended it as a point to which the survey might well proceed, and even construction, pending consideration of the exact line beyond. Up to a few months ago we know very little of the country between the proposed easterly and westerly swings. The country to the east and west was picked out not because it was any better, but because of the existence of known permanent waters. As a terminus Mataranka would be equally as futile from a profit-making point of view as for defence for many years to come. My reasons for advocating that the line should go via Mataranka and the Warlock Ponds near All Saints' Well are (1) Beyond Katherine Crossing, in the vicinity of Maude Creek, a considerable area of black soil country, which I have every reason to hope may be suitable for agriculture, would be served. (2) The line will be within twelve miles of Murrumbidgee Government Battery, the centre of a very large tin field at least twenty miles in extent, which has every prospect of proving a very valuable asset to the Territory. The battery is forty miles from Katherine by the road, which is very sandy. A railway within reasonable distance will do much to stimulate development, and by reducing cost of transport will enable the lower grade lodes to be worked. (3) Although the country beyond the Maude Creek, and especially beyond the King River, is poor in quality, there is a marked alteration noticeable as Mataranka Springs are approached. From about the junction of the Beswick and Roper Creeks onward to the source of the Roper River (Mataranka) there are considerable areas of black and red rich looking loamy soils near the latter creek, while permanent waters are fairly numerous. At Mataranka is the source of the Roper, which begins in a series of springs whence rushes a large never-failing and rarely varying body of beautiful looking clear water slightly above normal temperature of water in that region, very slightly alkaline but quite potable. The statements that this water is unfit for human consumption are, to my mind, unfounded. There have been no complaints from the sheep station, and my own experience is that people soon become quite accustomed to the slightly unusual taste. From the source to the junction with the Elsey Creek about two miles beyond in a direct line, the right bank of the winding Roper, is more or less bordered generally by an extensive black soil plain extending back-wards for from about one-half to two miles towards

higher and more sandy country, well grazed and with many edible shrubs. The same class of country, which is excellent for horse-breeding, extends intermittently along the Elsey Creek to the Warlock Ponds. This black soil is very rich. In the wet season surface water lies in places for short periods, but it is rarely really flooded. I have made myself thoroughly acquainted with this country under review by visits and examinations at different periods of the year and under all conditions, and am firmly of opinion it forms the most favorable site for an agricultural settlement, provided it is supplied with railway communication. Further, if any agricultural settlement here, with the proper class of settler, does not prove successful, attempts at agriculture may then be entirely abandoned by white people in the Territory. The situation is high, it is almost as far from the coast as is possible north of the 17th parallel, rainfall is good; water supply good, and inexhaustible. (4) The railway here would also tap the Roper River Valley, which contains the only extent of country so far classified by the Land Board as first-class pastoral country. (5) Again, a glance at the map will show that as the country is developed, Darwin will become more and more vulnerable as the capital on account of its situation, a more central administration point will become necessary. This, in addition to the fact that residence in the tropics should be as far from the seaboard as possible, especially for women and children. Despite other opinions expressed to the contrary, I am convinced that the neighbourhood of the Roper River source (Mataranka) offers the very best site available in the Territory for an inland city, because of altitude, its distance from the sea, central situation, good soil, and beautiful river. Darwin is on the coast, the climate is humid, and the water supply even now insufficient at times for requirements; there is no supply nearer than fifteen miles, and that supply would require to be pumped. It is understood, of course, that there is no suggestion of immediate change of capital site, but in considering a route for a Transcontinental railway it would seem obvious that the probability of some such central city becoming a necessity in the future should be considered, especially in a country badly watered by nature on the whole. Extension beyond Mataranka, between All Saints' Well, south of Mataranka, and Daly Waters feature surveys of the unoccupied country along the telegraph line, and on each side show it to be on the whole fair to medium grazing land without (excepting perhaps one or two very small permanent waters. To the west the unoccupied land is even poorer. Bordering the Birdum Creek there are areas of very rich black soil overlying limestone. The telegraph line follows the course of the Birdum Creek, which runs only after heavy rains. Between the Warlock Ponds (near All Saints' Well) and Daly Waters there are no permanent billabongs. Travelers are supplied by Government subs. Boring for sub-artesian water is now being prosecuted by the Mines Department along this route. The first bore has been sunk to a depth of 120 feet, resulting in a supply of at least 60,000 gallons per day of good potable water. There is no reason to anticipate less favorable results further along the line. Were that route adopted, which, I believe, has been already surveyed nearly as far as Daly Waters (although no official data are apparently available), and runs close to the overland telegraph line, railway construction employees will thus have ample water

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supply by wells and lodes. Another point against the central route is that if we leave the river we are less likely to get aridness at it. Between Daly Waters and North Newcastle Waters (except one small billabong near Daly Waters) there are again no permanent surface waters. Sturt's Plain, an enormous area of well-grassed undulating greyish black soil, which is first encountered on the overland route south of the Frews Ironstone Ponds (not permanent) stretches to the westward for about eighteen to twenty miles, and northward irregularly to the vicinity of Milner's Well on the west side of the line. This Sturt's Plain is of much the same nature, though not quite so rich, as the plains of the Barkly Tablelands with which it is continuous. At present, though portion is held on lease, it is unstocked because of the absence of surface water. There seems no reason to doubt, however, that, as elsewhere on the Barkly Tablelands, sub-artesian water will be procurable at comparatively shallow depths. North of Milner's Well, which is situated almost on the divide, the drainages of the country is towards the Roper by means of the Birdum. South of Milner's Well drainages is southwards via Newcastle Waters (permanent) to Lake Woods (not permanent). The Newcastle Waters country, including Sturt's Plain, marks the extreme western border of the great Barkly Tablelands. It rarely comes as far westward as the overland telegraph route south of South Newcastle Waters, about ten miles below the homestead, although about Renner Springs, which is a small spring, permanent bays, so to speak, of the tableland plains come again within a few miles of the line. Beyond a small plain of grey black soil with tableland characteristics, north of Banksia Banksia Station, this class of country is not again met with on the telegraph line. To the west of the overland telegraph line, between approximately the 18th and 22nd parallels, the country is poor and sandy, with the exception of scattered patches, although the numerous native wells indicate that water is probably procurable at shallow depth. The line traverses on the whole, till the Burt Plain, near Anthony's Well, is reached, similar country with no permanent waters. Eastward of the telegraph line the country improves. The Davenport Ranges comprise good pastoral country fairly well watered. The tableland does not quite reach to that latitude, however, so far as our surveys indicate. Between the Davenport and Macdonnell Ranges the country is good pastoral land, devoid of permanent waters, but with every indication that water may be found at shallow depths by bores or even wells. It will be noted from the above that a railway following the present telegraph line between the South Newcastle Waters and the Burt Plains above the Macdonnell Ranges would at most only skirt any good, or even fair, pastoral country whatever, i.e., over a distance of about 300 miles by the route partially surveyed. I may remind you at this juncture that the railway systems of two States have at present termini within a hundred miles of the Territory borders, viz., Selwyn (Duchess Mine) in Queensland, and Oodnadatta in South Australia. Extension of the Queensland line will help to tap the richest pastoral country of the Territory—Barkly Tablelands. This extension is gradually being proceeded with, I understand. The Barkly Tableland roughly forms a rhomboid with its western boundary at or near the telegraph line, between 17th and 19th parallel; its eastern boundary, so far as it lies in the Territory, being the border

between the north of latitude 18 deg. and 21 deg.; and the northern and southern boundaries being irregular but fairly direct lines between the respective northern and southern points. *A priori* it would seem reasonable to anticipate that any transcontinental railway would endeavour to simply and develop this large tract trending south-east and improving as it extends, instead of simply skirting its least valuable border and thereafter for hundreds of miles immediately serving nothing of any great practical value. For these reasons I suggest the railway leave the telegraph line somewhere between Milner's Well and Newcastle Waters, at a point to be determined later, thence via Anthony's Lagoon, Brunette and Alroy stations, or thereabouts. From Alroy Station to Camooval is but 160 miles. The Townsville western railway is being extended, and will probably reach Camooval or some other point on the Georgina at no distant date. A junction between the two systems would then probably become imperative. The gap in the coastal railway system of Queensland would doubtless be filled in, even if the proposed linking up of the inland terminus were not completed, and a definite rail connexion between the populous south and the empty and dangerous north established. Between the Barkly Tableland and the Macdonnell Ranges from the evidence available, there is only one length of about sixty to seventy miles of poor sandy country between the better pastoral countries through which the railway would require to cross were it again to turn southward to meet the telegraph line about Alice Springs or Anthony's Well. The feature survey of this has not yet, however, been completed, but the maps in the Department of External Affairs, prepared by Chief Surveyor Day, show the class of country below the Davenport Ranges. It will be understood that the points mentioned above are but very approximate; the map is very empty of names, and what is thereon must be utilized. Briefly, what I suggest for serious consideration, whatever northern "swing" may be decided upon, is a deviation of not more than 120 miles to the east of the telegraph line at any point. Such a deviation, it seems obvious, will possess the following advantages:—1. The Katherine-Oodnadatta route would not be materially lengthened, and the swing will present absolutely no engineering problems. I question if one bridge will be required. I believe the only bridge likely to be required from Katherine to Newcastle Waters is that over the King River. 2. It will miss a long stretch of poor pastoral country, which the more direct route would traverse. 3. It will help tremendously to develop our best tracts of country nearly equal in extent to half the area of the State of Victoria. 4. Eventually a very short branch will link up with the Queensland railway system. 5. The total amount of railway construction if the branch to Queensland from Alroy be included will not be greater than without the swing plus the already proposed branch to Queensland from Newcastle Waters. In other words, supposing we were to construct a direct line and then put in a branch from Newcastle Waters to Camooval and compare that with the line I suggest and its branch, my suggested line would be a little shorter, but I contend that even if it were a little longer, it would be justified. 6. A branch from longer, it would be justified. 6. A branch from Anthony Lagoon may in the future be able to Port McArthur, where information available indicates it is likely to be found the best situation in the whole Gulf of Carpentaria for a good harbor to accommodate deep sea boats. I believe there

would be no difficulty in constructing a line right on to the island—in fact the natives walk across to the island now at low tide. Given such conditions, Port McArthur would be the natural outlet for much of the north-west of Queensland. 7. The early linking up with the Queensland system and railway connexion with the populous south will be expedited. 8. This scheme will not appreciably retard the extension of the main railway system to the Macdonnell Ranges and Oodnadatta. 9. Assuming the branch to the Gulf later on, then the natural outlet for export produce from north of the Macdonnell Ranges, east of the telegraph line may prove the northern port and not the southern, judging by the distance on the map. If so, this will again assist in peopling the north. In the foregoing I have not attempted to criticize the opinions of others. I have merely endeavoured to set before you what appeals to me as being the best route for the railway to take, bearing in mind the different factors. During the past four years I have seized every opportunity of becoming personally acquainted with the Territory, and I have not confined myself to the periods of the year when travelling is most comfortable, or when the country looks best. I have discussed the difficulties and possibilities with practically every one who has wide and long experience of the country, hence should be in a position to form a fairly just opinion. I do not propose that the swing I suggest should extend further than about 120 miles east of the overland telegraph. It would lengthen the line by perhaps 160 miles, but as against that it cuts down the length of the branch to connect with Queensland. There is this further to be said that, although the distance is increased by 160 miles, the cost would not be increased proportionately, because the overland line goes through rough country which would require bridges, whereas going further to the east you would avoid the necessity for a single bridge.

179. To Mr. Sampson.—Treating the line as purely one for local development I am satisfied that a careful consideration of the stock route would lead to the abandonment of the proposed swing to Wilbaroo. I do not think that country is associated with a main line proposition. As regards a suggested capital at Mataranka, I would mention that most of the capitals of Europe are inland, although it is admitted they are mostly on rivers. I would add, however, that even the Roper could be made navigable for small boats at comparatively little expense. In the future I believe it would be possible to ultimately bring small boats from the mouth of the Roper, by a series of locks, to Mataranka. My support of the extension of the present line to Bitter Springs is absolutely associated with a further extension to tap the Barkly Tableland. Connexion with the Barkly Tableland is vital to its development, but a connexion with the Victoria River country is not vital to its development, of course with the qualification that any railway communication with an area vastly improves it. I want, however, to go through the tableland so that it may be developed with sheep. I have also thought of the possibility of growing wheat in some of the sandy loam areas. I ordered some samples of Indian wheat from the Punjab, but they arrived too late for last season. I intend to get fresh samples next year. I have always been of the opinion that the construction of railways might be made to proceed hand in hand with the settlement of the country by insisting that the

APPENDIX.

EXTRACTS FROM REPORT BY MR. L. N. MARRANBOY, MANAGER OF THE MARRANBOY BATTERY, DATED 17TH JUNE, 1910, ADDRESSED TO THE DIRECTOR OF MINES.

THE MARRANBOY FIELD GENERALLY AND THE PROSPECTS OF THE CLAIMS.

The Marranboy field as at present prospected and opened up covers an approximate area of 8 miles by 5. At the end of May there were held about 33 mineral reef claims and ten approved protected leases. The former were held or controlled by about 23 individuals, or parties, and the latter eight.

Speaking broadly, the amount of developmental work carried out is practically nil, for only in one or two instances has any attempt been made to open up and prove the lodes in depth and values in anything approaching a workable manner.

Differently situated and endowed with a more energetic and capable class of prospector, this field could be regarded as a so-called "poor man's field," but under conditions existing the lodes are large, but values were picked up on the surface, and in the majority of cases have been proved to continue and improve in value and also with sinking.

If the time spent scratching about the surface for the over elusive leader, fluster, &c., had been spent in sinking on the fine, well-defined lodes, which are an almost self-evident fact in evidence, Marranboy would most assuredly have had a more prosperous and brighter record to-day. The almost general method adopted has been to pick up from the surface and scratch out very shallow depths, which, in the deluded opinion of the owner, would go something from 25 to 50 per cent tin oxide. The best specimens would be placed nicely on the outside and most conspicuous positions in the heap, which would be pointed out to all comers, either out of ignorance or a desire to mislead, as a heap of stone sampled and proved to go oftentimes three or four hundred per cent. its actual percentage in tin oxide. This practice accounts largely for the number of us reporting crushings without any more than a few ounces of tin, whereas those who over estimated were naturally disappointed notwithstanding many warnings to the effect that all which glitters is not gold. The result of this system of working on a number of the claims varies from 20 to 27 tons for a period up to three years.

There are two or three progressive parties and individuals on the field who have turned the limited capital at their disposal to good account. For such a proposition as Marranboy the scope of these is altogether too limited. Capital—real capital—and confidence are required to push things as they should be pushed.

One party has a shaft down 75 and another 35 feet, both in well defined lode carrying good values. Although sufficient has not yet been done to finally determine the width of the lode, it ranges approximately from 7 to 14 feet. A parcel of 315 tons, being the result of sinking, &c., was treated for a yield of 13.1 tons of tin oxide. This, together with a small piece of parcel which yielded 3.3 tons oxide, the whole valued at £1,700, returned a net profit of £850. This instance the dividend—percentage—of a claim or claims worked on legitimate lines.

There are at present six lode systems proved. The central system has been opened up at intervals in a few claims for a distance of 2 miles, and is carrying good values throughout. On the same lode a small piece about 35 feet, and it has produced to date about 270 tons of ore, which should average over 5 per cent. Close by is a shaft over 100 feet in depth, and from it are now awaiting treatment about 150 tons of very fair grade ore.

The lode next but one in a southerly direction has been opened up for about the same length, but not to the same depth. A shaft 60 feet just off the lode has been sunk, and from this and other sources there have been raised about 300 tons of stone. On the adjoining three blocks the lode has been exposed at intervals, and ranges from 4 to 8 and 10 feet in width. On the same lode, and further to the west, there is a specimen for about 150 feet in width, between walls ranging from 4 to 8 feet, the bulk carrying very fair tin.

To sum up, the prospects of most of the claims are distinctly good, that is, worked in an enterprising manner on progressive lines, and in my opinion this may be expected to be brought about only by the introduction of capital judiciously expended.

majority of the employees should bring their wives and families with them. I would encourage them to take up holdings so that their families could remain there while they were working on the railway, and thus form the beginning of a permanent settlement. At the present time the vast majority of the men employed are single, and even if they are saving money it is obvious the money they save is of no use to the Territory. Some men work for six or eight months or a year, and then go away and take their savings with them. Last year I made inquiries amongst the men and did not find anybody who had any intention of settling in the Territory—that applies to people brought from overseas as well as those from the south. If a railway were connected with the south I do not think there is any possibility of settling this country with the overflow from the south, because they do not want to come so far away from their homes. Every State is bidding against every other State to keep its own people, and in any case I do not think it is good policy to encourage the transfer of people from one part of an empty country to another part of an empty country. I would recommend the system of encouraging people from European countries to come here by guaranteeing them work if suitable by paying their passages in part and making it a debt against them until they were settled in the country. I would hold out hopes that if they settled permanently the debt might be foregone, but I would insist on either the people being married and bringing their wives and families with them or being closely related to those already here. I find it is not so much the attraction of his country which takes a man away, but the attraction of his friends and relations. In regard to land settlement I think there are certain areas that, with the expenditure of money could in the future be made available for irrigation areas, but one should be sure that those areas which can be developed without that expenditure should be developed first. I would apply the same condition to applicants for land, i.e., that they be married.

180. To Senator Story.—I think the proposed extension to Mataranka would materially assist in developing Marranboy. Then the question arises as to whether Marranboy should be connected by a 2-ft. gauge tramway with the railway, but I would favour a rough line of the same gauge as the railway and intended solely for the conveyance of ore and goods; that would save the cost of handling. If the railway be brought within twelve miles of Marranboy I think it would induce the introduction of capital for the development of the field. I do not think Marranboy capable of development without companies. My opinion is that to be a permanent field it will require the expenditure of a considerable amount of capital, particularly when they get down a little further. Labour-saving machinery could be used and should be used to the fullest extent. I think that even if the railway does not give complete connexion to Marranboy, the fact that it goes within twelve miles of it will tend to bring people there and add to the possibility of the railway becoming more valuable. There is every indication of Marranboy being one of the biggest tin fields in the world. There is no sign of it petering out as greater depths are reached, but of course one cannot speak definitely until the thing is proved. Even the country rock, I am told, shows traces of tin, and there is every reason to believe that other mineral fields may extend beyond the present confines of Marranboy.

THE CLASS OF CLAIM-HOLDERS.

During the past few weeks there has begun what appears to be a gradual displacement of the undesirable type of original claim-holder by a seemingly more capable and energetic class, but they have not yet become established. Five lease applications, and two or three mineral reef claims applications have been received during the past week or so from applicants apparently of this improved type.

The present claim and lease holders may be divided into at least three classes.

1 First of all we have the individuals or parties who are really confident in the prospects of their holdings and the field generally, and who may be reasonably expected to expend a liberal portion of their returns from stone raised in developing their property. This class, from time to time as finances allow, employs labour irrespective of the limited number required to comply with the regulation labour conditions. The work done by most of those included under this head has been briefly dealt with elsewhere.

2 Secondly there are the claim holders with, for the most part, good holdings, but who, through lack of enterprise and often confidence, and in many cases a keen desire to sell, are merely working their claims with the idea of breaking sufficient stone to return the equivalent of good wages.

Whilst the price of tin remains high, a fair amount of stone may be expected from those; immediately the price begins to drop, however, production will for the most part cease.

3 The third class of claim holder may be altogether ignored both from a battery-supporting stand point and a field developing point of view. The time is not far distant when the convenient and easily obtained surface stone will cease to exist, and when the energy and time expended will not be sufficient to produce enough to satisfy the demands of the most lenient creditor.

THE CLIMATE AND GENERAL LIVING CONDITIONS

Considering the tropical situation of Marranboy the climatic conditions may be proclaimed excellent. The months of May, June, July, and August could be classed as temperate, the temperature falling as low as 44 degrees during the early morning, rising gradually and reaching the maximum shortly after noon. Of the twelve, the months of February, March, November, and December are the most trying, the temperature ranging from 75 during the night and early morning to about 101 at midday. The months of January, April, September, and October are pleasant.

An occasional shower in November and early part of December heralds the approach of the "wet" season. The heavy rain, falling in frequent showers, falls during the latter part of December, January, and early February, when the heavy showers give place to lighter ones falling occasionally, and often as late as May. The registered rainfall for the "wet" season 1915-16 was 47 inches.

During the 1914-15 rainy season, owing to transport facilities being inadequate, stores of all kinds were exceedingly scarce. This, coupled with the fact that little or no provision had been made in the way of kitchen gardens, and the consequent lack of vegetables, combined with the absence of fresh beef, gave rise to general ill-health. This state of affairs has, to a great extent, been remedied, and during the last spell of wet

weather locally grown vegetables, such as pumpkins, water melons, and beans, with a more plentiful supply of fresh beef, resulted in a much improved state of general health.

THE CLASS OF LABOUR.

Generally speaking, men of long residence in the Territory employed here during the construction and subsequently performing physical labour, may be classed as slow and inefficient. This may be due largely to a process of assimilation whereby the better class of labour is able to remain amid congenial surroundings, leaving the unfit to satisfy outside requirements. The more recent arrivals from other States are a decided improvement, but very few could be classed as good. Labour of any class is very scarce locally.

THE EFFICIENCY OF THE CONCENTRATING MILL.

Crushing operations at the battery were begun on the 2nd January of this year after a preliminary run of a few hours during the last day or so of December. Since starting the mill has run one shift continuously, with the exception of ten and a half days lost during May owing to shortage of ore. The production of ore has not at any time warranted the running of more than one shift.

The efficiency of the plant is exceedingly high as regards recoveries, which range from 90 per cent, to 97 per cent extraction. The capacity has been moderate, owing to the extraordinarily hard and tough nature of most of the ores treated, and the necessity of very fine grinding. The capacity ranges from 0.7 to 20 tons per shift, according to the class of ore treated. If necessary, the capacity could be slightly increased by bringing into requisition the third grinding pan, two only being in use at present.

CONCLUSIONS.

Based on the work done and values proved, there is every reason to believe that the Marranboy field will be permanent. I am not aware of a single instance where, in the main lode channels, the lodes or values have cut out at depth although this may not be taken as final until the continuity of the lodes has been proved to a depth of at least two or three hundred feet; also the length of the shoots of tin has yet to be proved, for so far no driving has been done underground for a greater length than about 50 feet. To sum up, developmental work has been so far that it is impossible to state definitely the permanency or otherwise of the field, but the indications are very favorable.

The life of the field will depend on the economical working and handling of the lower grade ores, say, from 11 per cent to 4 per cent. This means capital and efficient labour.

At present the progress of the field depends almost wholly upon class one of the claim holders; classes two and three are not likely to produce much ore.

I anticipate difficulty in running one shift continuously, as I find that at the present rate of crushing and ore production the mill will catch up to the miners in about three months. During the past five months all ore raised has been treated, together with about 1,100 tons of that which had accumulated prior to the commencement of the crushing operations. I estimate the ore at grass at 500 tons, this including the ore already carted and awaiting treatment.